

STEADY HEAT CONDUCTION

In heat transfer analysis, we are often interested in the rate of heat transfer through a medium under steady conditions and surface temperatures. Such problems can be solved easily without involving any differential equations by the introduction of the *thermal resistance concept* in a manner analogous to electrical circuit problems. In this case, the thermal resistance corresponds to electrical resistance, temperature difference corresponds to voltage, and the heat transfer rate corresponds to electric current.

We start this chapter with *one-dimensional steady heat conduction* in a plane wall, a cylinder, and a sphere, and we develop relations for *thermal resistances* in these geometries. We also develop thermal resistance relations for convection and radiation conditions at the boundaries. We apply this concept to heat conduction problems in *multilayer* plane walls, cylinders, and spheres and generalize it to systems that involve heat transfer in two or three dimensions. We also discuss the *thermal contact resistance* and the *overall heat transfer coefficient* and develop relations for the critical radius of insulation for a cylinder and a sphere. Finally, we discuss steady heat transfer from *finned surfaces* and some complex geometries commonly encountered in practice through the use of *conduction shape factors*.



OBJECTIVES

When you finish studying this chapter, you should be able to:

- Understand the concept of thermal resistance and its limitations, and develop thermal resistance networks for practical heat conduction problems,
- Solve steady conduction problems that involve multilayer rectangular, cylindrical, or spherical geometries,
- Develop an intuitive understanding of thermal contact resistance and circumstances under which it may be significant,
- Identify applications in which insulation may actually increase heat transfer,
- Analyze finned surfaces, and assess how efficiently and effectively fins enhance heat transfer, and
- Solve multidimensional practical heat conduction problems using conduction shape factors.

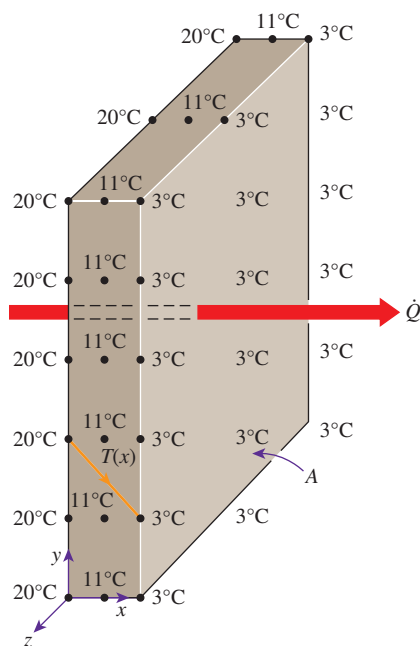


FIGURE 3-1

Heat transfer through a wall is one-dimensional when the temperature of the wall varies in one direction only.

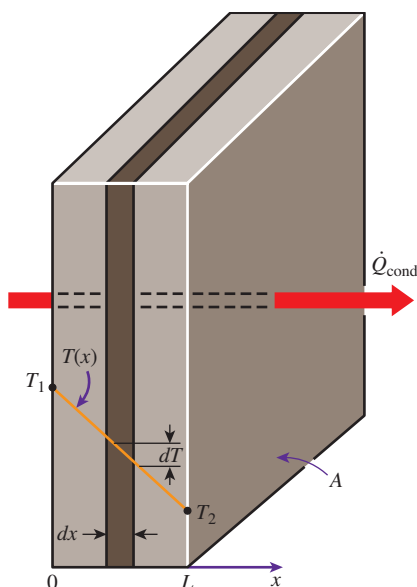


FIGURE 3-2

Under steady conditions, the temperature distribution in a plane wall is a straight line.

3-1 ■ STEADY HEAT CONDUCTION IN PLANE WALLS

Consider steady heat conduction through the walls of a house during a winter day. We know that heat is continuously lost to the outdoors through the wall. We intuitively feel that heat transfer through the wall is in the *normal direction* to the wall surface, and no significant heat transfer takes place in the wall in other directions (Fig. 3-1).

Recall that heat transfer in a certain direction is driven by the *temperature gradient* in that direction. There is no heat transfer in a direction in which there is no change in temperature. Temperature measurements at several locations on the inner or outer wall surface will confirm that a wall surface is nearly *isothermal*. That is, the temperatures at the top and bottom of a wall surface as well as at the right and left ends are almost the same. Therefore, there is no heat transfer through the wall from the top to the bottom, or from left to right, but there is considerable temperature difference between the inner and the outer surfaces of the wall, and thus significant heat transfer in the direction from the inner surface to the outer one.

The small thickness of the wall causes the temperature gradient in that direction to be large. Further, if the air temperatures inside and outside the house remain constant, then heat transfer through the wall of a house can be modeled as *steady* and *one-dimensional*. The temperature of the wall in this case depends on one direction only (say the x -direction) and can be expressed as $T(x)$.

Noting that heat transfer is the only energy interaction involved in this case and there is no heat generation, the *energy balance* for the wall can be expressed as

$$\left(\begin{array}{c} \text{Rate of} \\ \text{heat transfer} \\ \text{into the wall} \end{array} \right) - \left(\begin{array}{c} \text{Rate of} \\ \text{heat transfer} \\ \text{out of the wall} \end{array} \right) = \left(\begin{array}{c} \text{Rate of change} \\ \text{of the energy} \\ \text{of the wall} \end{array} \right)$$

or

$$\dot{Q}_{\text{in}} - \dot{Q}_{\text{out}} = \frac{dE_{\text{wall}}}{dt} \quad (3-1)$$

But $dE_{\text{wall}}/dt = 0$ for *steady* operation, since there is no change in the temperature of the wall with time at any point. Therefore, the rate of heat transfer into the wall must be equal to the rate of heat transfer out of it. In other words, *the rate of heat transfer through the wall must be constant*, $\dot{Q}_{\text{cond, wall}} = \text{constant}$.

Consider a plane wall of thickness L and average thermal conductivity k . The two surfaces of the wall are maintained at constant temperatures of T_1 and T_2 . For one-dimensional steady heat conduction through the wall, we have $T(x)$. Then Fourier's law of heat conduction for the wall can be expressed as

$$\dot{Q}_{\text{cond, wall}} = -kA \frac{dT}{dx} \quad (\text{W}) \quad (3-2)$$

where the rate of conduction heat transfer $\dot{Q}_{\text{cond, wall}}$ and the wall area A are constant. Thus $dT/dx = \text{constant}$, which means that *the temperature through the wall varies linearly with x* . That is, the temperature distribution in the wall under steady conditions is a *straight line* (Fig. 3-2).

Separating the variables in the preceding equation and integrating from $x = 0$, where $T(0) = T_1$, to $x = L$, where $T(L) = T_2$, we get

$$\int_{x=0}^L \dot{Q}_{\text{cond, wall}} dx = - \int_{T=T_1}^{T_2} kA dT$$

Performing the integrations and rearranging gives

$$\dot{Q}_{\text{cond, wall}} = kA \frac{T_1 - T_2}{L} \quad (\text{W}) \quad (3-3)$$

which is identical to Eq. 1–21. Again, *the rate of heat conduction through a plane wall is proportional to the average thermal conductivity, the wall area, and the temperature difference, but is inversely proportional to the wall thickness.* Also, once the rate of heat conduction is available, the temperature $T(x)$ at any location x can be determined by replacing T_2 in Eq. 3–3 with T and L with x .

Thermal Resistance Concept

Equation 3–3 for heat conduction through a plane wall can be rearranged as

$$\dot{Q}_{\text{cond, wall}} = \frac{T_1 - T_2}{R_{\text{wall}}} \quad (\text{W}) \quad (3-4)$$

where

$$R_{\text{wall}} = \frac{L}{kA} \quad (\text{K/W}) \quad (3-5)$$

is the *thermal resistance* of the wall against heat conduction or simply the **conduction resistance** of the wall. Note that the thermal resistance of a medium depends on the *geometry* and the *thermal properties* of the medium. Note that thermal resistance can also be expressed as $R_{\text{wall}} = \Delta T / \dot{Q}_{\text{cond, wall}}$, which is the ratio of the driving potential ΔT to the corresponding transfer rate $\dot{Q}_{\text{cond, wall}}$.

This equation for heat transfer is analogous to the relation for *electric current flow* I , expressed as

$$I = \frac{V_1 - V_2}{R_e} \quad (3-6)$$

where $R_e = L / \sigma_e A$ is the *electrical resistance* and $V_1 - V_2$ is the *voltage difference* across the resistance (σ_e is the electrical conductivity). Thus, the *rate of heat transfer* through a layer corresponds to the *electric current*, the *thermal resistance* corresponds to *electrical resistance*, and the *temperature difference* corresponds to *voltage difference* across the layer (Fig. 3–3).

Consider convection heat transfer from a solid surface of area A_s and temperature T_s to a fluid whose temperature sufficiently far from the surface is T_∞ , with a convection heat transfer coefficient h . Newton's law of cooling for convection heat transfer rate $\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty)$ can be rearranged as

$$\dot{Q}_{\text{conv}} = \frac{T_s - T_\infty}{R_{\text{conv}}} \quad (\text{W}) \quad (3-7)$$

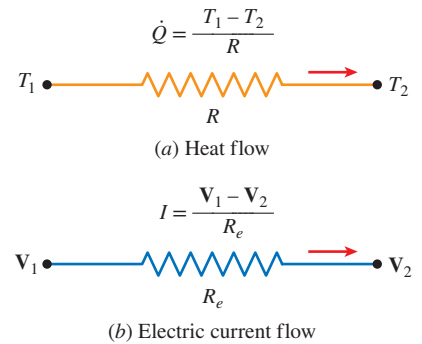


FIGURE 3–3

Analogy between thermal and electrical resistance concepts.

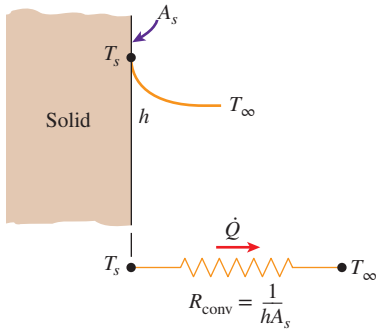


FIGURE 3-4

Schematic for convection resistance at a surface.

where

$$R_{\text{conv}} = \frac{1}{hA_s} \quad (\text{K/W}) \quad (3-8)$$

is the *thermal resistance* of the surface against heat convection, or simply the **convection resistance** of the surface (Fig. 3-4). Note that when the convection heat transfer coefficient is very large ($h \rightarrow \infty$), the convection resistance becomes *zero* and $T_s \approx T_\infty$. That is, the surface offers *no resistance to convection*, and thus it does not slow down the heat transfer process. This situation is approached in practice at surfaces where boiling and condensation occur. Also note that the surface does not have to be a plane surface. Equation 3-8 for convection resistance is valid for surfaces of any shape, provided that the assumption of $h = \text{constant}$ and uniform is reasonable.

When the wall is surrounded by a gas, the *radiation effects*, which we have ignored so far, can be significant and may need to be considered. The rate of radiation heat transfer between a surface of emissivity ϵ and area A_s at temperature T_s and the surrounding surfaces at some average temperature T_{surr} can be expressed as

$$\dot{Q}_{\text{rad}} = \epsilon \sigma A_s (T_s^4 - T_{\text{surr}}^4) = h_{\text{rad}} A_s (T_s - T_{\text{surr}}) = \frac{T_s - T_{\text{surr}}}{R_{\text{rad}}} \quad (\text{W}) \quad (3-9)$$

where

$$R_{\text{rad}} = \frac{1}{h_{\text{rad}} A_s} \quad (\text{K/W}) \quad (3-10)$$

is the *thermal resistance* of a surface against radiation, or the **radiation resistance**, and

$$h_{\text{rad}} = \frac{\dot{Q}_{\text{rad}}}{A_s (T_s - T_{\text{surr}})} = \epsilon \sigma (T_s^2 + T_{\text{surr}}^2)(T_s + T_{\text{surr}}) \quad (\text{W/m}^2 \cdot \text{K}) \quad (3-11)$$

is the **radiation heat transfer coefficient**. Note that both T_s and T_{surr} *must* be in K in the evaluation of h_{rad} . The definition of the radiation heat transfer coefficient enables us to express radiation conveniently in a manner analogous to convection in terms of a temperature difference. But h_{rad} depends strongly on temperature, while h_{conv} usually does not.

A surface exposed to the surrounding air involves convection and radiation simultaneously, and the total heat transfer at the surface is determined by adding (or subtracting, if in the opposite direction) the radiation and convection components. The convection and radiation resistances are parallel to each other, as shown in Fig. 3-5, and they may cause some complication in the thermal resistance network. When $T_{\text{surr}} \approx T_\infty$, the radiation effect can properly be accounted for by replacing h in the convection resistance relation (Eq. 3-8) with

$$h_{\text{combined}} = h_{\text{conv}} + h_{\text{rad}} \quad (\text{W/m}^2 \cdot \text{K}) \quad (3-12)$$

where h_{combined} is the **combined heat transfer coefficient** discussed in Chap. 1. This way all complications associated with radiation are avoided.

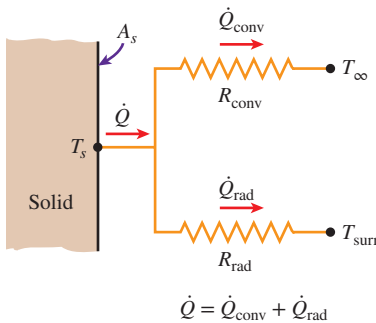


FIGURE 3-5

Schematic for convection and radiation resistances at a surface.

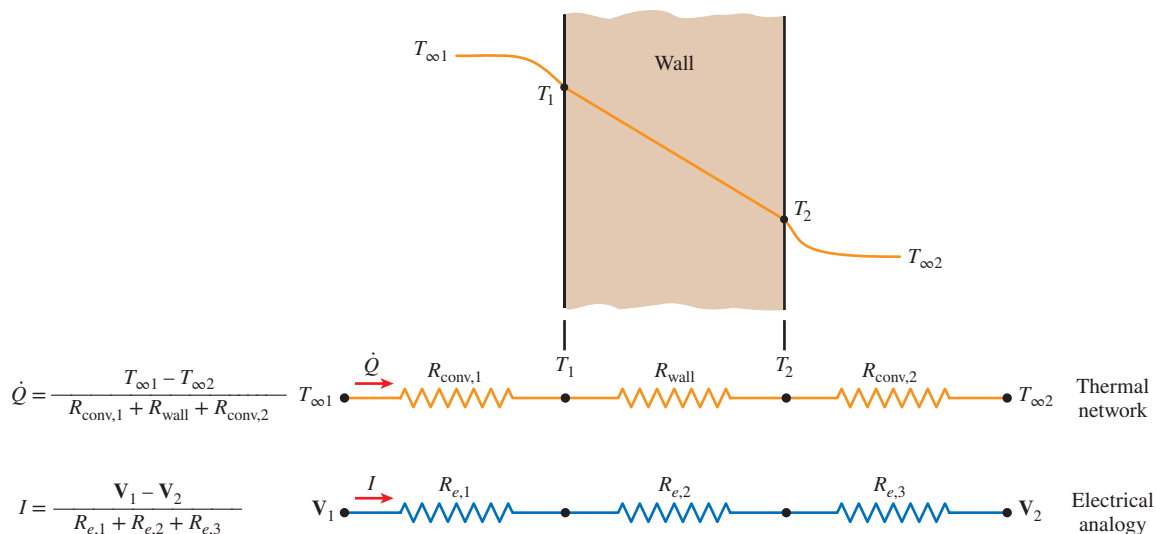


FIGURE 3-6

The thermal resistance network for heat transfer through a plane wall subjected to convection on both sides, and the electrical analogy.

Thermal Resistance Network

Now consider steady one-dimensional heat transfer through a plane wall of thickness L , area A , and thermal conductivity k that is exposed to convection on both sides to fluids at temperatures $T_{\infty 1}$ and $T_{\infty 2}$ with heat transfer coefficients h_1 and h_2 , respectively, as shown in Fig. 3–6. Assuming $T_{\infty 2} < T_{\infty 1}$, the variation of temperature will be as shown in the figure. Note that the temperature varies linearly in the wall and asymptotically approaches $T_{\infty 1}$ and $T_{\infty 2}$ in the fluids as we move away from the wall.

Under steady conditions we have

$$\left(\begin{array}{c} \text{Rate of} \\ \text{heat convection} \\ \text{into the wall} \end{array} \right) = \left(\begin{array}{c} \text{Rate of} \\ \text{heat conduction} \\ \text{through the wall} \end{array} \right) = \left(\begin{array}{c} \text{Rate of} \\ \text{heat convection} \\ \text{from the wall} \end{array} \right)$$

or

$$\dot{Q} = h_1 A (T_{\infty 1} - T_1) = kA \frac{T_1 - T_2}{L} = h_2 A (T_2 - T_{\infty 2}) \quad (3-13)$$

which can be rearranged as

$$\begin{aligned} \dot{Q} &= \frac{T_{\infty 1} - T_1}{1/h_1 A} = \frac{T_1 - T_2}{L/kA} = \frac{T_2 - T_{\infty 2}}{1/h_2 A} \\ &= \frac{T_{\infty 1} - T_1}{R_{\text{conv},1}} = \frac{T_1 - T_2}{R_{\text{wall}}} = \frac{T_2 - T_{\infty 2}}{R_{\text{conv},2}} \end{aligned} \quad (3-14)$$

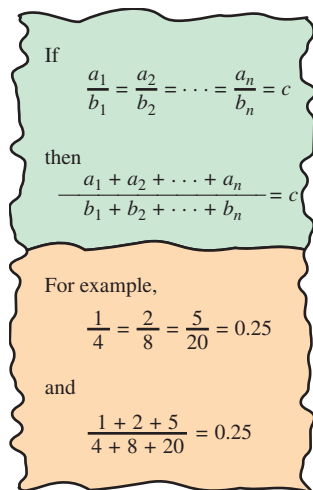


FIGURE 3-7

A useful mathematical identity.

Once the rate of heat transfer is calculated, Eq. 3-14 can also be used to determine the intermediate temperatures T_1 or T_2 . Adding the numerators and denominators yields (Fig. 3-7)

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{\text{total}}} \quad (\text{W}) \quad (3-15)$$

where

$$R_{\text{total}} = R_{\text{conv},1} + R_{\text{wall}} + R_{\text{conv},2} = \frac{1}{h_1 A} + \frac{L}{kA} + \frac{1}{h_2 A} \quad (\text{K/W}) \quad (3-16)$$

Note that the heat transfer area A is constant for a plane wall, and the rate of heat transfer through a wall separating two media is equal to the overall temperature difference $(T_{\infty 1} - T_{\infty 2})$ divided by the total thermal resistance between the media. Also note that the thermal resistances are in *series*, and the equivalent thermal resistance is determined by simply *adding* the individual resistances, just like the electrical resistances connected in series. Thus, the electrical analogy still applies. We summarize this as *the rate of steady heat transfer between two surfaces is equal to the temperature difference divided by the total thermal resistance between those two surfaces*.

Another observation that can be made from Eq. 3-15 is that the ratio of the temperature drop to the thermal resistance across any layer is constant, and thus the temperature drop across any layer is proportional to the thermal resistance of the layer. The larger the resistance, the larger the temperature drop. In fact, the equation $\dot{Q} = \Delta T/R$ can be rearranged as

$$\Delta T = \dot{Q} R \quad (^\circ\text{C}) \quad (3-17)$$

which indicates that the *temperature drop* across any layer is equal to the *rate of heat transfer* times the *thermal resistance* across that layer (Fig. 3-8). You may recall that this is also true for voltage drop across an electrical resistance when the electric current is constant.

It is sometimes convenient to express heat transfer through a medium in a manner analogous to Newton's law of cooling as

$$\dot{Q} = UA \Delta T \quad (\text{W}) \quad (3-18)$$

where U is the **overall heat transfer coefficient** with the unit $\text{W/m}^2 \cdot \text{K}$. The overall heat transfer coefficient is usually used in heat transfer calculations associated with heat exchangers (Chap. 11). It is also used in heat transfer calculations through windows (Chap. 9), commonly referred to as *U-factor*. A comparison of Eqs. 3-15 and 3-18 reveals that

$$UA = \frac{1}{R_{\text{total}}} \quad (\text{W/K}) \quad (3-19)$$

Therefore, for a unit area, the overall heat transfer coefficient is equal to the inverse of the total thermal resistance.

Note that we do not need to know the surface temperatures of the wall in order to evaluate the rate of steady heat transfer through it. All we need to know is the convection heat transfer coefficients and the fluid temperatures

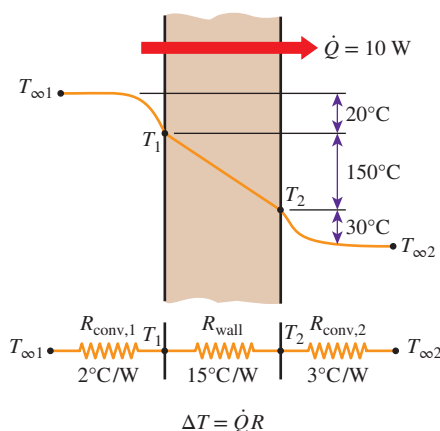


FIGURE 3-8

The temperature drop across a layer is proportional to its thermal resistance.

on both sides of the wall. The *surface temperature* of the wall can be determined as described above using the thermal resistance concept, but by taking the surface at which the temperature is to be determined as one of the terminal surfaces. For example, once $\dot{Q}$ is evaluated, the surface temperature T_1 can be determined from

$$\dot{Q} = \frac{T_{\infty 1} - T_1}{R_{\text{conv}, 1}} = \frac{T_{\infty 1} - T_1}{1/h_1 A} \quad (3-20)$$

Multilayer Plane Walls

In practice, we often encounter plane walls that consist of several layers of different materials. The thermal resistance concept can still be used to determine the rate of steady heat transfer through such *composite* walls. As you may have already guessed, this is done by simply noting that the conduction resistance of each wall is L/kA connected in series, and using the electrical analogy—that is, by dividing the *temperature difference* between two surfaces at known temperatures by the *total thermal resistance* between them.

Consider a plane wall that consists of two layers (such as a brick wall with a layer of insulation). The rate of steady heat transfer through this two-layer composite wall can be expressed as (Fig. 3–9)

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{\text{total}}} \quad (3-21)$$

where R_{total} is the *total thermal resistance*, expressed as

$$\begin{aligned} R_{\text{total}} &= R_{\text{conv}, 1} + R_{\text{wall}, 1} + R_{\text{wall}, 2} + R_{\text{conv}, 2} \\ &= \frac{1}{h_1 A} + \frac{L_1}{k_1 A} + \frac{L_2}{k_2 A} + \frac{1}{h_2 A} \end{aligned} \quad (3-22)$$

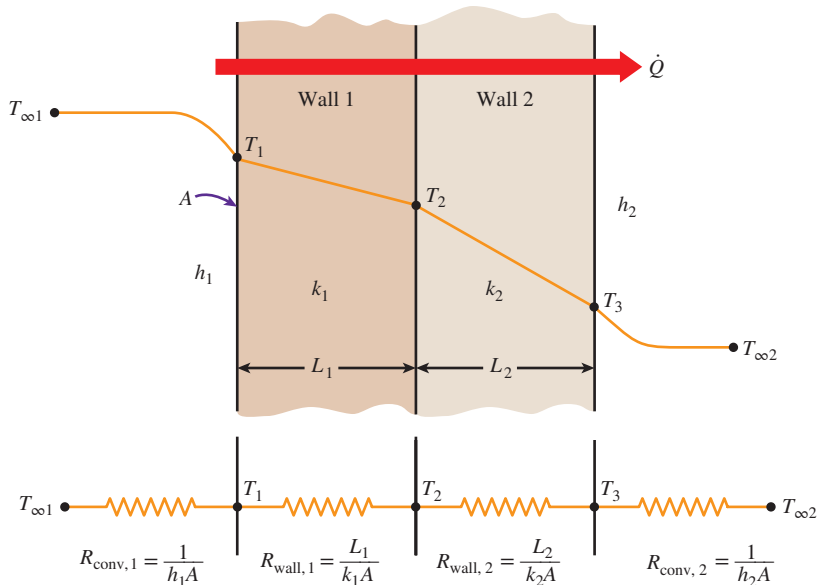
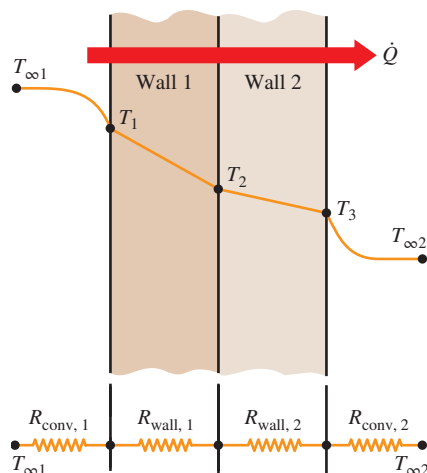


FIGURE 3–9

The thermal resistance network for heat transfer through a two-layer plane wall subjected to convection on both sides.



$$\text{To find } T_1: \dot{Q} = \frac{T_{\infty 1} - T_1}{R_{\text{conv}, 1}}$$

$$\text{To find } T_2: \dot{Q} = \frac{T_{\infty 1} - T_2}{R_{\text{conv}, 1} + R_{\text{wall}, 1}}$$

$$\text{To find } T_3: \dot{Q} = \frac{T_3 - T_{\infty 2}}{R_{\text{conv}, 2}}$$

FIGURE 3-10

The evaluation of the surface and interface temperatures when $T_{\infty 1}$ and $T_{\infty 2}$ are given and $\dot{Q}$ is calculated.

The subscripts 1 and 2 in the R_{wall} relations indicate the first and the second layers, respectively. We could also obtain this result by following the approach already used for the single-layer case by noting that the rate of steady heat transfer $\dot{Q}$ through a multilayer medium is constant, and thus it must be the same through each layer. Note from the thermal resistance network that the resistances are *in series*, and thus the *total thermal resistance* is simply the *arithmetic sum* of the individual thermal resistances in the path of heat transfer.

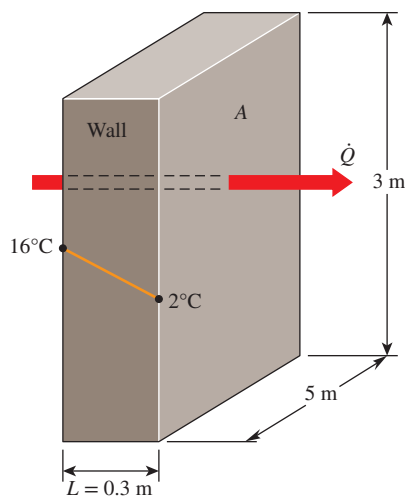
This result for the *two-layer* case is analogous to the *single-layer* case, except that an *additional resistance* is added for the *additional layer*. This result can be extended to plane walls that consist of *three or more layers* by adding an *additional resistance* for each *additional layer*.

Once $\dot{Q}$ is known, an unknown surface temperature T_j at any surface or interface j can be determined from

$$\dot{Q} = \frac{T_i - T_j}{R_{\text{total}, i-j}} \quad (3-23)$$

where T_i is a *known* temperature at location i and $R_{\text{total}, i-j}$ is the total thermal resistance between locations i and j . For example, when the fluid temperatures $T_{\infty 1}$ and $T_{\infty 2}$ for the two-layer case shown in Fig. 3-9 are available and $\dot{Q}$ is calculated from Eq. 3-21, the interface temperature T_2 between the two walls can be determined from (Fig. 3-10)

$$\dot{Q} = \frac{T_{\infty 1} - T_2}{R_{\text{conv}, 1} + R_{\text{wall}, 1}} = \frac{T_{\infty 1} - T_2}{\frac{1}{h_1 A} + \frac{L_1}{k_1 A}} \quad (3-24)$$

**FIGURE 3-11**

Schematic for Example 3-1.

The temperature drop across a layer is easily determined from Eq. 3-17 by multiplying $\dot{Q}$ by the thermal resistance of that layer.

The thermal resistance concept is widely used in practice because it is intuitively easy to understand and it has proven to be a powerful tool in the solution of a wide range of heat transfer problems. But its use is limited to systems through which the rate of heat transfer $\dot{Q}$ remains *constant*; that is, to systems involving *steady* heat transfer with *no heat generation* (such as resistance heating or chemical reactions) within the medium.

EXAMPLE 3-1 Heat Loss Through a Wall

Consider a 3-m-high, 5-m-wide, and 0.3-m-thick wall whose thermal conductivity is $k = 0.9 \text{ W/m}\cdot\text{K}$ (Fig. 3-11). On a certain day, the temperatures of the inner and the outer surfaces of the wall are measured to be 16°C and 2°C , respectively. Determine the rate of heat loss through the wall on that day.

SOLUTION The two surfaces of a wall are maintained at specified temperatures. The rate of heat loss through the wall is to be determined.

Assumptions 1 Heat transfer through the wall is steady since the surface temperatures remain constant at the specified values. 2 Heat transfer is one-dimensional since any significant temperature gradients exist in the direction from the indoors to the outdoors. 3 Thermal conductivity is constant.

Properties The thermal conductivity is given to be $k = 0.9 \text{ W/m}\cdot\text{K}$.

Analysis Noting that heat transfer through the wall is by conduction and the area of the wall is $A = 3 \text{ m} \times 5 \text{ m} = 15 \text{ m}^2$, the steady rate of heat transfer through the wall can be determined from Eq. 3–3 to be

$$\dot{Q} = kA \frac{T_1 - T_2}{L} = (0.9 \text{ W/m}\cdot\text{C})(15 \text{ m}^2) \frac{(16 - 2)^\circ\text{C}}{0.3 \text{ m}} = \mathbf{630 \text{ W}}$$

We could also determine the steady rate of heat transfer through the wall by making use of the thermal resistance concept from

$$\dot{Q} = \frac{\Delta T_{\text{wall}}}{R_{\text{wall}}}$$

where

$$R_{\text{wall}} = \frac{L}{kA} = \frac{0.3 \text{ m}}{(0.9 \text{ W/m}\cdot\text{C})(15 \text{ m}^2)} = 0.02222^\circ\text{C/W}$$

Substituting, we get

$$\dot{Q} = \frac{(16 - 2)^\circ\text{C}}{0.02222^\circ\text{C/W}} = 630 \text{ W}$$

Discussion This is the same result obtained earlier. Note that heat conduction through a plane wall with specified surface temperatures can be determined directly and easily without utilizing the thermal resistance concept. However, the thermal resistance concept serves as a valuable tool in more complex heat transfer problems, as you will see in the following examples. Also, the units $\text{W/m}\cdot\text{C}$ and $\text{W/m}\cdot\text{K}$ for thermal conductivity are equivalent, and thus interchangeable. This is also the case for $^\circ\text{C}$ and K for temperature differences.

EXAMPLE 3–2 Heat Loss Through a Single-Pane Window

Consider a 0.8-m-high and 1.5-m-wide glass window with a thickness of 8 mm and a thermal conductivity of $k = 0.78 \text{ W/m}\cdot\text{K}$. Determine the steady rate of heat transfer through this glass window and the temperature of its inner surface for a day during which the room is maintained at 20°C while the temperature of the outdoors is -10°C . Take the heat transfer coefficients on the inner and outer surfaces of the window to be $h_1 = 10 \text{ W/m}^2\cdot\text{K}$ and $h_2 = 40 \text{ W/m}^2\cdot\text{K}$, which includes the effects of radiation.

SOLUTION Heat loss through a window glass is considered. The rate of heat transfer through the window and the inner surface temperature are to be determined.

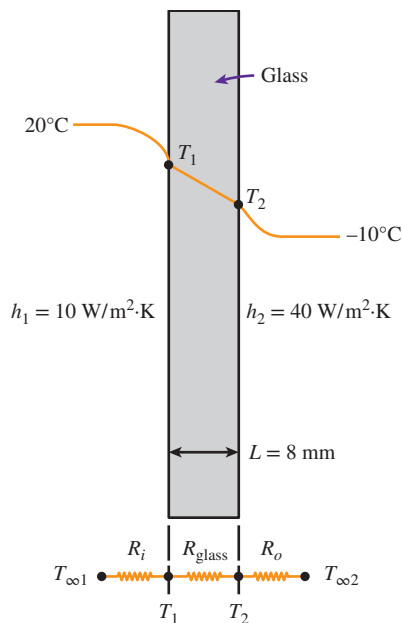


FIGURE 3–12

Schematic for Example 3–2.

Assumptions 1 Heat transfer through the window is steady since the surface temperatures remain constant at the specified values. 2 Heat transfer through the wall is one-dimensional since any significant temperature gradients exist in the direction from the indoors to the outdoors. 3 Thermal conductivity is constant.

Properties The thermal conductivity is given to be $k = 0.78 \text{ W/m}\cdot\text{K}$.

Analysis This problem involves conduction through the glass window and convection at its surfaces, and it can best be handled by making use of the thermal resistance concept and drawing the thermal resistance network, as shown in Fig. 3–12. Noting that the area of the window is $A = 0.8 \text{ m} \times 1.5 \text{ m} = 1.2 \text{ m}^2$, the individual resistances are evaluated from their definitions to be

$$R_i = R_{\text{conv}, 1} = \frac{1}{h_1 A} = \frac{1}{(10 \text{ W/m}^2\cdot\text{K})(1.2 \text{ m}^2)} = 0.08333^\circ\text{C/W}$$

$$R_{\text{glass}} = \frac{L}{kA} = \frac{0.008 \text{ m}}{(0.78 \text{ W/m}\cdot\text{K})(1.2 \text{ m}^2)} = 0.00855^\circ\text{C/W}$$

$$R_o = R_{\text{conv}, 2} = \frac{1}{h_2 A} = \frac{1}{(40 \text{ W/m}^2\cdot\text{K})(1.2 \text{ m}^2)} = 0.02083^\circ\text{C/W}$$

Noting that all three resistances are in series, the total resistance is

$$R_{\text{total}} = R_{\text{conv}, 1} + R_{\text{glass}} + R_{\text{conv}, 2} = 0.08333 + 0.00855 + 0.02083 = 0.1127^\circ\text{C/W}$$

Then the steady rate of heat transfer through the window becomes

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{\text{total}}} = \frac{[20 - (-10)]^\circ\text{C}}{0.1127^\circ\text{C/W}} = \mathbf{266 \text{ W}}$$

Knowing the rate of heat transfer, the inner surface temperature of the window glass can be determined from

$$\begin{aligned} \dot{Q} &= \frac{T_{\infty 1} - T_1}{R_{\text{conv}, 1}} \rightarrow T_1 = T_{\infty 1} - \dot{Q} R_{\text{conv}, 1} \\ &= 20^\circ\text{C} - (266 \text{ W})(0.08333^\circ\text{C/W}) \\ &= \mathbf{-2.2^\circ\text{C}} \end{aligned}$$

Discussion Note that the inner surface temperature of the window glass is -2.2°C even though the temperature of the air in the room is maintained at 20°C . Such low surface temperatures are highly undesirable since they cause the formation of fog or even frost on the inner surfaces of the glass when the humidity in the room is high.

EXAMPLE 3–3 Heat Loss Through Double-Pane Windows

Consider a 0.8-m-high and 1.5-m-wide double-pane window consisting of two 4-mm-thick layers of glass ($k = 0.78 \text{ W/m}\cdot\text{K}$) separated by a 10-mm-wide stagnant airspace ($k = 0.026 \text{ W/m}\cdot\text{K}$). Determine the steady rate of heat transfer through this

double-pane window and the temperature of its inner surface for a day during which the room is maintained at 20°C while the temperature of the outdoors is -10°C . Take the convection heat transfer coefficients on the inner and outer surfaces of the window to be $h_1 = 10 \text{ W/m}^2\cdot\text{K}$ and $h_2 = 40 \text{ W/m}^2\cdot\text{K}$, which includes the effects of radiation.

SOLUTION A double-pane window is considered. The rate of heat transfer through the window and the inner surface temperature are to be determined.

Analysis This example problem is identical to the previous one except that the single 8-mm-thick window glass is replaced by two 4-mm-thick glasses that enclose a 10-mm-wide stagnant airspace. Therefore, the thermal resistance network of this problem involves two additional conduction resistances corresponding to the two additional layers, as shown in Fig. 3–13. Noting that the area of the window is again $A = 0.8 \text{ m} \times 1.5 \text{ m} = 1.2 \text{ m}^2$, the individual resistances are evaluated from their definitions to be

$$R_i = R_{\text{conv}, 1} = \frac{1}{h_1 A} = \frac{1}{(10 \text{ W/m}^2\cdot\text{K})(1.2 \text{ m}^2)} = 0.08333^{\circ}\text{C/W}$$

$$R_1 = R_3 = R_{\text{glass}} = \frac{L_1}{k_1 A} = \frac{0.004 \text{ m}}{(0.78 \text{ W/m}\cdot\text{K})(1.2 \text{ m}^2)} = 0.00427^{\circ}\text{C/W}$$

$$R_2 = R_{\text{air}} = \frac{L_2}{k_2 A} = \frac{0.01 \text{ m}}{(0.026 \text{ W/m}\cdot\text{K})(1.2 \text{ m}^2)} = 0.3205^{\circ}\text{C/W}$$

$$R_o = R_{\text{conv}, 2} = \frac{1}{h_2 A} = \frac{1}{(40 \text{ W/m}^2\cdot\text{K})(1.2 \text{ m}^2)} = 0.02083^{\circ}\text{C/W}$$

Noting that all three resistances are in series, the total resistance is

$$\begin{aligned} R_{\text{total}} &= R_{\text{conv}, 1} + R_{\text{glass}, 1} + R_{\text{air}} + R_{\text{glass}, 2} + R_{\text{conv}, 2} \\ &= 0.08333 + 0.00427 + 0.3205 + 0.00427 + 0.02083 \\ &= 0.4332^{\circ}\text{C/W} \end{aligned}$$

Then the steady rate of heat transfer through the window becomes

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{\text{total}}} = \frac{[20 - (-10)]^{\circ}\text{C}}{0.4332^{\circ}\text{C/W}} = \mathbf{69.2 \text{ W}}$$

which is about one-fourth of the result obtained in Example 3–3. This explains the popularity of the double- and even triple-pane windows in cold climates. The drastic reduction in the heat transfer rate in this case is due to the large thermal resistance of the air layer between the glasses.

The inner surface temperature of the window in this case will be

$$T_1 = T_{\infty 1} - \dot{Q} R_{\text{conv}, 1} = 20^{\circ}\text{C} - (69.2 \text{ W})(0.08333^{\circ}\text{C/W}) = \mathbf{14.2^{\circ}\text{C}}$$

which is considerably higher than the -2.2°C obtained in Example 3–2. Therefore, a double-pane window will rarely get fogged. A double-pane window will also reduce the heat gain in summer and thus reduce the air-conditioning costs.

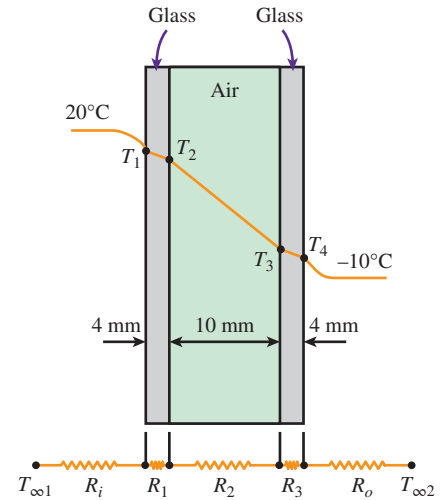
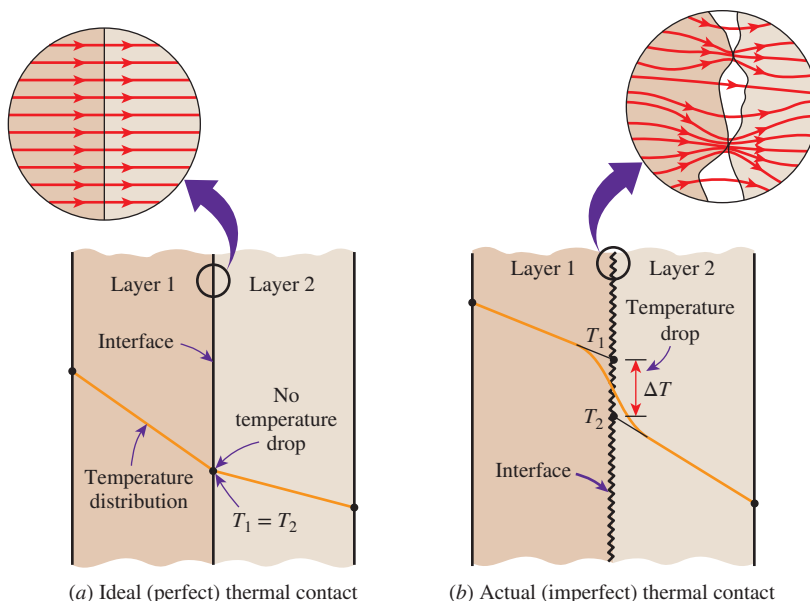


FIGURE 3–13
Schematic for Example 3–3.

FIGURE 3–14

Temperature distribution and heat flow lines along two solid plates pressed against each other for the case of perfect and imperfect contact.



3–2 ■ THERMAL CONTACT RESISTANCE

In the analysis of heat conduction through multilayer solids, we assumed “perfect contact” at the interface of two layers, and thus no temperature drop at the interface. This would be the case when the surfaces are perfectly smooth and they produce a perfect contact at each point. In reality, however, even flat surfaces that appear smooth to the eye turn out to be rather rough when examined under a microscope, as shown in Fig. 3–14, with numerous peaks and valleys. That is, a surface is *microscopically rough* no matter how smooth it appears to be.

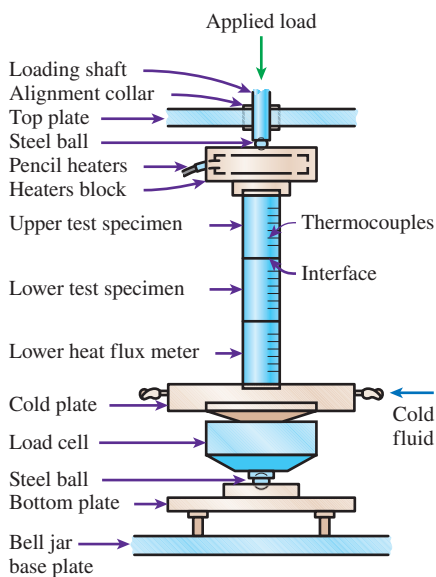
When two such surfaces are pressed against each other, the peaks form good material contact but the valleys form voids filled with air in most cases. As a result, an interface contains numerous *air gaps* of varying sizes that act as *insulation* because of the low thermal conductivity of air. Thus, an interface offers some resistance to heat transfer, and this resistance for a unit interface area is called the **thermal contact resistance**, R_c . The value of R_c is determined experimentally using a setup like the one shown in Fig. 3–15, and as expected, there is considerable scatter of data because of the difficulty in characterizing the surfaces.

Consider heat transfer through two metal rods of cross-sectional area A that are pressed against each other. Heat transfer through the interface of these two rods is the sum of the heat transfers through the *solid contact spots* (solid-to-solid conduction) and the *gaps* (conduction and/or radiation across the gaps) in the noncontact areas (which is a major contributor to heat transfer) and can be expressed as

$$\dot{Q} = \dot{Q}_{\text{contact}} + \dot{Q}_{\text{gap}} \quad (3-25)$$

It can also be expressed in a manner analogous to Newton’s law of cooling as

$$\dot{Q} = h_c A \Delta T_{\text{interface}} \quad (3-26)$$

**FIGURE 3–15**

A typical experimental setup for the determination of thermal contact resistance.

Source: Song et al., 1993.

where A is the apparent interface area (which is the same as the cross-sectional area of the rods) and $\Delta T_{\text{interface}}$ is the effective temperature difference at the interface. The quantity h_c , which corresponds to the convection heat transfer coefficient, is called the **thermal contact conductance** and is expressed as

$$h_c = \frac{\dot{Q}/A}{\Delta T_{\text{interface}}} \quad (\text{W/m}^2\cdot\text{K}) \quad (3-27)$$

It is related to thermal contact resistance by

$$R_c = \frac{1}{h_c} = \frac{\Delta T_{\text{interface}}}{\dot{Q}/A} \quad (\text{m}^2\cdot\text{K/W}) \quad (3-28)$$

That is, thermal contact resistance is the inverse of thermal contact conductance. Usually, thermal contact conductance is reported in the literature, but the concept of thermal contact resistance serves as a better vehicle for explaining the effect of interface on heat transfer. Note that R_c represents thermal contact resistance for a *unit area*. The thermal resistance for the entire interface is obtained by dividing R_c by the apparent interface area A .

The thermal contact resistance can be determined from Eq. 3-28 by measuring the temperature drop at the interface and dividing it by the heat flux under steady conditions. The value of thermal contact resistance depends on the *surface roughness* and the *material properties* as well as the *temperature* and *pressure* at the interface and the *type of fluid* trapped at the interface. The situation becomes more complex when plates are fastened by bolts, screws, or rivets since the interface pressure in this case is nonuniform. The thermal contact resistance in that case also depends on the plate thickness, the bolt radius, and the size of the contact zone. Thermal contact resistance is observed to *decrease* with *decreasing surface roughness* and *increasing interface pressure*, as expected. Most experimentally determined values of the thermal contact resistance fall between 0.000005 and 0.0005 $\text{m}^2\cdot\text{K/W}$ (the corresponding range of thermal contact conductance is 2000 to 200,000 $\text{W/m}^2\cdot\text{K}$).

When we analyze heat transfer in a medium consisting of two or more layers, the first thing we need to know is whether the thermal contact resistance is *significant* or not. We can answer this question by comparing the magnitudes of the thermal resistances of the layers with typical values of thermal contact resistance. For example, the thermal resistance of a 1-cm-thick layer of an insulating material for a unit surface area is

$$R_{c, \text{insulation}} = \frac{L}{k} = \frac{0.01 \text{ m}}{0.04 \text{ W/m}\cdot\text{K}} = 0.25 \text{ m}^2\cdot\text{K/W}$$

whereas for a 1-cm-thick layer of copper, it is

$$R_{c, \text{copper}} = \frac{L}{k} = \frac{0.01 \text{ m}}{386 \text{ W/m}\cdot\text{K}} = 0.000026 \text{ m}^2\cdot\text{K/W}$$

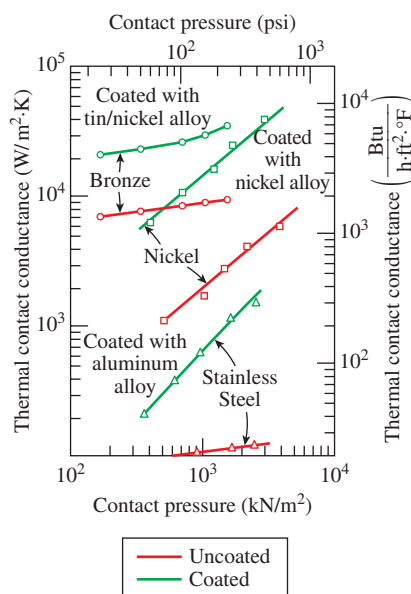
Comparing the values preceding with typical values of thermal contact resistance, we conclude that thermal contact resistance is significant and can even dominate the heat transfer for good heat conductors such as metals, but it can

TABLE 3–1

Thermal contact conductance for aluminum plates with different fluids at the interface for a surface roughness of 10 μm and interface pressure of 1 atm

Fluid at the Interface	Contact Conductance, h_c , $\text{W/m}^2\cdot\text{K}$
Air	3640
Helium	9520
Hydrogen	13,900
Silicone oil	19,000
Glycerin	37,700

Source: E. Fried, "Thermal Conduction Contribution to Heat Transfer at Contacts," Thermal Conductivity, vol. 2, ed. R. P. Tye. London: Academic Press, 1969.

**FIGURE 3–16**

Effect of metallic coatings on thermal contact conductance.

Source: Peterson, 1987.

be disregarded for poor heat conductors such as insulations. This is not surprising since insulating materials consist mostly of airspace just like the interface itself.

The thermal contact resistance can be minimized by applying a thermally conducting liquid called a *thermal grease* such as silicon oil on the surfaces before they are pressed against each other. This is commonly done when attaching electronic components such as power transistors to heat sinks. The thermal contact resistance can also be reduced by replacing the air at the interface with a *better conducting gas* such as helium or hydrogen, as shown in Table 3–1.

Another way to minimize the contact resistance is to insert a *soft metallic foil* such as tin, silver, copper, nickel, or aluminum between the two surfaces. Experimental studies show that the thermal contact resistance can be reduced by a factor of up to 7 by a metallic foil at the interface. For maximum effectiveness, the foils must be very thin. The effect of metallic coatings on thermal contact conductance is shown in Fig. 3–16 for various metal surfaces.

There is considerable uncertainty in the contact conductance data reported in the literature, and care should be exercised when using them. In Table 3–2, some experimental results are given for the contact conductance between similar and dissimilar metal surfaces for use in preliminary design calculations. Note that the *thermal contact conductance* is *highest* (and thus the contact resistance is lowest) for *soft metals* with *smooth surfaces* at *high pressure*.

EXAMPLE 3–4 Equivalent Thickness for Contact Resistance

The thermal contact conductance at the interface of two 1-cm-thick copper plates is measured to be 18,000 $\text{W/m}^2\cdot\text{K}$. Determine the thickness of the copper plate whose thermal resistance is equal to the thermal resistance of the interface between the plates (Fig. 3–17).

SOLUTION The thickness of the copper plate whose thermal resistance is equal to the thermal contact resistance is to be determined.

Properties The thermal conductivity of copper at room temperature is $k = 401 \text{ W/m}\cdot\text{K}$ (Table A–3).

Analysis Noting that thermal contact resistance is the inverse of thermal contact conductance, the thermal contact resistance is

$$R_c = \frac{1}{h_c} = \frac{1}{18,000 \text{ W/m}^2\cdot\text{K}} = 5.556 \times 10^{-5} \text{ m}^2\cdot\text{K/W}$$

For a unit surface area, the thermal resistance of a flat plate is defined as

$$R = \frac{L}{k}$$

where L is the thickness of the plate and k is the thermal conductivity. Setting $R = R_c$, the equivalent thickness is determined from the relation above to be

$$L = kR_c = (401 \text{ W/m}\cdot\text{K})(5.556 \times 10^{-5} \text{ m}^2\cdot\text{K/W}) = 0.0223 \text{ m} = \mathbf{2.23 \text{ cm}}$$

TABLE 3-2

Thermal contact conductance of some metal surfaces in air (from various sources)

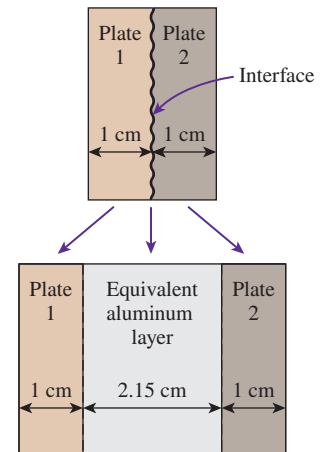
Material	Surface Condition	Roughness, μm	Temperature, $^{\circ}\text{C}$	Pressure, MPa	h_c^* $\text{W}/\text{m}^2\cdot\text{K}$
Identical Metal Pairs					
416 Stainless steel	Ground	2.54	90–200	0.17–2.5	3,800
304 Stainless steel	Ground	1.14	20	4–7	1,900
Aluminum	Ground	2.54	150	1.2–2.5	11,400
Copper	Ground	1.27	20	1.2–20	143,000
Copper	Milled	3.81	20	1–5	55,500
Copper (vacuum)	Milled	0.25	30	0.17–7	11,400
Dissimilar Metal Pairs					
Stainless steel– Aluminum		20–30	20	10	2,900
				20	3,600
Stainless steel– Aluminum		1.0–2.0	20	10	16,400
				20	20,800
Steel Ct-30– Aluminum	Ground	1.4–2.0	20	10	50,000
				15–35	59,000
Steel Ct-30– Aluminum	Milled	4.5–7.2	20	10	4,800
				30	8,300
Aluminum–Copper	Ground	1.17–1.4	20	5	42,000
				15	56,000
Aluminum–Copper	Milled	4.4–4.5	20	10	12,000
				20–35	12,000

*Divide the given values by 5.678 to convert to $\text{Btu}/\text{h}\cdot\text{ft}^2\cdot^{\circ}\text{F}$.

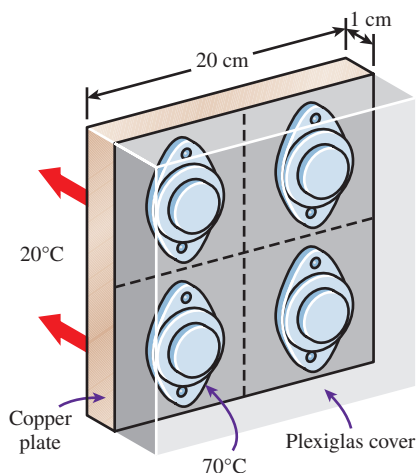
Discussion Note that the interface between the two plates offers as much resistance to heat transfer as a 2.23-cm-thick copper plate. It is interesting that the thermal contact resistance in this case is greater than the sum of the thermal resistances of both plates.

EXAMPLE 3-5 Contact Resistance of Transistors

Four identical power transistors with aluminum casing are attached on one side of a 1-cm-thick 20-cm $\times$ 20-cm square copper plate ($k = 386 \text{ W}/\text{m}\cdot\text{K}$) by screws that exert an average pressure of 6 MPa (Fig. 3–18). The base area of each transistor is 8 cm^2 , and each transistor is placed at the center of a 10-cm $\times$ 10-cm quarter section of the plate. The interface roughness is estimated to be about $1.5 \mu\text{m}$. All transistors are covered by a thick Plexiglas layer, which is a poor conductor of heat, and thus all the heat generated at the junction of the transistor must be dissipated to the ambient at 20°C through the back surface of the copper plate. The combined convection/radiation heat transfer coefficient at the back surface can be taken to be $25 \text{ W}/\text{m}^2\cdot\text{K}$. If the case temperature of the transistor is not to exceed 70°C , determine the maximum power each transistor can dissipate safely and the temperature jump at the case/plate interface.

**FIGURE 3-17**

Schematic for Example 3-4.

**FIGURE 3–18**

Schematic for Example 3–5.

SOLUTION Four identical power transistors are attached on a copper plate. For a maximum case temperature of 70°C , the maximum power dissipation and the temperature jump at the interface are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat transfer can be approximated as being one-dimensional, although it is recognized that heat conduction in some parts of the plate will be two-dimensional since the plate area is much larger than the base area of the transistor. But the large thermal conductivity of copper will minimize this effect. 3 All the heat generated at the junction is dissipated through the back surface of the plate since the transistors are covered by a thick Plexiglas layer. 4 Thermal conductivities are constant.

Properties The thermal conductivity of copper is given to be $k = 386 \text{ W/m}\cdot\text{K}$. The contact conductance is obtained from Table 3–2 to be $h_c = 42,000 \text{ W/m}^2\cdot\text{K}$, which corresponds to a copper/aluminum interface for the case of $1.17 - 1.4 \mu\text{m}$ roughness and 5 MPa pressure, which is sufficiently close to what we have.

Analysis The contact area between the case and the plate is given to be 8 cm^2 , and the plate area for each transistor is 100 cm^2 . The thermal resistance network of this problem consists of three resistances in series (interface, plate, and convection), which are determined to be

$$R_{\text{interface}} = \frac{1}{h_c A_c} = \frac{1}{(42,000 \text{ W/m}^2\cdot\text{K})(8 \times 10^{-4} \text{ m}^2)} = 0.030^\circ\text{C/W}$$

$$R_{\text{plate}} = \frac{L}{kA} = \frac{0.01 \text{ m}}{(386 \text{ W/m}\cdot\text{K})(0.01 \text{ m}^2)} = 0.0026^\circ\text{C/W}$$

$$R_{\text{conv}} = \frac{1}{h_o A} = \frac{1}{(25 \text{ W/m}^2\cdot\text{K})(0.01 \text{ m}^2)} = 4.0^\circ\text{C/W}$$

The total thermal resistance is then

$$R_{\text{total}} = R_{\text{interface}} + R_{\text{plate}} + R_{\text{ambient}} = 0.030 + 0.0026 + 4.0 = 4.0326^\circ\text{C/W}$$

Note that the thermal resistance of a copper plate is very small and can be ignored altogether. Then the rate of heat transfer is determined to be

$$\dot{Q} = \frac{\Delta T}{R_{\text{total}}} = \frac{(70 - 20)^\circ\text{C}}{4.0326^\circ\text{C/W}} = \mathbf{12.4 \text{ W}}$$

Therefore, the power transistor should not be operated at power levels greater than 12.4 W if the case temperature is not to exceed 70°C .

The temperature jump at the interface is determined from

$$\Delta T_{\text{interface}} = \dot{Q} R_{\text{interface}} = (12.4 \text{ W})(0.030^\circ\text{C/W}) = \mathbf{0.37^\circ\text{C}}$$

which is not very large. Therefore, even if we eliminate the thermal contact resistance at the interface completely, we lower the operating temperature of the transistor in this case by less than 0.4°C .

3-3 ■ GENERALIZED THERMAL RESISTANCE NETWORKS

The *thermal resistance* concept or the *electrical analogy* can also be used to solve steady heat transfer problems that involve parallel layers or combined series-parallel arrangements. Although such problems are often two- or even three-dimensional, approximate solutions can be obtained by assuming one-dimensional heat transfer and using the thermal resistance network.

Consider the composite wall shown in Fig. 3-19, which consists of two parallel layers. The thermal resistance network, which consists of two parallel resistances, can be represented as shown in the figure. Noting that the total heat transfer is the sum of the heat transfers through each layer, we have

$$\dot{Q} = \dot{Q}_1 + \dot{Q}_2 = \frac{T_1 - T_2}{R_1} + \frac{T_1 - T_2}{R_2} = (T_1 - T_2) \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \quad (3-29)$$

Utilizing the electrical analogy, we get

$$\dot{Q} = \frac{T_1 - T_2}{R_{\text{total}}} \quad (3-30)$$

where

$$\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} \rightarrow R_{\text{total}} = \frac{R_1 R_2}{R_1 + R_2} \quad (3-31)$$

since the resistances are in parallel.

Now consider the combined series-parallel arrangement shown in Fig. 3-20. The total rate of heat transfer through this composite system can again be expressed as

$$\dot{Q} = \frac{T_1 - T_\infty}{R_{\text{total}}} \quad (3-32)$$

where

$$R_{\text{total}} = R_{12} + R_3 + R_{\text{conv}} = \frac{R_1 R_2}{R_1 + R_2} + R_3 + R_{\text{conv}} \quad (3-33)$$

and

$$R_1 = \frac{L_1}{k_1 A_1} \quad R_2 = \frac{L_2}{k_2 A_2} \quad R_3 = \frac{L_3}{k_3 A_3} \quad R_{\text{conv}} = \frac{1}{h A_3} \quad (3-34)$$

Once the individual thermal resistances are evaluated, the total resistance and the total rate of heat transfer can easily be determined from the relations above.

The result obtained is somewhat approximate, since the surfaces of the third layer are probably not isothermal, and heat transfer between the first two layers is likely to occur.

Two assumptions commonly used in solving complex multidimensional heat transfer problems by treating them as one-dimensional (say, in the x -direction) using the thermal resistance network are (1) any plane wall normal to the x -axis is *isothermal* (i.e., assumes the temperature varies in

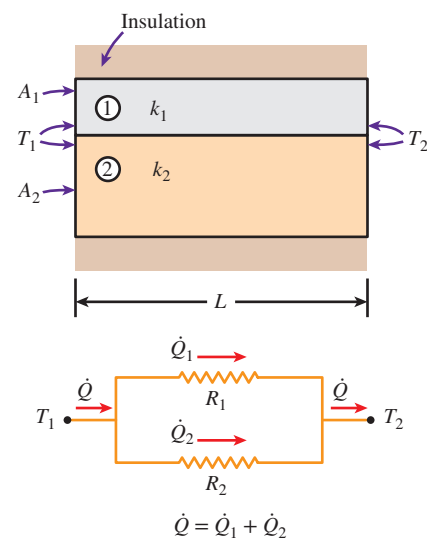


FIGURE 3-19

Thermal resistance network for two parallel layers.

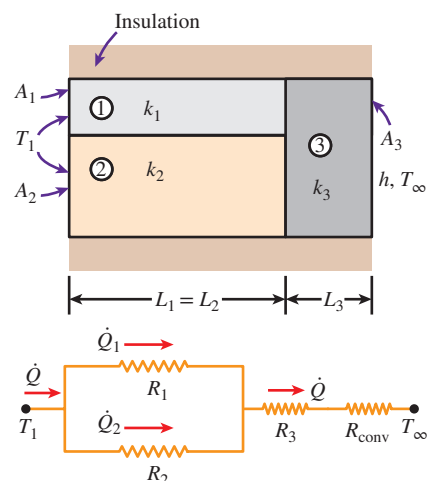


FIGURE 3-20

Thermal resistance network for combined series-parallel arrangement.

the x -direction only) and (2) any plane parallel to the x -axis is *adiabatic* (i.e., assumes heat transfer occurs in the x -direction only). These two assumptions result in different resistance networks, and thus different (but usually close) values for the total thermal resistance and thus heat transfer. The actual result lies between these two values. In geometries in which heat transfer occurs predominantly in one direction, either approach gives satisfactory results.

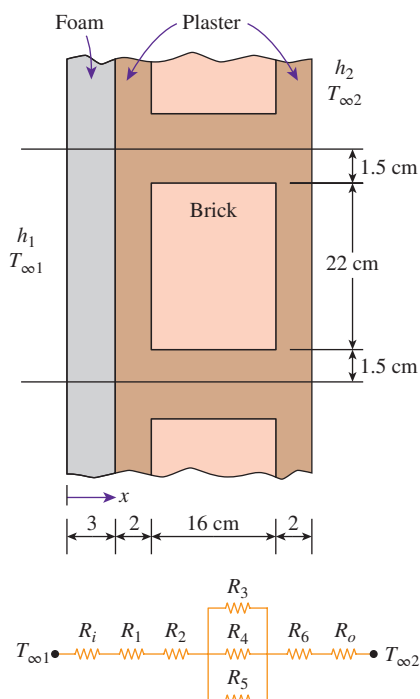


FIGURE 3-21

Schematic for Example 3-6.

EXAMPLE 3-6 Heat Loss Through a Composite Wall

A 3-m-high and 5-m-wide wall consists of long 16-cm $\times$ 22-cm cross section horizontal bricks ($k = 0.72$ W/m $\cdot$ K) separated by 3-cm-thick plaster layers ($k = 0.22$ W/m $\cdot$ K). There are also 2-cm-thick plaster layers on each side of the brick and a 3-cm-thick rigid foam ($k = 0.026$ W/m $\cdot$ K) on the inner side of the wall, as shown in Fig. 3-21. The indoor and the outdoor temperatures are 20°C and -10°C , respectively, and the convection heat transfer coefficients on the inner and the outer sides are $h_1 = 10$ W/m $^2\cdot$ K and $h_2 = 25$ W/m $^2\cdot$ K, respectively. Assuming one-dimensional heat transfer and disregarding radiation, determine the rate of heat transfer through the wall.

SOLUTION The composition of a composite wall is given. The rate of heat transfer through the wall is to be determined.

Assumptions 1 Heat transfer is steady since there is no indication of change with time. 2 Heat transfer can be approximated as being one-dimensional since it is predominantly in the x -direction. 3 Thermal conductivities are constant. 4 Heat transfer by radiation is negligible.

Properties The thermal conductivities are given to be $k = 0.72$ W/m $\cdot$ K for bricks, $k = 0.22$ W/m $\cdot$ K for plaster layers, and $k = 0.026$ W/m $\cdot$ K for the rigid foam.

Analysis There is a pattern in the construction of this wall that repeats itself every 25-cm distance in the vertical direction. There is no variation in the horizontal direction. Therefore, we consider a 1-m-deep and 0.25-m-high portion of the wall, since it is representative of the entire wall.

Assuming any cross section of the wall normal to the x -direction to be *isothermal*, the thermal resistance network for the representative section of the wall becomes as shown in Fig. 3-21. The individual resistances are evaluated as:

$$R_i = R_{\text{conv}, 1} = \frac{1}{h_1 A} = \frac{1}{(10 \text{ W/m}^2\cdot\text{K})(0.25 \times 1 \text{ m}^2)} = 0.40^\circ\text{C/W}$$

$$R_1 = R_{\text{foam}} = \frac{L}{kA} = \frac{0.03 \text{ m}}{(0.026 \text{ W/m}\cdot\text{K})(0.25 \times 1 \text{ m}^2)} = 4.62^\circ\text{C/W}$$

$$R_2 = R_6 = R_{\text{plaster, side}} = \frac{L}{kA} = \frac{0.02 \text{ m}}{(0.22 \text{ W/m}\cdot\text{K})(0.25 \times 1 \text{ m}^2)} = 0.36^\circ\text{C/W}$$

$$R_3 = R_5 = R_{\text{plaster, center}} = \frac{L}{kA} = \frac{0.16 \text{ m}}{(0.22 \text{ W/m}\cdot\text{K})(0.015 \times 1 \text{ m}^2)} = 48.48^\circ\text{C/W}$$

$$R_4 = R_{\text{brick}} = \frac{L}{kA} = \frac{0.16 \text{ m}}{(0.72 \text{ W/m}\cdot\text{K})(0.22 \times 1 \text{ m}^2)} = 1.01^\circ\text{C/W}$$

$$R_o = R_{\text{conv}, 2} = \frac{1}{h_2 A} = \frac{1}{(25 \text{ W/m}^2\cdot\text{K})(0.25 \times 1 \text{ m}^2)} = 0.16^\circ\text{C/W}$$

The three resistances R_3 , R_4 , and R_5 in the middle are parallel, and their equivalent resistance is determined from

$$\frac{1}{R_{\text{mid}}} = \frac{1}{R_3} + \frac{1}{R_4} + \frac{1}{R_5} = \frac{1}{48.48} + \frac{1}{1.01} + \frac{1}{48.48} = 1.03 \text{ W/}^\circ\text{C}$$

which gives

$$R_{\text{mid}} = 0.97^\circ\text{C/W}$$

Now all the resistances are in series, and the total resistance is

$$\begin{aligned} R_{\text{total}} &= R_i + R_1 + R_2 + R_{\text{mid}} + R_6 + R_o \\ &= 0.40 + 4.62 + 0.36 + 0.97 + 0.36 + 0.16 \\ &= 6.87^\circ\text{C/W} \end{aligned}$$

Then the steady rate of heat transfer through the wall becomes

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{\text{total}}} = \frac{[20 - (-10)]^\circ\text{C}}{6.87^\circ\text{C/W}} = 4.37 \text{ W} \quad (\text{per } 0.25 \text{ m}^2 \text{ surface area})$$

or $4.37/0.25 = 17.5 \text{ W per m}^2$ area. The total area of the wall is $A = 3 \text{ m} \times 5 \text{ m} = 15 \text{ m}^2$. Then the rate of heat transfer through the entire wall becomes

$$\dot{Q}_{\text{total}} = (17.5 \text{ W/m}^2)(15 \text{ m}^2) = \mathbf{263 \text{ W}}$$

Of course, this result is approximate, since we assumed the temperature within the wall to vary in one direction only and ignored any temperature change (and thus heat transfer) in the other two directions.

Discussion In the preceding solution, we assumed the temperature at any cross section of the wall normal to the x -direction was *isothermal*. We could also solve this problem by going to the other extreme and assuming the surfaces parallel to the x -direction are *adiabatic*. The thermal resistance network in this case will be as shown in Fig. 3–22. By following the approach outlined above, the total thermal resistance in this case is determined to be $R_{\text{total}} = 6.97^\circ\text{C/W}$, which is very close to the value 6.85°C/W obtained before. Thus, either approach gives roughly the same result in this case. This example demonstrates that either approach can be used in practice to obtain satisfactory results.

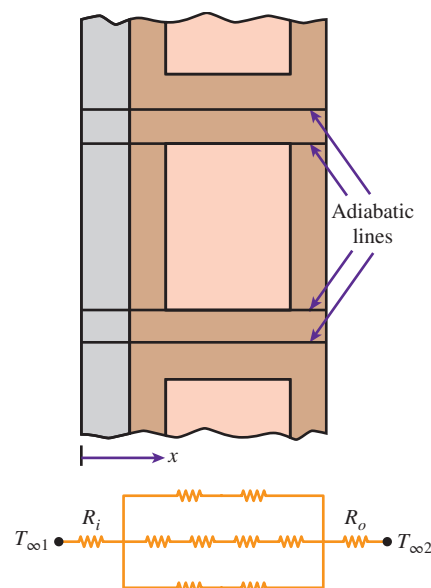
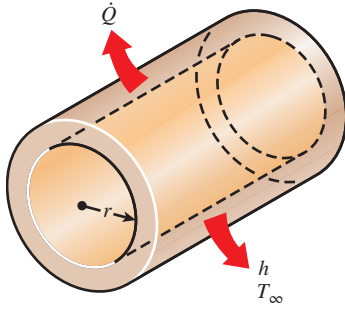


FIGURE 3–22

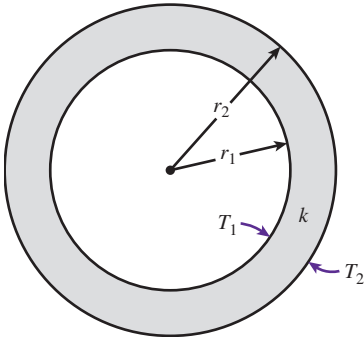
Alternative thermal resistance network for Example 3–6 for the case of surfaces parallel to the primary direction of heat transfer being adiabatic.

3–4 ■ HEAT CONDUCTION IN CYLINDERS AND SPHERES

Consider steady heat conduction through a hot-water pipe. Heat is continuously lost to the outdoors through the wall of the pipe, and we intuitively feel that heat transfer through the pipe is in the normal direction to the pipe

**FIGURE 3–23**

Heat is lost from a hot-water pipe to the air outside in the radial direction, and thus heat transfer from a long pipe is one-dimensional.

**FIGURE 3–24**

A long cylindrical pipe (or spherical shell) with specified inner and outer surface temperatures T_1 and T_2 .

surface and that no significant heat transfer takes place in the pipe in other directions (Fig. 3–23). The wall of the pipe, whose thickness is rather small, separates two fluids at different temperatures, and thus the temperature gradient in the radial direction is relatively large. Further, if the fluid temperatures inside and outside the pipe remain constant, then heat transfer through the pipe is *steady*. Thus heat transfer through the pipe can be modeled as *steady* and *one-dimensional*. The temperature of the pipe in this case depends on one direction only (the radial r -direction) and can be expressed as $T = T(r)$. The temperature is independent of the azimuthal angle or the axial distance. This situation is approximated in practice in long cylindrical pipes and spherical containers.

In *steady* operation, there is no change in the temperature of the pipe with time at any point. Therefore, the rate of heat transfer into the pipe must be equal to the rate of heat transfer out of it. In other words, heat transfer through the pipe must be constant, $\dot{Q}_{\text{cond, cyl}} = \text{constant}$.

Consider a long cylindrical layer (such as a circular pipe) of inner radius r_1 , outer radius r_2 , length L , and average thermal conductivity k (Fig. 3–24). The two surfaces of the cylindrical layer are maintained at constant temperatures T_1 and T_2 . There is no heat generation in the layer, and the thermal conductivity is constant. For one-dimensional heat conduction through the cylindrical layer, we have $T(r)$. Then Fourier's law of heat conduction for heat transfer through the cylindrical layer can be expressed as

$$\dot{Q}_{\text{cond, cyl}} = -kA \frac{dT}{dr} \quad (\text{W}) \quad (3-35)$$

where $A = 2\pi rL$ is the heat transfer area at location r . Note that A depends on r , and thus it *varies* in the direction of heat transfer. Separating the variables in Eq. 3–35 and integrating from $r = r_1$, where $T(r_1) = T_1$, to $r = r_2$, where $T(r_2) = T_2$, gives

$$\int_{r=r_1}^{r_2} \frac{\dot{Q}_{\text{cond, cyl}}}{A} dr = - \int_{T=T_1}^{T_2} k dT \quad (3-36)$$

Substituting $A = 2\pi rL$ and performing the integrations give

$$\dot{Q}_{\text{cond, cyl}} = 2\pi Lk \frac{T_1 - T_2}{\ln(r_2/r_1)} \quad (\text{W}) \quad (3-37)$$

since $\dot{Q}_{\text{cond, cyl}} = \text{constant}$. This equation can be rearranged as

$$\dot{Q}_{\text{cond, cyl}} = \frac{T_1 - T_2}{R_{\text{cyl}}} \quad (\text{W}) \quad (3-38)$$

where

$$R_{\text{cyl}} = \frac{\ln(r_2/r_1)}{2\pi Lk} = \frac{\ln(\text{Outer radius/Inner radius})}{2\pi \times \text{Length} \times \text{Thermal conductivity}} \quad (3-39)$$

is the *thermal resistance* of the cylindrical layer against heat conduction, or simply the *conduction resistance* of the cylinder layer. Note that Eq. 3–37 is identical to Eq. 2–59 which was obtained by using the “standard” approach by first solving the heat conduction equation in cylindrical coordinates, Eq. 2–29, to obtain the temperature distribution, Eq. 2–58, and then using the Fourier’s law to obtain the heat transfer rate. The method used in obtaining Eq. 3–37 can be considered an “alternative” approach. However, it is restricted to one-dimensional steady heat conduction with no heat generation.

We can repeat the analysis for a *spherical layer* by taking $A = 4\pi r^2$ and performing the integrations in Eq. 3–36. The result can be expressed as

$$\dot{Q}_{\text{cond, sph}} = \frac{T_1 - T_2}{R_{\text{sph}}} \quad (3-40)$$

where

$$R_{\text{sph}} = \frac{r_2 - r_1}{4\pi r_1 r_2 k} = \frac{\text{Outer radius} - \text{Inner radius}}{4\pi(\text{Outer radius})(\text{Inner radius})(\text{Thermal conductivity})} \quad (3-41)$$

is the *thermal resistance* of the spherical layer against heat conduction, or simply the *conduction resistance* of the spherical layer. Note also that Eq. 3–40 is identical to Eq. 2–61, which was obtained by solving the heat conduction equation in spherical coordinates.

Now consider steady one-dimensional heat transfer through a cylindrical or spherical layer that is exposed to convection on both sides to fluids at temperatures $T_{\infty 1}$ and $T_{\infty 2}$ with heat transfer coefficients h_1 and h_2 , respectively, as shown in Fig. 3–25. The thermal resistance network in this case consists of one conduction and two convection resistances in series, just like the one for the plane wall, and the rate of heat transfer under steady conditions can be expressed as

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{\text{total}}} \quad (3-42)$$

where

$$\begin{aligned} R_{\text{total}} &= R_{\text{conv, 1}} + R_{\text{cyl}} + R_{\text{conv, 2}} \\ &= \frac{1}{(2\pi r_1 L)h_1} + \frac{\ln r_2/r_1}{2\pi Lk} + \frac{1}{(2\pi r_2 L)h_2} \end{aligned} \quad (3-43)$$

for a *cylindrical* layer, and

$$\begin{aligned} R_{\text{total}} &= R_{\text{conv, 1}} + R_{\text{sph}} + R_{\text{conv, 2}} \\ &= \frac{1}{(4\pi r_1^2)h_1} + \frac{r_2 - r_1}{4\pi r_1 r_2 k} + \frac{1}{(4\pi r_2^2)h_2} \end{aligned} \quad (3-44)$$

for a *spherical* layer. Note that A in the convection resistance relation $R_{\text{conv}} = 1/hA$ is the *surface area at which convection occurs*. It is equal to $A = 2\pi rL$ for a cylindrical surface and $A = 4\pi r^2$ for a spherical surface of radius r . Also note that the thermal resistances are in series, and thus the total thermal resistance is determined by simply adding the individual resistances, just like the electrical resistances connected in series.

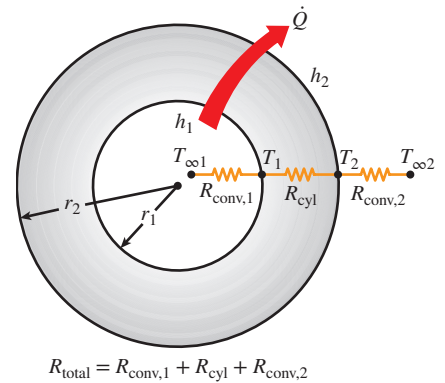


FIGURE 3–25

The thermal resistance network for a cylindrical (or spherical) shell subjected to convection from both the inner and the outer sides.

Multilayered Cylinders and Spheres

Steady heat transfer through multilayered cylindrical or spherical shells can be handled just like multilayered plane walls discussed earlier by simply adding an *additional resistance* in series for each *additional layer*. For example, the steady heat transfer rate through the three-layered composite cylinder of length L shown in Fig. 3–26 with convection on both sides can be expressed as

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{\text{total}}} \quad (3-45)$$

where R_{total} is the *total thermal resistance*, expressed as

$$\begin{aligned} R_{\text{total}} &= R_{\text{conv}, 1} + R_{\text{cyl}, 1} + R_{\text{cyl}, 2} + R_{\text{cyl}, 3} + R_{\text{conv}, 2} \\ &= \frac{1}{h_1 A_1} + \frac{\ln(r_2/r_1)}{2\pi L k_1} + \frac{\ln(r_3/r_2)}{2\pi L k_2} + \frac{\ln(r_4/r_3)}{2\pi L k_3} + \frac{1}{h_2 A_4} \end{aligned} \quad (3-46)$$

where $A_1 = 2\pi r_1 L$ and $A_4 = 2\pi r_4 L$. Equation 3–46 can also be used for a three-layered spherical shell by replacing the thermal resistances of cylindrical layers with the corresponding spherical ones. Again, note from the thermal resistance network that the resistances are in series, and thus the total thermal resistance is simply the *arithmetic sum* of the individual thermal resistances in the path of heat flow.

Once $\dot{Q}$ is known, we can determine any intermediate temperature T_j by applying the relation $\dot{Q} = (T_i - T_j)/R_{\text{total}, i-j}$ across any layer or layers such that T_i is a *known* temperature at location i and $R_{\text{total}, i-j}$ is the total thermal resistance between locations i and j (Fig. 3–27). For example, once $\dot{Q}$ has been calculated, the interface temperature T_2 between the first and second cylindrical layers can be determined from

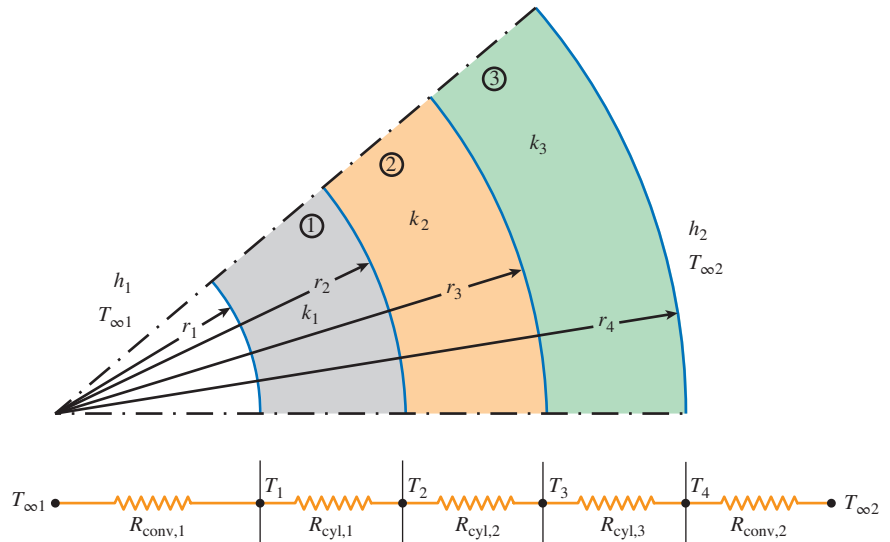


FIGURE 3–26

The thermal resistance network for heat transfer through a three-layered composite cylinder subjected to convection on both sides.

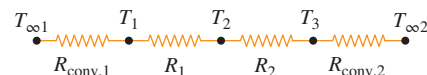
$$\dot{Q} = \frac{T_{\infty 1} - T_2}{R_{\text{conv},1} + R_{\text{cy},1}} = \frac{T_{\infty 1} - T_2}{\frac{1}{h_1(2\pi r_1 L)} + \frac{\ln(r_2/r_1)}{2\pi L k_1}} \quad (3-47)$$

We could also calculate T_2 from

$$\dot{Q} = \frac{T_2 - T_{\infty 2}}{R_2 + R_3 + R_{\text{conv},2}} = \frac{T_2 - T_{\infty 2}}{\frac{\ln(r_3/r_2)}{2\pi L k_2} + \frac{\ln(r_4/r_3)}{2\pi L k_3} + \frac{1}{h_o(2\pi r_4 L)}} \quad (3-48)$$

Although both relations give the same result, we prefer the first one since it involves fewer terms and thus less work.

The thermal resistance concept can also be used for *other geometries*, provided that the proper conduction resistances and the proper surface areas in convection resistances are used.



$$\begin{aligned} \dot{Q} &= \frac{T_{\infty 1} - T_1}{R_{\text{conv},1}} \\ &= \frac{T_{\infty 1} - T_2}{R_{\text{conv},1} + R_1} \\ &= \frac{T_1 - T_3}{R_1 + R_2} \\ &= \frac{T_2 - T_3}{R_2} \\ &= \frac{T_2 - T_{\infty 2}}{R_2 + R_{\text{conv},2}} \\ &= \dots \end{aligned}$$

FIGURE 3-27

The ratio $\Delta T/R$ across any layer is equal to $\dot{Q}$, which remains constant in one-dimensional steady conduction.

EXAMPLE 3-7 Heat Transfer to a Spherical Container

A 3-m internal diameter spherical tank made of 2-cm-thick stainless steel ($k = 15$ W/m·K) is used to store iced water at $T_{\infty 1} = 0^\circ\text{C}$. The tank is located in a room whose temperature is $T_{\infty 2} = 22^\circ\text{C}$. The walls of the room are also at 22°C . The outer surface of the tank is black, and heat transfer between the outer surface of the tank and the surroundings is by natural convection and radiation. The convection heat transfer coefficients at the inner and the outer surfaces of the tank are $h_1 = 80$ W/m²·K and $h_2 = 10$ W/m²·K, respectively. Determine (a) the rate of heat transfer to the iced water in the tank and (b) the amount of ice at 0°C that melts during a 24-h period.

SOLUTION A spherical container filled with iced water is subjected to convection and radiation heat transfer at its outer surface. The rate of heat transfer and the amount of ice that melts per day are to be determined.

Assumptions 1 Heat transfer is steady since the specified thermal conditions at the boundaries do not change with time. 2 Heat transfer is one-dimensional since there is thermal symmetry about the midpoint. 3 Thermal conductivity is constant.

Properties The thermal conductivity of steel is given to be $k = 15$ W/m·K. The heat of fusion of water at atmospheric pressure is $h_{if} = 333.7$ kJ/kg. The outer surface of the tank is black, and thus its emissivity is $\varepsilon = 1$.

Analysis (a) The thermal resistance network for this problem is given in Fig. 3-28. Noting that the inner diameter of the tank is $D_1 = 3$ m and the outer diameter is $D_2 = 3.04$ m, the inner and the outer surface areas of the tank are

$$A_1 = \pi D_1^2 = \pi(3 \text{ m})^2 = 28.2 \text{ m}^2$$

$$A_2 = \pi D_2^2 = \pi(3.04 \text{ m})^2 = 29.0 \text{ m}^2$$

Also, the radiation heat transfer coefficient is given by

$$h_{\text{rad}} = \varepsilon \sigma (T_2^2 + T_{\infty 2}^2)(T_2 + T_{\infty 2})$$

But we do not know the outer surface temperature T_2 of the tank, and thus we cannot calculate h_{rad} . Therefore, we need to assume a T_2 value now and check the accuracy

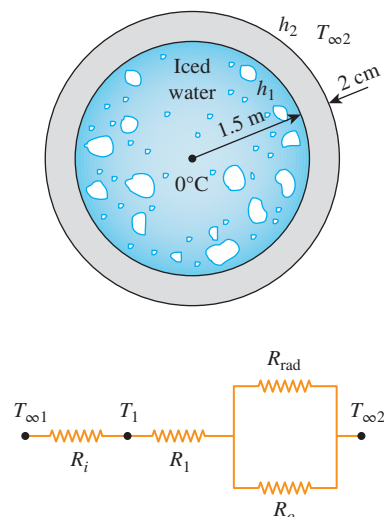


FIGURE 3-28

Schematic for Example 3-7.

of this assumption later. We will repeat the calculations if necessary using a revised value for T_2 .

We note that T_2 must be between 0°C and 22°C , but it must be closer to 0°C , since the heat transfer coefficient inside the tank is much larger. Taking $T_2 = 5^\circ\text{C} = 278\text{ K}$, the radiation heat transfer coefficient is determined to be

$$\begin{aligned} h_{\text{rad}} &= (1)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) [(295 \text{ K})^2 + (278 \text{ K})^2][(295 + 278) \text{ K}] \\ &= 5.34 \text{ W/m}^2 \cdot \text{K} = 5.34 \text{ W/m}^2 \cdot ^\circ\text{C} \end{aligned}$$

Then the individual thermal resistances become

$$R_i = R_{\text{conv},1} = \frac{1}{h_1 A_1} = \frac{1}{(80 \text{ W/m}^2 \cdot \text{K})(28.3 \text{ m}^2)} = 0.000442^\circ\text{C/W}$$

$$\begin{aligned} R_1 = R_{\text{sphere}} &= \frac{r_2 - r_1}{4\pi k r_1 r_2} = \frac{(1.52 - 1.50) \text{ m}}{4\pi(15 \text{ W/m} \cdot \text{K})(1.52 \text{ m})(1.50 \text{ m})} \\ &= 0.000047^\circ\text{C/W} \end{aligned}$$

$$R_o = R_{\text{conv},2} = \frac{1}{h_2 A_2} = \frac{1}{(10 \text{ W/m}^2 \cdot \text{K})(29.0 \text{ m}^2)} = 0.00345^\circ\text{C/W}$$

$$R_{\text{rad}} = \frac{1}{h_{\text{rad}} A_2} = \frac{1}{(5.34 \text{ W/m}^2 \cdot \text{K})(29.0 \text{ m}^2)} = 0.00646^\circ\text{C/W}$$

The two parallel resistances R_o and R_{rad} can be replaced by an equivalent resistance R_{equiv} determined from

$$\frac{1}{R_{\text{equiv}}} = \frac{1}{R_o} + \frac{1}{R_{\text{rad}}} = \frac{1}{0.00345} + \frac{1}{0.00646} = 444.7 \text{ W/}^\circ\text{C}$$

which gives

$$R_{\text{equiv}} = 0.00225^\circ\text{C/W}$$

Now all the resistances are in series, and the total resistance is

$$R_{\text{total}} = R_i + R_1 + R_{\text{equiv}} = 0.000442 + 0.000047 + 0.00225 = 0.00274^\circ\text{C/W}$$

Then the steady rate of heat transfer to the iced water becomes

$$\dot{Q} = \frac{T_{\infty 2} - T_{\infty 1}}{R_{\text{total}}} = \frac{(22 - 0)^\circ\text{C}}{0.00274^\circ\text{C/W}} = \mathbf{8029 \text{ W}} \quad (\text{or } \dot{Q} = 8.029 \text{ kJ/s})$$

To check the validity of our original assumption, we now determine the outer surface temperature from

$$\begin{aligned} \dot{Q} &= \frac{T_{\infty 2} - T_2}{R_{\text{equiv}}} \longrightarrow T_2 = T_{\infty 2} - \dot{Q} R_{\text{equiv}} \\ &= 22^\circ\text{C} - (8029 \text{ W})(0.00225^\circ\text{C/W}) = 4^\circ\text{C} \end{aligned}$$

which is sufficiently close to the 5°C assumed in the determination of the radiation heat transfer coefficient. Therefore, there is no need to repeat the calculations using 4°C for T_2 .

(b) The total amount of heat transfer during a 24-h period is

$$Q = \dot{Q} \Delta t = (8.029 \text{ kJ/s})(24 \times 3600 \text{ s}) = 693,700 \text{ kJ}$$

Noting that it takes 333.7 kJ of energy to melt 1 kg of ice at 0°C, the amount of ice that will melt during a 24-h period is

$$m_{\text{ice}} = \frac{Q}{h_{\text{if}}} = \frac{693,700 \text{ kJ}}{333.7 \text{ kJ/kg}} = \mathbf{2079 \text{ kg}}$$

Therefore, about 2 metric tons of ice will melt in the tank every day.

Discussion An easier way to deal with combined convection and radiation at a surface when the surrounding medium and surfaces are at the same temperature is to add the radiation and convection heat transfer coefficients and to treat the result as the convection heat transfer coefficient. That is, to take $h = 10 + 5.34 = 15.34 \text{ W/m}^2 \cdot \text{K}$ in this case. This way, we can ignore radiation since its contribution is accounted for in the convection heat transfer coefficient. The convection resistance of the outer surface in this case would be

$$R_{\text{combined}} = \frac{1}{h_{\text{combined}} A_2} = \frac{1}{(15.34 \text{ W/m}^2 \cdot \text{K})(29.0 \text{ m}^2)} = 0.00225^\circ \text{C/W}$$

which is identical to the value obtained for equivalent resistance for the parallel convection and the radiation resistances.

EXAMPLE 3–8 Heat Loss Through an Insulated Steam Pipe

Steam at $T_{\infty 1} = 320^\circ \text{C}$ flows in a cast iron pipe ($k = 80 \text{ W/m} \cdot \text{K}$) whose inner and outer diameters are $D_1 = 5 \text{ cm}$ and $D_2 = 5.5 \text{ cm}$, respectively. The pipe is covered with 3-cm-thick glass wool insulation with $k = 0.05 \text{ W/m} \cdot \text{K}$. Heat is lost to the surroundings at $T_{\infty 2} = 5^\circ \text{C}$ by natural convection and radiation, with a combined heat transfer coefficient of $h_2 = 18 \text{ W/m}^2 \cdot \text{K}$. Taking the heat transfer coefficient inside the pipe to be $h_1 = 60 \text{ W/m}^2 \cdot \text{K}$, determine the rate of heat loss from the steam per unit length of the pipe. Also determine the temperature drops across the pipe shell and the insulation.

SOLUTION A steam pipe covered with glass wool insulation is subjected to convection on its surfaces. The rate of heat transfer per unit length and the temperature drops across the pipe and the insulation are to be determined.

Assumptions **1** Heat transfer is steady since there is no indication of any change with time. **2** Heat transfer is one-dimensional since there is thermal symmetry about the centerline and no variation in the axial direction. **3** Thermal conductivities are constant. **4** The thermal contact resistance at the interface is negligible.

Properties The thermal conductivities are given to be $k = 80 \text{ W/m} \cdot \text{K}$ for cast iron and $k = 0.05 \text{ W/m} \cdot \text{K}$ for glass wool insulation.

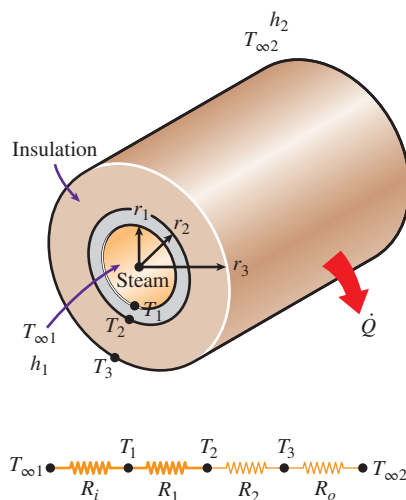


FIGURE 3–29

Schematic for Example 3–8.

Analysis The thermal resistance network for this problem involves four resistances in series and is given in Fig. 3–29. Taking $L = 1$ m, the areas of the surfaces exposed to convection are determined to be

$$A_1 = 2\pi r_1 L = 2\pi(0.025 \text{ m})(1 \text{ m}) = 0.157 \text{ m}^2$$

$$A_3 = 2\pi r_3 L = 2\pi(0.0575 \text{ m})(1 \text{ m}) = 0.361 \text{ m}^2$$

Then the individual thermal resistances become

$$R_i = R_{\text{conv}, 1} = \frac{1}{h_1 A_1} = \frac{1}{(60 \text{ W/m}^2 \cdot \text{K})(0.157 \text{ m}^2)} = 0.106^\circ\text{C/W}$$

$$R_1 = R_{\text{pipe}} = \frac{\ln(r_2/r_1)}{2\pi k_1 L} = \frac{\ln(2.75/2.5)}{2\pi(80 \text{ W/m} \cdot \text{K})(1 \text{ m})} = 0.0002^\circ\text{C/W}$$

$$R_2 = R_{\text{insulation}} = \frac{\ln(r_3/r_2)}{2\pi k_2 L} = \frac{\ln(5.75/2.75)}{2\pi(0.05 \text{ W/m} \cdot \text{K})(1 \text{ m})} = 2.35^\circ\text{C/W}$$

$$R_o = R_{\text{conv}, 2} = \frac{1}{h_2 A_3} = \frac{1}{(18 \text{ W/m}^2 \cdot \text{K})(0.361 \text{ m}^2)} = 0.154^\circ\text{C/W}$$

Noting that all resistances are in series, the total resistance is determined to be

$$R_{\text{total}} = R_i + R_1 + R_2 + R_o = 0.106 + 0.0002 + 2.35 + 0.154 = 2.61^\circ\text{C/W}$$

Then the steady rate of heat loss from the steam becomes

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{\text{total}}} = \frac{(320 - 5)^\circ\text{C}}{2.61^\circ\text{C/W}} = \mathbf{121 \text{ W}} \quad (\text{per m pipe length})$$

The heat loss for a given pipe length can be determined by multiplying the preceding quantity by the pipe length L .

The temperature drops across the pipe and the insulation are determined from Eq. 3–17 to be

$$\Delta T_{\text{pipe}} = \dot{Q} R_{\text{pipe}} = (121 \text{ W})(0.0002^\circ\text{C/W}) = \mathbf{0.02^\circ\text{C}}$$

$$\Delta T_{\text{insulation}} = \dot{Q} R_{\text{insulation}} = (121 \text{ W})(2.35^\circ\text{C/W}) = \mathbf{284^\circ\text{C}}$$

That is, the temperatures between the inner and the outer surfaces of the pipe differ by 0.02°C , whereas the temperatures between the inner and the outer surfaces of the insulation differ by 284°C .

Discussion Note that the thermal resistance of the pipe is too small relative to the other resistances and can be neglected without causing any significant error. Also note that the temperature drop across the pipe is practically zero, and thus the pipe can be assumed to be isothermal. The resistance to heat flow in insulated pipes is primarily due to insulation.

3–5 ■ CRITICAL RADIUS OF INSULATION

We know that adding more insulation to a wall or to the attic always decreases heat transfer. The thicker the insulation, the lower the heat transfer rate. This is expected, since the heat transfer area A is constant, and adding insulation always increases the thermal resistance of the wall without increasing the convection resistance.

Adding insulation to a cylindrical pipe or a spherical shell, however, is a different matter. The additional insulation increases the conduction resistance

of the insulation layer but decreases the convection resistance of the surface because of the increase in the outer surface area for convection. The heat transfer from the pipe may increase or decrease, depending on which effect dominates.

Consider a cylindrical pipe of outer radius r_1 whose outer surface temperature T_1 is kept constant (Fig. 3–30). The pipe is now insulated with a material whose thermal conductivity is k and whose outer radius is r_2 . Heat is lost from the pipe to the surrounding medium at temperature T_∞ , with a convection heat transfer coefficient h . The rate of heat transfer from the insulated pipe to the surrounding air can be expressed as (Fig. 3–31)

$$\dot{Q} = \frac{T_1 - T_\infty}{R_{\text{ins}} + R_{\text{conv}}} = \frac{T_1 - T_\infty}{\frac{\ln(r_2/r_1)}{2\pi Lk} + \frac{1}{h(2\pi r_2 L)}} \quad (3-49)$$

The variation of $\dot{Q}$ with the outer radius of the insulation r_2 is plotted in Fig. 3–31. The value of r_2 at which $\dot{Q}$ reaches a maximum is determined from the requirement that $d\dot{Q}/dr_2 = 0$ (zero slope). Performing the differentiation and solving for r_2 yields the **critical radius of insulation** for a cylindrical body:

$$r_{\text{cr, cylinder}} = \frac{k}{h} \quad (\text{m}) \quad (3-50)$$

Note that the critical radius of insulation depends on the thermal conductivity of the insulation k and the external convection heat transfer coefficient h . The rate of heat transfer from the cylinder increases with the addition of insulation for $r_2 < r_{\text{cr}}$, reaches a maximum when $r_2 = r_{\text{cr}}$, and starts to decrease for $r_2 > r_{\text{cr}}$. Thus, insulating the pipe may actually increase the rate of heat transfer from the pipe instead of decreasing it when $r_2 < r_{\text{cr}}$.

The important question to answer at this point is whether we need to be concerned about the critical radius of insulation when insulating hot-water pipes or even hot-water tanks. Should we always check and make sure that the outer radius of insulation sufficiently exceeds the critical radius before we install any insulation? Probably not, as explained here.

The value of the critical radius r_{cr} is largest when k is large and h is small. Noting that the lowest value of h encountered in practice is about $5 \text{ W/m}^2\cdot\text{K}$ for the case of natural convection of gases, and that the thermal conductivity of common insulating materials is about $0.05 \text{ W/m}\cdot\text{K}$, the largest value of the critical radius we are likely to encounter is

$$r_{\text{cr, max}} = \frac{k_{\text{max, insulation}}}{h_{\text{min}}} \approx \frac{0.05 \text{ W/m}\cdot\text{K}}{5 \text{ W/m}^2\cdot\text{K}} = 0.01 \text{ m} = 1 \text{ cm}$$

This value would be even smaller when the radiation effects are considered. The critical radius would be much less in forced convection, often less than 1 mm, because of much larger h values associated with forced convection. Therefore, we can insulate hot-water or steam pipes freely without worrying about the possibility of increasing the heat transfer by insulating the pipes.

The radius of electric wires may be smaller than the critical radius. Therefore, the plastic electrical insulation may actually *enhance* the heat

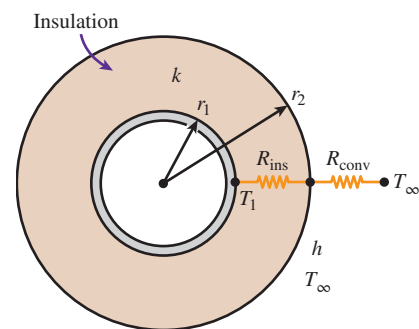


FIGURE 3-30

An insulated cylindrical pipe exposed to convection from the outer surface and the thermal resistance network associated with it.

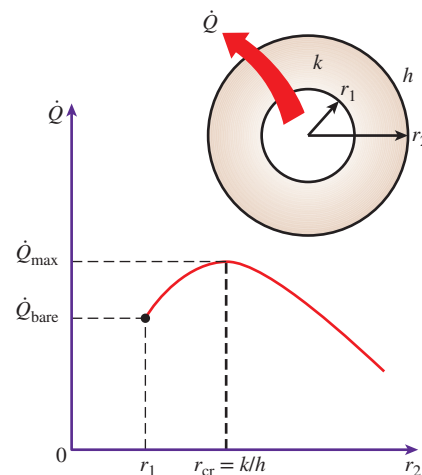


FIGURE 3-31

The variation of heat transfer rate with the outer radius of the insulation r_2 when $r_1 < r_{\text{cr}}$.

transfer from electric wires and thus keep their steady operating temperatures at lower and thus safer levels.

The discussions above can be repeated for a sphere, and it can be shown in a similar manner that the critical radius of insulation for a spherical shell is

$$r_{cr, sphere} = \frac{2k}{h} \quad (3-51)$$

where k is the thermal conductivity of the insulation and h is the convection heat transfer coefficient on the outer surface.

EXAMPLE 3-9 Heat Loss from an Insulated Electric Wire

A 3-mm-diameter and 5-m-long electric wire is tightly wrapped with a 2-mm-thick plastic cover whose thermal conductivity is $k = 0.15 \text{ W/m}\cdot\text{K}$. Electrical measurements indicate that a current of 10 A passes through the wire, and there is a voltage drop of 8 V along the wire. If the insulated wire is exposed to a medium at $T_\infty = 30^\circ\text{C}$ with a heat transfer coefficient of $h = 12 \text{ W/m}^2\cdot\text{K}$, determine the temperature at the interface of the wire and the plastic cover in steady operation. Also determine whether doubling the thickness of the plastic cover will increase or decrease this interface temperature.

SOLUTION An electric wire is tightly wrapped with a plastic cover. The interface temperature and the effect of doubling the thickness of the plastic cover on the interface temperature are to be determined.

Assumptions 1 Heat transfer is steady since there is no indication of any change with time. 2 Heat transfer is one-dimensional since there is thermal symmetry about the centerline and no variation in the axial direction. 3 Thermal conductivities are constant. 4 The thermal contact resistance at the interface is negligible. 5 Heat transfer coefficient incorporates the radiation effects, if any.

Properties The thermal conductivity of plastic is given to be $k = 0.15 \text{ W/m}\cdot\text{K}$.

Analysis Heat is generated in the wire, and its temperature rises as a result of resistance heating. We assume heat is generated uniformly throughout the wire and is transferred to the surrounding medium in the radial direction. In steady operation, the rate of heat transfer becomes equal to the heat generated within the wire, which is determined to be

$$\dot{Q} = \dot{W}_e = VI = (8 \text{ V})(10 \text{ A}) = 80 \text{ W}$$

The thermal resistance network for this problem involves a conduction resistance for the plastic cover and a convection resistance for the outer surface in series, as shown in Fig. 3-32. The values of these two resistances are

$$A_2 = (2\pi r_2) L = 2\pi(0.0035 \text{ m})(5 \text{ m}) = 0.110 \text{ m}^2$$

$$R_{\text{conv}} = \frac{1}{hA_2} = \frac{1}{(12 \text{ W/m}^2\cdot\text{K})(0.110 \text{ m}^2)} = 0.76^\circ\text{C/W}$$

$$R_{\text{plastic}} = \frac{\ln(r_2/r_1)}{2\pi kL} = \frac{\ln(3.5/1.5)}{2\pi(0.15 \text{ W/m}\cdot\text{K})(5 \text{ m})} = 0.18^\circ\text{C/W}$$

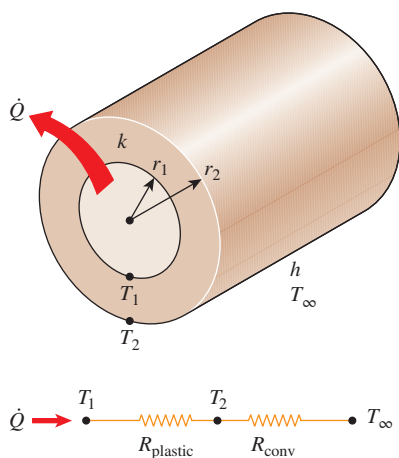


FIGURE 3-32

Schematic for Example 3-9.

and therefore

$$R_{\text{total}} = R_{\text{plastic}} + R_{\text{conv}} = 0.76 + 0.18 = 0.94^\circ\text{C/W}$$

Then the interface temperature can be determined from

$$\begin{aligned}\dot{Q} = \frac{T_1 - T_\infty}{R_{\text{total}}} \quad \longrightarrow \quad T_1 &= T_\infty + \dot{Q} R_{\text{total}} \\ &= 30^\circ\text{C} + (80 \text{ W})(0.94^\circ\text{C/W}) = \mathbf{105^\circ\text{C}}\end{aligned}$$

Note that we did not involve the electrical wire directly in the thermal resistance network, since the wire involves heat generation.

To answer the second part of the question, we need to know the critical radius of insulation of the plastic cover. It is determined from Eq. 3–50 to be

$$r_{\text{cr}} = \frac{k}{h} = \frac{0.15 \text{ W/m}\cdot\text{K}}{12 \text{ W/m}^2\cdot\text{K}} = 0.0125 \text{ m} = 12.5 \text{ mm}$$

which is larger than the radius of the plastic cover. Therefore, increasing the thickness of the plastic cover will *enhance* heat transfer until the outer radius of the cover reaches 12.5 mm. As a result, the rate of heat transfer $\dot{Q}$ will *increase* when the interface temperature T_1 is held constant, or T_1 will *decrease* when $\dot{Q}$ is held constant, which is the case here.

Discussion It can be shown by repeating the calculations for a 4-mm-thick plastic cover that the interface temperature drops to 90.6°C when the thickness of the plastic cover is doubled. It can also be shown in a similar manner that the interface reaches a minimum temperature of 83°C when the outer radius of the plastic cover equals the critical radius.

3–6 ■ HEAT TRANSFER FROM FINNED SURFACES

The rate of heat transfer from a surface at a temperature T_s to the surrounding medium at T_∞ is given by Newton's law of cooling as

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty)$$

where A_s is the heat transfer surface area and h is the convection heat transfer coefficient. When the temperatures T_s and T_∞ are fixed by design considerations, as is often the case, there are *two ways* to increase the rate of heat transfer: to increase the *convection heat transfer coefficient* h or to increase the *surface area* A_s . Increasing h may require the installation of a pump or fan, or replacing the existing one with a larger one, but this approach may or may not be practical. Besides, it may not be adequate. The alternative is to increase the surface area by attaching to the surface *extended surfaces* called *fins* made of highly conductive materials such as aluminum. Finned surfaces are manufactured by extruding, welding, or wrapping a thin metal sheet on a surface. Fins enhance heat transfer from a surface by exposing a larger surface area to convection and radiation.

An interesting application of fins from about 150 million years ago, the Jurassic era, is shown in Fig. 3–33. The dinosaur stegosaurus lived during



FIGURE 3–33

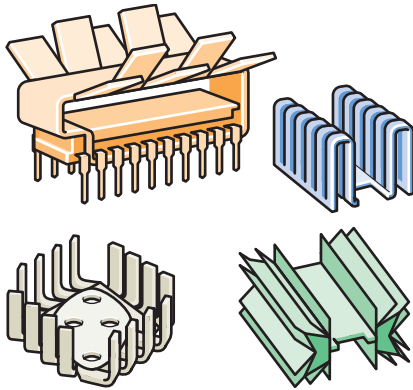
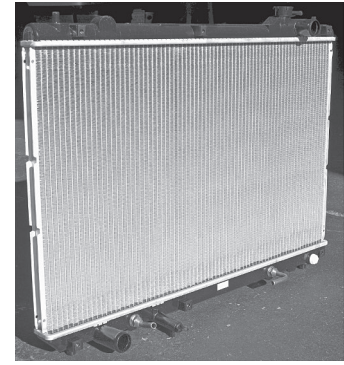
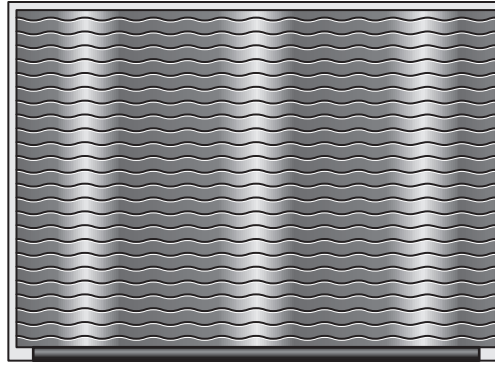
Presumed cooling fins on dinosaur stegosaurus.

©Christian Darkin/Alamy

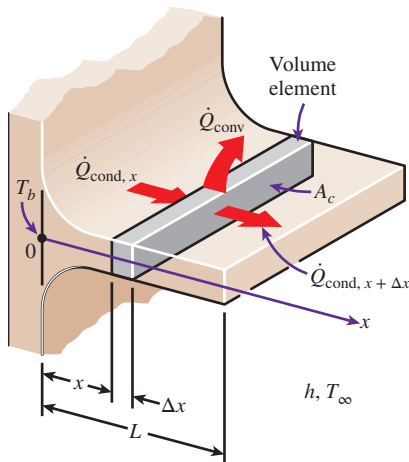
FIGURE 3–34

The thin plate fins of a car radiator greatly increase the rate of heat transfer to the air.

Left: ©James Kleiser/Yunus Cengel
right: ©McGraw-Hill Education/Christopher Kerrigan, photographer

**FIGURE 3–35**

Some innovative fin designs.

**FIGURE 3–36**

Volume element of a fin at location x having a length of Δx , cross-sectional area of A_c , and perimeter of p .

this era, and it had two rows of big (and bizarre) bony plates down its back. For a long time, scientists thought that the plates were some kind of armor to protect the vegetarian from predators. We now know that a lot of blood flowed through the plates, and they may have acted like a car radiator. The heart pumped blood through the plates, and the plates acted like cooling fins to cool the blood down.

Finned surfaces are commonly used in practice to enhance heat transfer, and they often increase the rate of heat transfer from a surface severalfold. The car radiator shown in Fig. 3–34 is an example of a finned surface. The closely packed thin metal sheets attached to the hot-water tubes increase the surface area for convection and thus the rate of convection heat transfer from the tubes to the air many times. There are a variety of innovative fin designs available in the market, and they seem to be limited only by imagination (Fig. 3–35).

In the analysis of fins, we consider *steady* operation with *no heat generation* in the fin, and we assume the thermal conductivity k of the material to remain constant. We also assume the convection heat transfer coefficient h to be *constant* and *uniform* over the entire surface of the fin for convenience in the analysis. We recognize that the convection heat transfer coefficient h , in general, varies along the fin as well as along its circumference, and its value at a point is a strong function of the *fluid motion* at that point. The value of h is usually much lower at the *fin base* than it is at the *fin tip* because the fluid is surrounded by solid surfaces near the base, which seriously disrupt its motion to the point of “suffocating” it, while the fluid near the fin tip has little contact with a solid surface and thus encounters little resistance to flow. Therefore, adding too many fins on a surface may actually decrease the overall heat transfer when the decrease in h offsets any gain resulting from the increase in the surface area.

Fin Equation

Consider a volume element of a fin at location x having a length of Δx , a cross-sectional area of A_c , and a perimeter of p , as shown in Fig. 3–36. Under steady conditions, the energy balance on this volume element can be expressed as

$$\left(\begin{array}{c} \text{Rate of heat} \\ \text{conduction into} \\ \text{the element at } x \end{array} \right) = \left(\begin{array}{c} \text{Rate of heat} \\ \text{conduction from the} \\ \text{element at } x + \Delta x \end{array} \right) + \left(\begin{array}{c} \text{Rate of heat} \\ \text{convection from} \\ \text{the element} \end{array} \right)$$

or

$$\dot{Q}_{\text{cond},x} = \dot{Q}_{\text{cond},x+\Delta x} + \dot{Q}_{\text{conv}}$$

where

$$\dot{Q}_{\text{conv}} = h(p \Delta x)(T - T_{\infty})$$

Substituting and dividing by Δx , we obtain

$$\frac{\dot{Q}_{\text{cond},x+\Delta x} - \dot{Q}_{\text{cond},x}}{\Delta x} + hp(T - T_{\infty}) = 0 \quad (3-52)$$

Taking the limit as $\Delta x \rightarrow 0$ gives

$$\frac{d\dot{Q}_{\text{cond}}}{dx} + hp(T - T_{\infty}) = 0 \quad (3-53)$$

From Fourier's law of heat conduction we have

$$\dot{Q}_{\text{cond}} = -kA_c \frac{dT}{dx} \quad (3-54)$$

where A_c is the cross-sectional area of the fin at location x . Substitution of this relation into Eq. 3-53 gives the differential equation governing heat transfer in fins,

$$\frac{d}{dx} \left(kA_c \frac{dT}{dx} \right) - hp(T - T_{\infty}) = 0 \quad (3-55)$$

In general, the cross-sectional area A_c and the perimeter p of a fin vary with x , which makes this differential equation difficult to solve. In the special case of *constant cross section* and *constant thermal conductivity*, the differential equation Eq. 3-55 reduces to

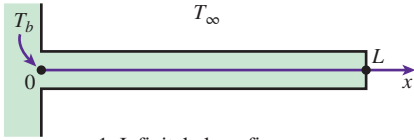
$$\frac{d^2T}{dx^2} - \frac{hp}{kA_c}(T - T_{\infty}) = 0 \quad \text{or} \quad \frac{d^2\theta}{dx^2} - m^2\theta = 0 \quad (3-56)$$

where

$$m^2 = \frac{hp}{kA_c} \quad (3-57)$$

and $\theta = T - T_{\infty}$ is the *temperature excess*. At the fin base we have $\theta_b = T_b - T_{\infty}$.

Equation 3-56 is a linear, homogeneous, second-order differential equation with constant coefficients. A fundamental theory of differential equations states that such an equation has two linearly independent solution functions, and its general solution is the linear combination of those two solution functions. A careful examination of the differential equation reveals that subtracting a constant multiple of the solution function θ from its second derivative yields zero. Thus we conclude that the function θ and its second derivative must be *constant multiples* of each other. The only functions whose derivatives are constant multiples of the functions themselves are the *exponential functions* (or a linear combination of exponential functions such as sine and cosine hyperbolic functions). Therefore, the solution functions of the preceding differential equation are the exponential functions e^{-mx} or e^{mx} or constant multiples of them. This can be verified by direct substitution. For example, the second derivative of e^{-mx} is m^2e^{-mx} , and its substitution into Eq. 3-56



1. Infinitely long fin
2. Negligible heat loss (adiabatic tip)
3. Specified temperature
4. Convection

FIGURE 3-37

Boundary conditions at the fin base and the fin tip.

yields zero. Therefore, the general solution of the differential equation Eq. 3-56 is

$$\theta(x) = C_1 e^{mx} + C_2 e^{-mx} \quad (3-58)$$

where C_1 and C_2 are arbitrary constants whose values are to be determined from the boundary conditions at the base and at the tip of the fin. Note that we need only two conditions to determine C_1 and C_2 uniquely.

The temperature of the plate to which the fins are attached is normally known in advance. Therefore, at the fin base we have a *specified temperature* boundary condition, expressed as

$$\text{Boundary condition at fin base:} \quad \theta(0) = \theta_b = T_b - T_\infty \quad (3-59)$$

At the fin tip we have several possibilities, including infinitely long fins, negligible heat loss (idealized as an adiabatic tip), specified temperature, and convection (Fig. 3-37). Next, we consider each case separately.

1 Infinitely Long Fin ($T_{\text{fin tip}} = T_\infty$)

For a sufficiently long fin of *uniform* cross section ($A_c = \text{constant}$), the temperature of the fin at the fin tip approaches the environment temperature T_∞ and thus θ approaches zero. That is,

$$\text{Boundary condition at fin tip:} \quad \theta(L) = T(L) - T_\infty = 0 \quad \text{as} \quad L \rightarrow \infty$$

This condition is satisfied by the function e^{-mx} , but not by the other prospective solution function e^{mx} since it tends to infinity as x gets larger. Therefore, the general solution in this case will consist of a constant multiple of e^{-mx} . The value of the constant multiple is determined from the requirement that at the fin base where $x = 0$, the value of θ is θ_b . Noting that $e^{-mx} = e^0 = 1$, the proper value of the constant is θ_b , and the solution function we are looking for is $\theta(x) = \theta_b e^{-mx}$. This function satisfies the differential equation as well as the requirements that the solution reduce to θ_b at the fin base and approach zero at the fin tip for large x . Noting that $\theta = T - T_\infty$ and $m = \sqrt{hp/kA_c}$, the variation of temperature along the fin in this case can be expressed as

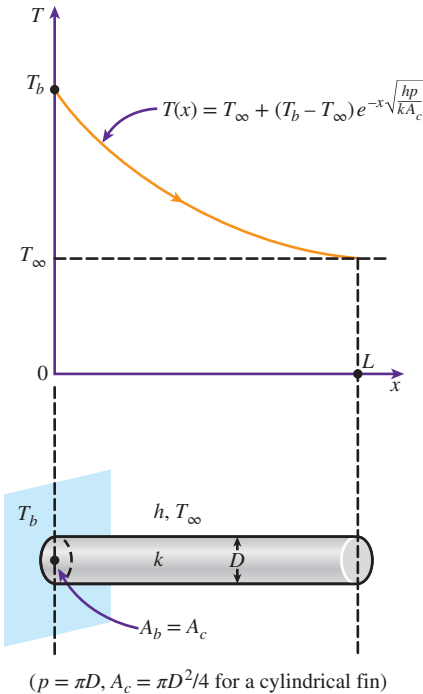
$$\text{Very long fin:} \quad \frac{T(x) - T_\infty}{T_b - T_\infty} = e^{-mx} = e^{-x\sqrt{hp/kA_c}} \quad (3-60)$$

Note that the temperature along the fin in this case decreases *exponentially* from T_b to T_∞ , as shown in Fig. 3-38. The steady rate of *heat transfer* from the entire fin can be determined from Fourier's law of heat conduction

$$\text{Very long fin:} \quad \dot{Q}_{\text{long fin}} = -kA_c \left. \frac{dT}{dx} \right|_{x=0} = \sqrt{hpkA_c} (T_b - T_\infty) \quad (3-61)$$

where p is the perimeter, A_c is the cross-sectional area of the fin, and x is the distance from the fin base. Alternatively, the rate of heat transfer from the fin could also be determined by considering heat transfer from a differential volume element of the fin and integrating it over the entire surface of the fin:

$$\dot{Q}_{\text{fin}} = \int_{A_{\text{fin}}} h[T(x) - T_\infty] dA_{\text{fin}} = \int_{A_{\text{fin}}} h\theta(x) dA_{\text{fin}} \quad (3-62)$$



($p = \pi D$, $A_c = \pi D^2/4$ for a cylindrical fin)

FIGURE 3-38

A long circular fin of uniform cross section and the variation of temperature along it.

The two approaches described are equivalent and give the same result since, under steady conditions, the heat transfer from the exposed surfaces of the fin is equal to the heat transfer to the fin at the base (Fig. 3–39).

2 Negligible Heat Loss from the Fin Tip (Adiabatic fin tip, $\dot{Q}_{\text{fin tip}} = 0$)

Fins are not likely to be so long that their temperature approaches the surrounding temperature at the tip. A more realistic situation is for heat transfer from the fin tip to be negligible since the heat transfer from the fin is proportional to its surface area, and the surface area of the fin tip is usually a negligible fraction of the total fin area. Then the fin tip can be assumed to be adiabatic, and the condition at the fin tip can be expressed as

$$\text{Boundary condition at fin tip: } \left. \frac{d\theta}{dx} \right|_{x=L} = 0 \quad (3-63)$$

The condition at the fin base remains the same as expressed in Eq. 3–59. The application of the boundary conditions given by Eqs. 3–59 and 3–63 on the general solution (Eq. 3–58) requires that $\theta(0) = \theta_b = C_1 + C_2$ and $mC_1e^{mL} - mC_2e^{-mL} = 0$, respectively. Solving these two equations simultaneously for C_1 and C_2 yields $C_1 = \theta_b/(1 + e^{2mL})$ and $C_2 = \theta_b/(1 + e^{-2mL})$. Substituting the relations for C_1 and C_2 into Eq. 3–58 and using the definition of the hyperbolic cosine function $\cosh x = (e^x + e^{-x})/2$ gives the desired relation for the temperature distribution:

$$\text{Adiabatic fin tip: } \frac{T(x) - T_\infty}{T_b - T_\infty} = \frac{\cosh m(L - x)}{\cosh mL} \quad (3-64)$$

The rate of heat transfer from the fin can be determined again from Fourier's law of heat conduction:

$$\text{Adiabatic fin tip: } \dot{Q}_{\text{adiabatic tip}} = -kA_c \left. \frac{dT}{dx} \right|_{x=0} = \sqrt{hpkA_c} (T_b - T_\infty) \tanh mL \quad (3-65)$$

where the equation for the hyperbolic tangent function is

$$\tanh x = \sinh x / \cosh x = (e^x - e^{-x}) / (e^x + e^{-x}).$$

Note that the heat transfer relations for the very long fin and the fin with negligible heat loss at the tip differ by the factor $\tanh mL$, which approaches 1 as L becomes very large.

3 Specified Temperature ($T_{\text{fin, tip}} = T_L$)

In this case the temperature at the end of the fin (the fin tip) is fixed at a specified temperature T_L . This case could be considered as a generalization of the case of *Infinitely Long Fin* where the fin tip temperature was fixed at T_∞ . The condition at the fin tip for this case is

$$\text{Boundary condition at fin tip: } \theta(L) = \theta_L = T_L - T_\infty \quad (3-66)$$

The fin base boundary condition remains the same as given in Eq. 3–59. Applying the boundary conditions given by Eqs. 3–59 and 3–66 on the general solution (Eq. 3–58) gives, after some lengthy algebra and using the

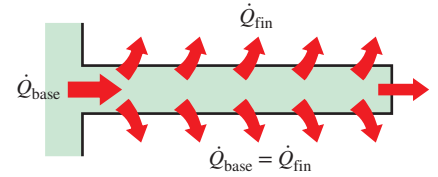


FIGURE 3–39

Under steady conditions, heat transfer from the exposed surfaces of the fin is equal to heat conduction to the fin at the base.

definition of the hyperbolic sine function, $\sinh x = (e^x - e^{-x})/2$, the desired temperature distribution:

Specified fin tip temperature:

$$\frac{T(x) - T_\infty}{T_b - T_\infty} = \frac{[(T_L - T_\infty)/(T_b - T_\infty)] \sinh mx + \sinh m(L - x)}{\sinh mL} \quad (3-67)$$

Using the Fourier's law of heat conduction, the *rate of heat transfer* from the fin is

Specified fin tip temperature:

$$\begin{aligned} \dot{Q}_{\text{specified temp.}} &= -kA_c \left. \frac{dT}{dx} \right|_{x=0} \\ &= \sqrt{hpkA_c}(T_b - T_\infty) \frac{\cosh mL - [(T_L - T_\infty)/(T_b - T_\infty)]}{\sinh mL} \end{aligned} \quad (3-68)$$

Note that Eqs. 3-67 and 3-68 reduce to Eqs. 3-60 and 3-61 for the case of *infinitely long fin* ($L \rightarrow \infty$).

4 Convection from Fin Tip

The fin tips, in practice, are exposed to the surroundings, and thus the proper boundary condition for the fin tip is convection that may also include the effects of radiation. Consider the case of convection only at the tip. The condition at the fin tip can be obtained from an energy balance at the fin tip ($\dot{Q}_{\text{cond}} = \dot{Q}_{\text{conv}}$). That is,

$$\text{Boundary condition at fin tip:} \quad -kA_c \left. \frac{dT}{dx} \right|_{x=L} = hA_c[T(L) - T_\infty] \quad (3-69)$$

The boundary condition at the fin base is Eq. 3-59, which is the same as the three previous cases. Substituting the two boundary conditions given by Eqs. 3-59 and 3-69 into the general solution (Eq. 3-58), it may be shown after some lengthy manipulation that the temperature distribution is

$$\text{Convection from fin tip:} \quad \frac{T(x) - T_\infty}{T_b - T_\infty} = \frac{\cosh m(L - x) + (h/mk) \sinh m(L - x)}{\cosh mL + (h/mk) \sinh mL} \quad (3-70)$$

The *rate of heat transfer* from the fin can be found by substituting the temperature gradient at the base of the fin, obtained from Eq. 3-70, into Fourier's law of heat conduction. The result is

Convection from fin tip:

$$\begin{aligned} \dot{Q}_{\text{convection}} &= -kA_c \left. \frac{dT}{dx} \right|_{x=0} \\ &= \sqrt{hpkA_c}(T_b - T_\infty) \frac{\sinh mL + (h/mk) \cosh mL}{\cosh mL + (h/mk) \sinh mL} \end{aligned} \quad (3-71)$$

The solution to the general fin equation for the case of *convection from fin tip* is rather complex. An approximate, yet practical and accurate, way of accounting for the loss from the fin tip is to replace the *fin length* L in the relation for the *insulated tip* case with a **corrected fin length** defined as (Fig. 3-40)

$$\text{Corrected fin length:} \quad L_c = L + \frac{A_c}{p} \quad (3-72)$$

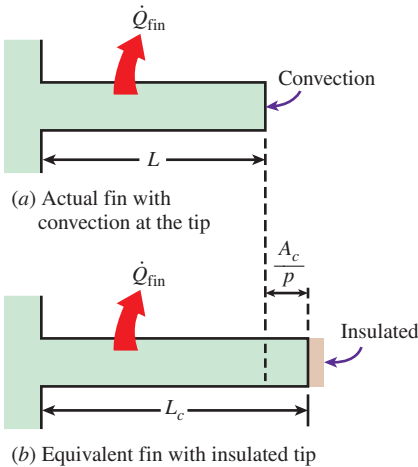


FIGURE 3-40

Corrected fin length L_c is defined such that heat transfer from a fin of length L_c with insulated tip is equal to heat transfer from the actual fin of length L with convection at the fin tip.

where A_c is the cross-sectional area and p is the perimeter of the fin at the tip. Multiplying the preceding relation by the perimeter gives $A_{\text{corrected}} = A_{\text{fin(lateral)}} + A_{\text{tip}}$, which indicates that the fin area determined using the corrected length is equivalent to the sum of the lateral fin area plus the fin tip area.

The corrected length approximation gives very good results when the variation of temperature near the fin tip is small (which is the case when $mL \geq 1$) and the heat transfer coefficient at the fin tip is about the same as that at the lateral surface of the fin. Therefore, *pins subjected to convection at their tips can be treated as fins with insulated tips by replacing the actual fin length with the corrected length in Eqs. 3–64 and 3–65.*

Using the proper relations for A_c and p , the corrected lengths for rectangular and cylindrical fins are easily determined to be

$$L_{c, \text{rectangular fin}} = L + \frac{t}{2} \quad \text{and} \quad L_{c, \text{cylindrical fin}} = L + \frac{D}{4}$$

where t is the thickness of the rectangular fins and D is the diameter of the cylindrical fins.

Fin Efficiency

Consider the surface of a *plane wall* at temperature T_b exposed to a medium at temperature T_∞ . Heat is lost from the surface to the surrounding medium by convection with a heat transfer coefficient of h . Disregarding radiation or accounting for its contribution in the convection coefficient h , heat transfer from a surface area A_s is expressed as $\dot{Q} = hA_s(T_b - T_\infty)$.

Now let us consider a fin of constant cross-sectional area $A_c = A_b$ and length L that is attached to the surface with a perfect contact (Fig. 3–41). This time heat is transferred from the surface to the fin *by conduction* and from the fin to the surrounding medium *by convection* with the same heat transfer coefficient h . The temperature of the fin is T_b at the fin base and gradually decreases toward the fin tip. Convection from the fin surface causes the temperature at any cross section to drop somewhat from the midsection toward the outer surfaces. However, the cross-sectional area of the fins is usually very small, and thus the temperature at any cross section can be considered to be uniform. Also, the fin tip can be assumed for convenience and simplicity to be adiabatic by using the corrected length for the fin instead of the actual length.

In the limiting case of *zero thermal resistance* or *infinite thermal conductivity* ($k \rightarrow \infty$), the temperature of the fin is uniform at the base value of T_b . The heat transfer from the fin is *maximum* in this case and can be expressed as

$$\dot{Q}_{\text{fin, max}} = hA_{\text{fin}}(T_b - T_\infty) \quad (3-73)$$

In reality, however, the temperature of the fin drops along the fin, and thus the heat transfer from the fin is less because of the decreasing temperature difference $T(x) - T_\infty$ toward the fin tip, as shown in Fig. 3–42. To account for the effect of this decrease in temperature on heat transfer, we define a **fin efficiency** as

$$\eta_{\text{fin}} = \frac{\dot{Q}_{\text{fin}}}{\dot{Q}_{\text{fin, max}}} = \frac{\text{Actual heat transfer rate from the fin}}{\text{Ideal heat transfer rate from the fin if the entire fin were at base temperature}} \quad (3-74)$$

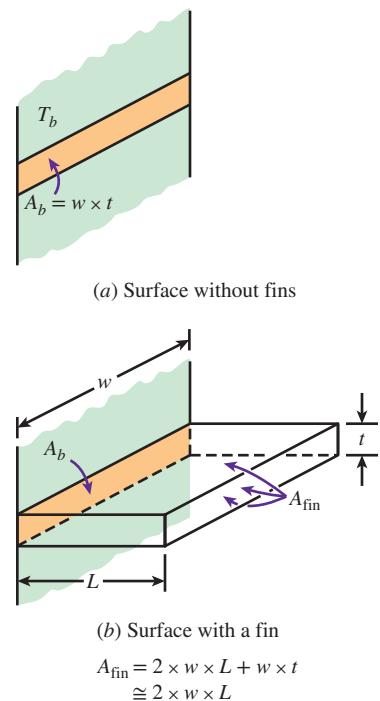


FIGURE 3–41

Fins enhance heat transfer from a surface by enhancing surface area.

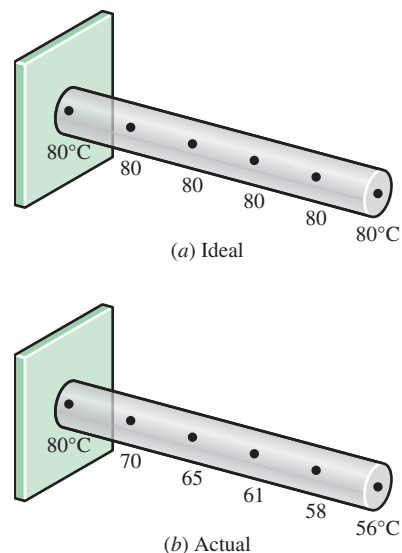


FIGURE 3–42

Ideal and actual temperature distribution along a fin.

TABLE 3-3

Efficiency and surface areas of common fin configurations

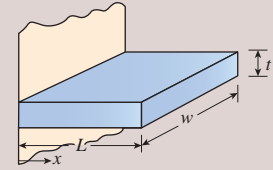
Straight rectangular fins

$$m = \sqrt{2h/kt}$$

$$L_c = L + t/2$$

$$A_{\text{fin}} = 2wL_c$$

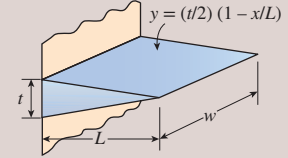
$$\eta_{\text{fin}} = \frac{\tanh mL_c}{mL_c}$$

**Straight triangular fins**

$$m = \sqrt{2h/kt}$$

$$A_{\text{fin}} = 2w\sqrt{L^2 + (t/2)^2}$$

$$\eta_{\text{fin}} = \frac{1}{mL} \frac{I_1(2mL)}{I_0(2mL)}$$

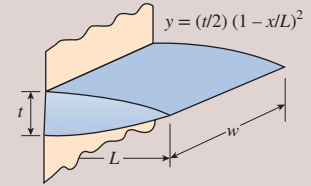
**Straight parabolic fins**

$$m = \sqrt{2h/kt}$$

$$A_{\text{fin}} = wL[C_1 + (L/t) \ln(t/L + C_1)]$$

$$C_1 = \sqrt{1 + (t/L)^2}$$

$$\eta_{\text{fin}} = \frac{2}{1 + \sqrt{(2mL)^2 + 1}}$$

**Circular fins of rectangular profile**

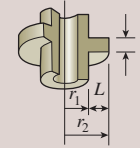
$$m = \sqrt{2h/kt}$$

$$r_{2c} = r_2 + t/2$$

$$A_{\text{fin}} = 2\pi(r_{2c}^2 - r_1^2)$$

$$\eta_{\text{fin}} = C_2 \frac{K_1(mr_1)I_1(mr_{2c}) - I_1(mr_1)K_1(mr_{2c})}{I_0(mr_1)K_1(mr_{2c}) + K_0(mr_1)I_1(mr_{2c})}$$

$$C_2 = \frac{2r_1/m}{r_{2c}^2 - r_1^2}$$

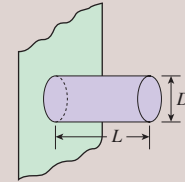
**Pin fins of rectangular profile**

$$m = \sqrt{4h/kD}$$

$$L_c = L + D/4$$

$$A_{\text{fin}} = \pi D L_c$$

$$\eta_{\text{fin}} = \frac{\tanh mL_c}{mL_c}$$

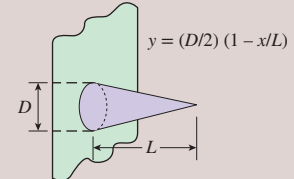
**Pin fins of triangular profile**

$$m = \sqrt{4h/kD}$$

$$A_{\text{fin}} = \frac{\pi D}{2} \sqrt{L^2 + (D/2)^2}$$

$$\eta_{\text{fin}} = \frac{2}{mL} \frac{I_2(2mL)}{I_1(2mL)}$$

$$I_2(x) = I_0(x) - (2/x) I_1(x) \text{ where } x = 2mL$$

**Pin fins of parabolic profile**

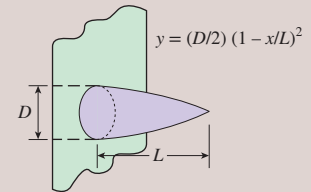
$$m = \sqrt{4h/kD}$$

$$A_{\text{fin}} = \frac{\pi L^3}{8D} \left[C_3 C_4 - \frac{L}{2D} \ln[(2DC_4/L) + C_3] \right]$$

$$C_3 = 1 + 2(D/L)^2$$

$$C_4 = \sqrt{1 + (D/L)^2}$$

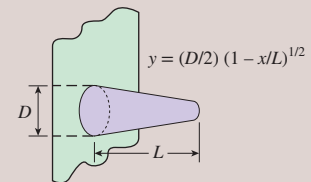
$$\eta_{\text{fin}} = \frac{2}{1 + \sqrt{(2mL/3)^2 + 1}}$$

**Pin fins of parabolic profile (blunt tip)**

$$m = \sqrt{4h/kD}$$

$$A_{\text{fin}} = \frac{\pi D^4}{96L^2} \{ [16(L/D)^2 + 1]^{3/2} - 1 \}$$

$$\eta_{\text{fin}} = \frac{3}{2mL} \frac{I_1(4mL/3)}{I_0(4mL/3)}$$



or

$$\dot{Q}_{\text{fin}} = \eta_{\text{fin}} \dot{Q}_{\text{fin, max}} = \eta_{\text{fin}} h A_{\text{fin}} (T_b - T_{\infty}) \quad (3-75)$$

where A_{fin} is the total surface area of the fin. This relation enables us to determine the heat transfer from a fin when its efficiency is known. For the cases of constant cross section of *very long fins* and *fins with adiabatic tips*, the fin efficiency can be expressed as

$$\eta_{\text{long fin}} = \frac{\dot{Q}_{\text{fin}}}{\dot{Q}_{\text{fin, max}}} = \frac{\sqrt{hpkA_c}(T_b - T_{\infty})}{hA_{\text{fin}}(T_b - T_{\infty})} = \frac{1}{L} \sqrt{\frac{kA_c}{hp}} = \frac{1}{mL} \quad (3-76)$$

and

$$\eta_{\text{adiabatic tip}} = \frac{\dot{Q}_{\text{fin}}}{\dot{Q}_{\text{fin, max}}} = \frac{\sqrt{hpkA_c}(T_b - T_{\infty}) \tanh mL}{hA_{\text{fin}}(T_b - T_{\infty})} = \frac{\tanh mL}{mL} \quad (3-77)$$

since $A_{\text{fin}} = pL$ for fins with constant cross section. Equation 3-77 can also be used for fins subjected to convection provided that the fin length L is replaced by the corrected length L_c .

Table 3-3 provides fin efficiency relations for fins with uniform and nonuniform cross section. For fins with nonuniform profile, Eq. 3-56 is no longer valid, and the general form of the differential equation governing heat transfer in fins of arbitrary shape, Eq. 3-55, must be used. For these cases the solution is no longer in the form of simple exponential or hyperbolic functions. The mathematical functions I and K that appear in some of these relations are the *modified Bessel functions*, and their values are given in Table 3-4. Efficiencies are plotted in Fig. 3-43 for fins on a *plain surface* and in Fig. 3-44 for *circular fins* of constant thickness. For most fins of constant thickness encountered in practice, the fin thickness t is too small relative to the fin length L , and thus the fin tip area is negligible.

Note that fins with triangular and parabolic profiles contain less material and are more efficient than the ones with rectangular profiles, and thus are more suitable for applications requiring minimum weight such as space applications.

An important consideration in the design of finned surfaces is the selection of the proper *fin length* L . Normally the *longer* the fin, the *larger* the heat transfer area and thus the *higher* the rate of heat transfer from the fin. But also the larger the fin, the bigger the mass, the higher the price, and the larger the fluid friction. Therefore, increasing the length of the fin beyond a certain value cannot be justified unless the added benefits outweigh the added cost. Also, the fin efficiency decreases with increasing fin length because of the decrease in fin temperature with length. Fin lengths that cause the fin efficiency to drop below 60 percent usually cannot be justified economically and should be avoided. The efficiency of most fins used in practice is above 90 percent.

Fin Effectiveness

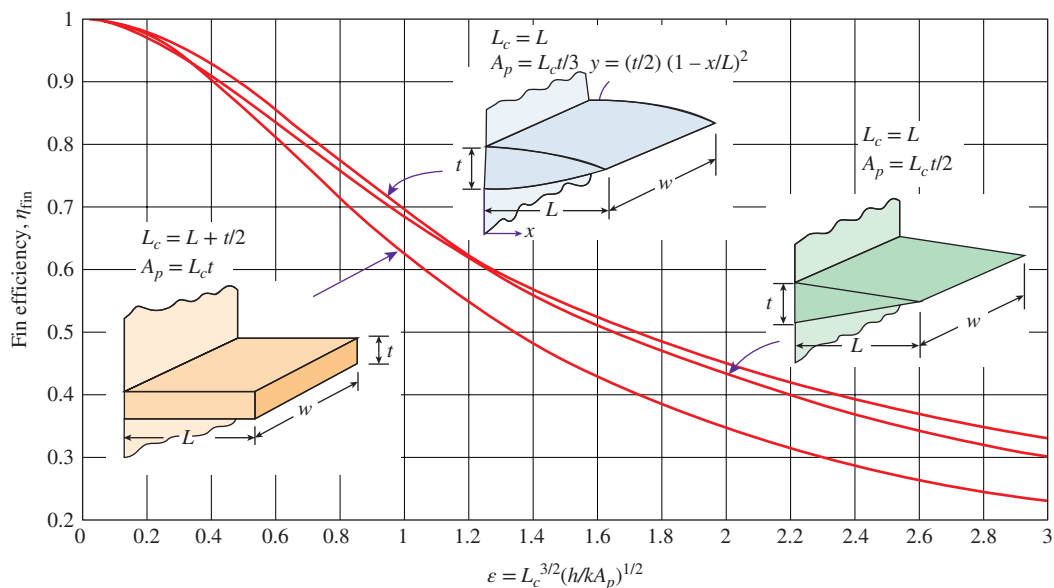
Fins are used to *enhance* heat transfer, and the use of fins on a surface cannot be recommended unless the enhancement in heat transfer justifies the added cost and complexity associated with the fins. In fact, there is no assurance that

TABLE 3-4

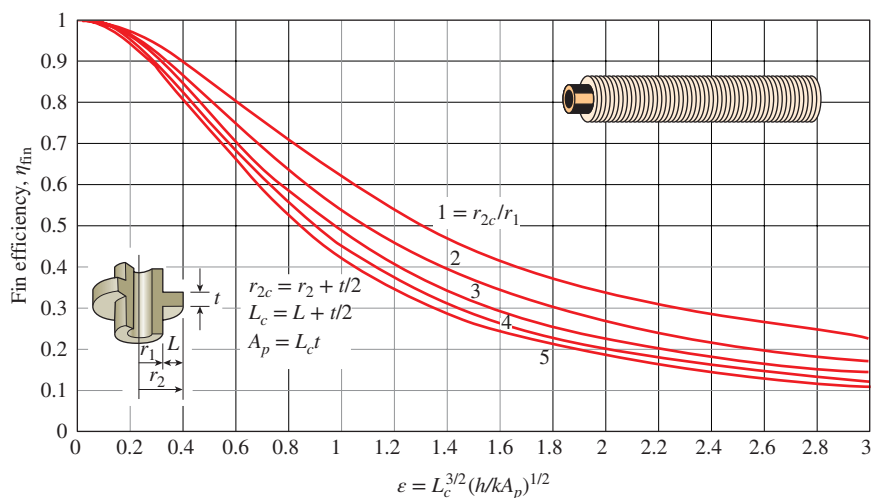
Modified Bessel functions of the first and second kinds*

x	$e^{-x}I_0(x)$	$e^{-x}I_1(x)$	$e^{-x}K_0(x)$	$e^{-x}K_1(x)$
0.0	1.0000	0.0000	∞	∞
0.2	0.8269	0.0823	2.1408	5.8334
0.4	0.6974	0.1368	1.6627	3.2587
0.6	0.5993	0.1722	1.4167	2.3739
0.8	0.5241	0.1945	1.2582	1.9179
1.0	0.4658	0.2079	1.1445	1.6362
1.2	0.4198	0.2153	1.0575	1.4429
1.4	0.3831	0.2185	0.9881	1.3011
1.6	0.3533	0.2190	0.9309	1.1919
1.8	0.3289	0.2177	0.8828	1.1048
2.0	0.3085	0.2153	0.8416	1.0335
2.2	0.2913	0.2121	0.8057	0.9738
2.4	0.2766	0.2085	0.7740	0.9229
2.6	0.2639	0.2047	0.7459	0.8790
2.8	0.2528	0.2007	0.7206	0.8405
3.0	0.2430	0.1968	0.6978	0.8066
3.2	0.2343	0.1930	0.6770	0.7763
3.4	0.2264	0.1892	0.6580	0.7491
3.6	0.2193	0.1856	0.6405	0.7245
3.8	0.2129	0.1821	0.6243	0.7021
4.0	0.2070	0.1788	0.6093	0.6816
4.2	0.2016	0.1755	0.5953	0.6627
4.4	0.1966	0.1725	0.5823	0.6454
4.6	0.1919	0.1695	0.5701	0.6292
4.8	0.1876	0.1667	0.5586	0.6143
5.0	0.1835	0.1640	0.5478	0.6003
5.2	0.1797	0.1614	0.5376	0.5872
5.4	0.1762	0.1589	0.5280	0.5749
5.6	0.1728	0.1565	0.5188	0.5634
5.8	0.1697	0.1542	0.5101	0.5525
6.0	0.1667	0.1521	0.5019	0.5422
6.5	0.1598	0.1469	0.4828	0.5187
7.0	0.1537	0.1423	0.4658	0.4981
7.5	0.1483	0.1380	0.4505	0.4797
8.0	0.1434	0.1341	0.4366	0.4631
8.5	0.1390	0.1305	0.4239	0.4482
9.0	0.1350	0.1272	0.4123	0.4346
9.5	0.1313	0.1241	0.4016	0.4222
10.0	0.1278	0.1213	0.3916	0.4108

*Evaluated from EES using the mathematical functions Bessel_I(x) and Bessel_K(x)

**FIGURE 3-43**

Efficiency of straight fins of rectangular, triangular, and parabolic profiles.

**FIGURE 3-44**

Efficiency of annular fins of constant thickness t .

adding fins on a surface will *enhance* heat transfer. The performance of the fins is judged on the basis of the enhancement in heat transfer relative to the no-fin case. The performance of fins is expressed in terms of the *fin effectiveness* ε_{fin} defined as (Fig. 3-45)

$$\varepsilon_{fin} = \frac{\dot{Q}_{fin}}{\dot{Q}_{no\ fin}} = \frac{\dot{Q}_{fin}}{hA_b(T_b - T_\infty)} = \frac{\text{Heat transfer rate from the fin of base area } A_b}{\text{Heat transfer rate from the surface of area } A_b} \quad (3-78)$$

Hence, A_b is the cross-sectional area of the fin at the base and $\dot{Q}_{\text{no fin}}$ represents the rate of heat transfer from this area if no fins are attached to the surface. An effectiveness of $\epsilon_{\text{fin}} = 1$ indicates that the addition of fins to the surface does not affect heat transfer at all. That is, heat conducted to the fin through the base area A_b is equal to the heat transferred from the same area A_b to the surrounding medium. An effectiveness of $\epsilon_{\text{fin}} < 1$ indicates that the fin actually acts as *insulation*, slowing down the heat transfer from the surface. This situation can occur when fins made of low thermal conductivity materials are used. An effectiveness of $\epsilon_{\text{fin}} > 1$ indicates that fins are *enhancing* heat transfer from the surface, as they should. However, the use of fins cannot be justified unless ϵ_{fin} is sufficiently larger than 1. Finned surfaces are designed on the basis of *maximizing* effectiveness for a specified cost or *minimizing* cost for a desired effectiveness.

Note that both the fin efficiency and fin effectiveness are related to the performance of the fin, but they are different quantities. However, they are related to each other by

$$\epsilon_{\text{fin}} = \frac{\dot{Q}_{\text{fin}}}{\dot{Q}_{\text{no fin}}} = \frac{\dot{Q}_{\text{fin}}}{hA_b(T_b - T_{\infty})} = \frac{\eta_{\text{fin}} hA_{\text{fin}}(T_b - T_{\infty})}{hA_b(T_b - T_{\infty})} = \frac{A_{\text{fin}}}{A_b} \eta_{\text{fin}} \quad (3-79)$$

Therefore, the fin effectiveness can be determined easily when the fin efficiency is known, or vice versa.

The rate of heat transfer from a sufficiently *long* fin of *uniform* cross section under steady conditions is given by Eq. 3-61. Substituting this relation into Eq. 3-78, the effectiveness of such a long fin is determined to be

$$\epsilon_{\text{long fin}} = \frac{\dot{Q}_{\text{fin}}}{\dot{Q}_{\text{no fin}}} = \frac{\sqrt{hpkA_c}(T_b - T_{\infty})}{hA_b(T_b - T_{\infty})} = \sqrt{\frac{kp}{hA_c}} \quad (3-80)$$

since $A_c = A_b$ in this case. We can draw several important conclusions from the fin effectiveness relation above for consideration in the design and selection of the fins:

- The *thermal conductivity* k of the fin material should be as high as possible. Thus it is no coincidence that fins are made from metals, with copper, aluminum, and iron being the most common ones. Perhaps the most widely used fins are made of aluminum because of its low cost and weight and its resistance to corrosion.
- The ratio of the *perimeter* to the *cross-sectional area* of the fin p/A_c should be as high as possible. This criterion is satisfied by *thin* plate fins and *slender* pin fins.
- The use of fins is *most effective* in applications involving a *low convection heat transfer coefficient*. Thus, the use of fins is more easily justified when the medium is a *gas* instead of a liquid and the heat transfer is by *natural convection* instead of by forced convection. Therefore, it is no coincidence that in liquid-to-gas heat exchangers such as the car radiator, fins are placed on the *gas* side.

When determining the rate of heat transfer from a finned surface, we must consider the *unfinned portion* of the surface as well as the *fins*. Therefore, the rate of heat transfer for a surface containing n fins can be expressed as

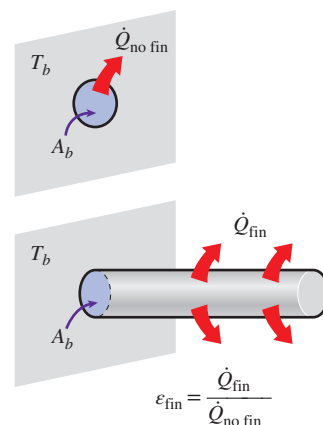
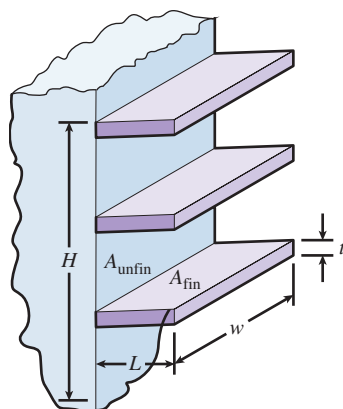


FIGURE 3-45

The effectiveness of a fin.



$$\begin{aligned} A_{\text{no fin}} &= w \times H \\ A_{\text{unfin}} &= w \times H - 3 \times (t \times w) \\ A_{\text{fin}} &= 2 \times L \times w + t \times w \\ &\cong 2 \times L \times w \text{ (one fin)} \end{aligned}$$

FIGURE 3-46

Various surface areas associated with a rectangular surface with three fins.

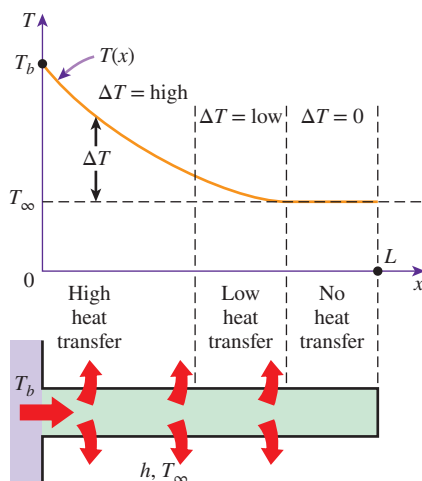


FIGURE 3-47

Because of the gradual temperature drop along the fin, the region near the fin tip makes little or no contribution to heat transfer.

$$\begin{aligned} \dot{Q}_{\text{total, fin}} &= \dot{Q}_{\text{unfin}} + \dot{Q}_{\text{fin}} \\ &= hA_{\text{unfin}}(T_b - T_{\infty}) + \eta_{\text{fin}}hA_{\text{fin}}(T_b - T_{\infty}) \\ &= h(A_{\text{unfin}} + \eta_{\text{fin}}A_{\text{fin}})(T_b - T_{\infty}) \end{aligned} \quad (3-81)$$

We can also define an **overall effectiveness** for a finned surface as the ratio of the total heat transfer from the finned surface to the heat transfer from the same surface if there were no fins,

$$\varepsilon_{\text{fin, overall}} = \frac{\dot{Q}_{\text{total, fin}}}{\dot{Q}_{\text{total, no fin}}} = \frac{h(A_{\text{unfin}} + \eta_{\text{fin}}A_{\text{fin}})(T_b - T_{\infty})}{hA_{\text{no fin}}(T_b - T_{\infty})} = \frac{A_{\text{unfin}} + \eta_{\text{fin}}A_{\text{fin}}}{A_{\text{no fin}}} \quad (3-82)$$

where $A_{\text{no fin}}$ is the area of the surface when there are no fins, A_{fin} is the total surface area of all the fins on the surface, and A_{unfin} is the area of the unfinned portion of the surface (Fig. 3-46). Note that the overall fin effectiveness depends on the fin density (number of fins per unit length) as well as the effectiveness of the individual fins. The overall effectiveness is a better measure of the performance of a finned surface than the effectiveness of the individual fins.

Proper Length of a Fin

An important step in the design of a fin is the determination of the appropriate length of the fin once the fin material and the fin cross section are specified. You may be tempted to think that the longer the fin, the larger the surface area and thus the higher the rate of heat transfer. Therefore, for maximum heat transfer, the fin should be infinitely long. However, the temperature drops along the fin exponentially and reaches the environment temperature at some length. The part of the fin beyond this length does not contribute to heat transfer since it is at the temperature of the environment, as shown in Fig. 3-47. Therefore, designing such an “extra long” fin is out of the question since it results in material waste, excessive weight, and increased size and thus increased cost with no benefit in return (in fact, such a long fin will hurt performance since it will suppress fluid motion and thus reduce the convection heat transfer coefficient). Fins that are so long that the temperature approaches the environment temperature cannot be recommended either since the little increase in heat transfer at the tip region cannot justify the disproportionate increase in the weight and cost.

To get a sense of the proper length of a fin, we compare heat transfer from a fin of finite length to heat transfer from an infinitely long fin under the same conditions. The ratio of these two heat transfers is

$$\text{Heat transfer ratio: } \frac{\dot{Q}_{\text{fin}}}{\dot{Q}_{\text{long fin}}} = \frac{\sqrt{hpkA_c}(T_b - T_{\infty}) \tanh mL}{\sqrt{hpkA_c}(T_b - T_{\infty})} = \tanh mL \quad (3-83)$$

Using a hand calculator, the values of $\tanh mL$ are evaluated for some values of mL , and the results are given in Table 3-5. We observe from the table that heat transfer from a fin increases with mL almost linearly at first, but the curve reaches a plateau later and reaches a value for the infinitely long fin at about $mL = 5$. Therefore, a fin whose length is $L = 5/m$ can be considered to be an infinitely long fin. We also observe that reducing the fin length by half in that case (from $mL = 5$ to $mL = 2.5$) causes a drop of just 1 percent in heat transfer. We certainly would not hesitate sacrificing 1 percent in heat transfer performance in return for 50 percent reduction in the size and possibly

the cost of the fin. In practice, a fin length that corresponds to about $mL = 1$ will transfer 76.2 percent of the heat that can be transferred by an infinitely long fin, and thus it should offer a good compromise between heat transfer performance and the fin size.

A common approximation used in the analysis of fins is to assume the fin temperature to vary in one direction only (along the fin length) and the temperature variation along other directions is negligible. Perhaps you are wondering if this one-dimensional approximation is a reasonable one. This is certainly the case for fins made of thin metal sheets such as the fins on a car radiator, but we wouldn't be so sure for fins made of thick materials. Studies have shown that the error involved in one-dimensional fin analysis is negligible (less than about 1 percent) when

$$\frac{h\delta}{k} < 0.2$$

where δ is the characteristic thickness of the fin, which is taken to be the plate thickness t for rectangular fins and the diameter D for cylindrical ones.

Specially designed finned surfaces called *heat sinks*, which are commonly used in the cooling of electronic equipment, involve one-of-a-kind complex geometries, as shown in Table 3–6. The heat transfer performance of heat sinks is usually expressed in terms of their *thermal resistances* R in $^{\circ}\text{C}/\text{W}$, which is defined as

$$\dot{Q}_{\text{fin}} = \frac{T_b - T_{\infty}}{R} = hA_{\text{fin}}\eta_{\text{fin}}(T_b - T_{\infty}) \quad (3-84)$$

A small value of thermal resistance indicates a small temperature drop across the heat sink, and thus a high fin efficiency.

TABLE 3–5

The variation of heat transfer from a fin relative to that from an infinitely long fin

mL	$\frac{\dot{Q}_{\text{fin}}}{\dot{Q}_{\text{long fin}}} = \tanh mL$
0.1	0.100
0.2	0.197
0.5	0.462
1.0	0.762
1.5	0.905
2.0	0.964
2.5	0.987
3.0	0.995
4.0	0.999
5.0	1.000

EXAMPLE 3–10 Maximum Power Dissipation of a Transistor

Power transistors that are commonly used in electronic devices consume large amounts of electric power. The failure rate of electronic components increases almost exponentially with operating temperature. As a rule of thumb, the failure rate of electronic components is halved for each 10°C reduction in the junction operating temperature. Therefore, the operating temperature of electronic components is kept below a safe level to minimize the risk of failure.

The sensitive electronic circuitry of a power transistor at the junction is protected by its case, which is a rigid metal enclosure. Heat transfer characteristics of a power transistor are usually specified by the manufacturer in terms of the case-to-ambient thermal resistance, which accounts for both the natural convection and radiation heat transfers.

The case-to-ambient thermal resistance of a power transistor that has a maximum power rating of 10 W is given to be $20^{\circ}\text{C}/\text{W}$. If the case temperature of the transistor is not to exceed 85°C , determine the power at which this transistor can be operated safely in an environment at 25°C .

SOLUTION The maximum power rating of a transistor whose case temperature is not to exceed 85°C is to be determined.

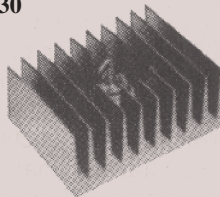
Assumptions 1 Steady operating conditions exist. 2 The transistor case is isothermal at 85°C .

Properties The case-to-ambient thermal resistance is given to be $20^{\circ}\text{C}/\text{W}$.

(Continued on page 193)

TABLE 3-6

Combined natural convection and radiation thermal resistance of various heat sinks used in the cooling of electronic devices between the heat sink and the surroundings. All fins are made of aluminum 6063T-5, are black anodized, and are 76 mm (3 in) long.

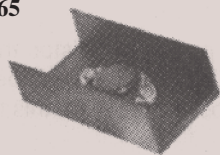
HS 5030

$$R = 0.9^{\circ}\text{C/W (vertical)}$$

$$R = 1.2^{\circ}\text{C/W (horizontal)}$$

Dimensions: 76 mm $\times$ 105 mm $\times$ 44 mm

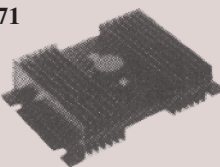
Surface area: 677 cm²

HS 6065

$$R = 5^{\circ}\text{C/W}$$

Dimensions: 76 mm $\times$ 38 mm $\times$ 24 mm

Surface area: 387 cm²

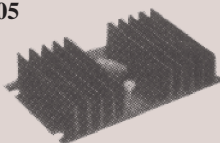
HS 6071

$$R = 1.4^{\circ}\text{C/W (vertical)}$$

$$R = 1.8^{\circ}\text{C/W (horizontal)}$$

Dimensions: 76 mm $\times$ 92 mm $\times$ 26 mm

Surface area: 968 cm²

HS 6105

$$R = 1.8^{\circ}\text{C/W (vertical)}$$

$$R = 2.1^{\circ}\text{C/W (horizontal)}$$

Dimensions: 76 mm $\times$ 127 mm $\times$ 91 mm

Surface area: 677 cm²

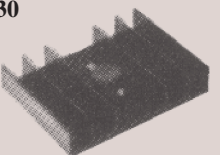
HS 6115

$$R = 1.1^{\circ}\text{C/W (vertical)}$$

$$R = 1.3^{\circ}\text{C/W (horizontal)}$$

Dimensions: 76 mm $\times$ 102 mm $\times$ 25 mm

Surface area: 929 cm²

HS 7030

$$R = 2.9^{\circ}\text{C/W (vertical)}$$

$$R = 3.1^{\circ}\text{C/W (horizontal)}$$

Dimensions: 76 mm $\times$ 97 mm $\times$ 19 mm

Surface area: 290 cm²

Analysis The power transistor and the thermal resistance network associated with it are shown in Fig. 3–48. We notice from the thermal resistance network that there is a single resistance of $20^\circ\text{C}/\text{W}$ between the case at $T_c = 85^\circ\text{C}$ and the ambient at $T_\infty = 22^\circ\text{C}$, and thus the rate of heat transfer is

$$\dot{Q} = \left(\frac{\Delta T}{R} \right)_{\text{case-ambient}} = \frac{T_c - T_\infty}{R_{\text{case-ambient}}} = \frac{(85 - 25)^\circ\text{C}}{20^\circ\text{C}/\text{W}} = 3 \text{ W}$$

Therefore, this power transistor should not be operated at power levels above 3 W if its case temperature is not to exceed 85°C .

Discussion This transistor can be used at higher power levels by attaching it to a heat sink (which lowers the thermal resistance by increasing the heat transfer surface area, as discussed in the next example) or by using a fan (which lowers the thermal resistance by increasing the convection heat transfer coefficient).

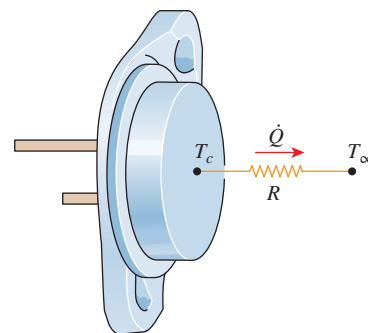


FIGURE 3–48
Schematic for Example 3–10.

EXAMPLE 3–11 Selecting a Heat Sink for a Transistor

A 25-W power transistor is to be cooled by attaching it to one of the commercially available heat sinks shown in Table 3–6. Select a heat sink that will allow the case temperature of the transistor not to exceed 55°C in the ambient air at 18°C .

SOLUTION A commercially available heat sink from Table 3–6 is to be selected to keep the case temperature of a transistor below 55°C .

Assumptions 1 Steady operating conditions exist. 2 The transistor case is isothermal at 55°C . 3 The contact resistance between the transistor and the heat sink is negligible.

Analysis The rate of heat transfer from a 25-W transistor at full power is $\dot{Q} = 25 \text{ W}$. The thermal resistance between the transistor attached to the heat sink and the ambient air for the specified temperature difference is determined to be

$$\dot{Q} = \frac{\Delta T}{R} \longrightarrow R = \frac{\Delta T}{\dot{Q}} = \frac{(55 - 18)^\circ\text{C}}{25 \text{ W}} = 1.5^\circ\text{C}/\text{W}$$

Therefore, the thermal resistance of the heat sink should be below $1.5^\circ\text{C}/\text{W}$. An examination of Table 3–6 reveals that the HS 5030 in both horizontal and vertical positions, HS 6071 in vertical position, and HS 6115 in both horizontal and vertical positions can be selected.

EXAMPLE 3–12 Heat Transfer from Fins of Variable Cross Section

Aluminum pin fins of parabolic profile with blunt tips are attached on a plane wall with surface temperature of 200°C (Fig. 3–49). Each fin has a length of 20 mm and a base diameter of 5 mm. The fins are exposed to an ambient air condition of 25°C and the convection heat transfer coefficient is $50 \text{ W}/\text{m}^2\cdot\text{K}$. If the thermal conductivity of the fins is $240 \text{ W}/\text{m}\cdot\text{K}$, determine the efficiency, heat transfer rate, and effectiveness of each fin.

SOLUTION The efficiency, heat transfer rate, and effectiveness of a pin fin of parabolic profile with a blunt tip is to be determined.

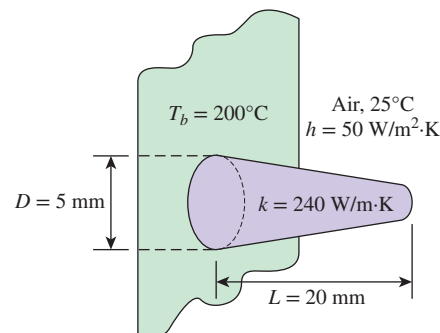


FIGURE 3–49
Schematic for Example 3–12.

Assumptions 1 Heat conduction is steady and one-dimensional. 2 Thermal properties are constant. 3 Heat transfer by radiation is negligible.

Properties The thermal conductivity of the fin is given as 240 W/m·K.

Analysis From Table 3–3, for pin fins of parabolic profile (blunt tip), we have

$$mL = \sqrt{\frac{4h}{kD}} L = \sqrt{\frac{4(50 \text{ W/m}^2\cdot\text{K})}{(240 \text{ W/m}\cdot\text{K})(0.005 \text{ m})}} (0.020 \text{ m}) = 0.2582$$

$$A_{\text{fin}} = \frac{\pi D^4}{96L^2} \left\{ \left[16 \left(\frac{L}{D} \right)^2 + 1 \right]^{3/2} - 1 \right\} = \frac{\pi(0.005 \text{ m})^4}{96(0.020 \text{ m})^2} \left\{ \left[16 \left(\frac{0.020 \text{ m}}{0.005 \text{ m}} \right)^2 + 1 \right]^{3/2} - 1 \right\}$$

$$= 2.106 \times 10^{-4} \text{ m}^2$$

$$\eta_{\text{fin}} = \frac{3}{2mL} \frac{I_1(4mL/3)}{I_0(4mL/3)} = \frac{3}{2(0.2582)} \frac{I_1[4(0.2582)/3]}{I_0[4(0.2582)/3]} = 5.8095 \frac{I_1[0.3443]}{I_0[0.3443]}$$

The values of the Bessel functions corresponding to $x = 0.3443$ are determined from Table 3–4 to be $I_0 = 1.0350$ and $I_1 = 0.1716$. Substituting, the fin efficiency is determined to be

$$\eta_{\text{fin}} = 5.8095 \frac{0.1716}{1.0350} = \mathbf{0.9632}$$

The heat transfer rate for a single fin is

$$\dot{Q}_{\text{fin}} = \eta_{\text{fin}} h A_{\text{fin}} (T_b - T_{\infty})$$

$$= (0.9632)(50 \text{ W/m}^2\cdot\text{K})(2.106 \times 10^{-4} \text{ m}^2)(200 - 25)^{\circ}\text{C} = \mathbf{1.77 \text{ W}}$$

The fin effectiveness is

$$\epsilon_{\text{fin}} = \frac{\dot{Q}_{\text{fin}}}{h A_b (T_b - T_{\infty})} = \frac{\dot{Q}_{\text{fin}}}{h (\pi D^2/4) (T_b - T_{\infty})}$$

$$= \frac{1.77 \text{ W}}{(50 \text{ W/m}^2\cdot\text{K}) [\pi (0.005 \text{ m})^2/4] (200 - 25)^{\circ}\text{C}}$$

$$= \mathbf{10.3}$$

That is, over a tenfold increase in heat transfer is achieved by using a pin fin in this case.

Discussion The fin efficiency can be determined more accurately by avoiding the interpolation error by using an equation solver with built-in mathematical functions such as EES. Copying the line

$$\text{eta_fin} = 3/(2*0.2582)*\text{Bessel_I}[1(4*0.2582/3)]/\text{Bessel_I}[0(4*0.2582/3)]$$

on a blank EES screen and hitting the ‘solve’ button gives the fin efficiency to be $\eta_{\text{fin}} = 0.9855$, which is about 2 percent higher than the result obtained above using the tables.

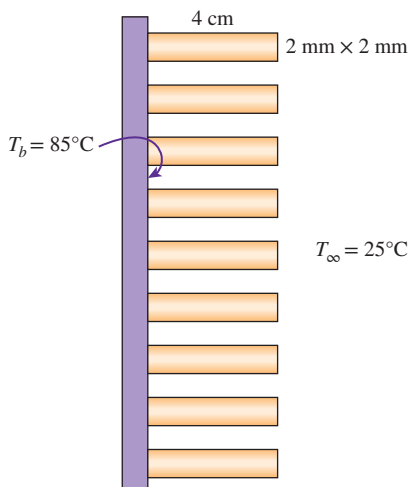


FIGURE 3–50

Schematic for Example 3–13.



EXAMPLE 3–13

Preventing Circuit Board Surface from Overheating

A 15-cm × 20-cm integrated circuit board is to be cooled by attaching 4-cm-long aluminum ($k = 237 \text{ W/m}\cdot\text{K}$) fins on one side of it (Fig. 3–50). Each fin has a 2-mm × 2-mm square cross section. The surrounding ambient temperature is 25°C and the convection heat transfer coefficient on each fin surface is 20 W/m²·K. To prevent the circuit board

from overheating, the upper surface of the circuit board needs to be at 85°C or cooler. Design a finned surface having the appropriate number of fins, with an overall effectiveness of 3 that can keep the circuit board surface from overheating.

SOLUTION In this example, the concepts of Prevention through Design (PtD) are applied in conjunction with the fin analysis. An integrated circuit board is to be cooled by attaching aluminum fins of square cross section on one side. The number of fins needed to keep the circuit board surface cooler than 85°C, while having an overall effectiveness of 3, is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat conduction is one-dimensional. 3 Heat transfer from the fin tips is negligible. 4 Fins are very long. 5 The heat transfer coefficient is constant and uniform over the entire fin surface. 6 The thermal properties of the fins are constant.

Properties The thermal conductivity of the aluminum fins is given to be $k = 237 \text{ W/m}\cdot\text{K}$.

Analysis Noting that the cross-sectional areas of the fins are constant, the efficiency of the square cross-section fins that are assumed to be very long with adiabatic tips can be determined to be

$$m = \sqrt{\frac{hp}{kA_c}} = \sqrt{\frac{4ha}{ka^2}} = \sqrt{\frac{4(20 \text{ W/m}^2\cdot\text{K})(0.02 \text{ m})}{(237 \text{ W/m}\cdot\text{K})(0.02 \text{ m})^2}} = 12.99 \text{ m}^{-1}$$

where a is the length of each side of the square fin.

$$\eta_{\text{fin}} = \frac{\tanh mL}{mL} = \frac{\tanh(12.99 \text{ m}^{-1} \times 0.04 \text{ m})}{12.99 \text{ m}^{-1} \times 0.04 \text{ m}} = 0.919$$

The finned and unfinned surface areas, and heat transfer rates from these areas are

$$\begin{aligned} A_{\text{fin}} &= n_{\text{fin}} \times 4 \times (0.002 \text{ m})(0.04 \text{ m}) = 0.00032 n_{\text{fin}} \text{ m}^2 \\ A_{\text{unfinned}} &= (0.15 \text{ m})(0.20 \text{ m}) - n_{\text{fin}}(0.002 \text{ m})(0.002 \text{ m}) \\ &= 0.03 - 0.000004 n_{\text{fin}} \text{ m}^2 \\ \dot{Q}_{\text{finned}} &= \eta_{\text{fin}} \dot{Q}_{\text{fin, max}} = \eta_{\text{fin}} h A_{\text{fin}} (T_b - T_{\infty}) \\ &= 0.919(20 \text{ W/m}^2\cdot\text{K})(0.00032 n_{\text{fin}} \text{ m}^2)(85 - 25)^{\circ}\text{C} \\ &= 0.3529 n_{\text{fin}} \text{ W} \\ \dot{Q}_{\text{unfinned}} &= h A_{\text{unfinned}} (T_b - T_{\infty}) = (20 \text{ W/m}^2\cdot\text{K})(0.03 - 0.000004 n_{\text{fin}} \text{ m}^2)(85 - 25)^{\circ}\text{C} \\ &= 36 - 0.0048 n_{\text{fin}} \text{ W} \end{aligned}$$

Then the total heat transfer from the finned surface (circuit board) becomes

$$\dot{Q}_{\text{total, fin}} = \dot{Q}_{\text{finned}} + \dot{Q}_{\text{unfinned}} = 0.3529 n_{\text{fin}} + 36 - 0.0048 n_{\text{fin}} \text{ W}$$

The rate of heat transfer if there were no fins attached to the plate would be

$$\begin{aligned} A_{\text{no fin}} &= (0.15 \text{ m})(0.20 \text{ m}) = 0.03 \text{ m}^2 \\ \dot{Q}_{\text{no fin}} &= h A_{\text{no fin}} (T_b - T_{\infty}) = (20 \text{ W/m}^2\cdot\text{K})(0.03 \text{ m}^2)(85 - 25)^{\circ}\text{C} = 36 \text{ W} \end{aligned}$$

The number of fins can be determined from the overall fin effectiveness equation

$$\begin{aligned} \epsilon_{\text{fin}} &= \frac{\dot{Q}_{\text{fin}}}{\dot{Q}_{\text{no fin}}} \\ 3 &= \frac{0.3529 n_{\text{fin}} + 36 - 0.0048 n_{\text{fin}}}{36} \\ n_{\text{fin}} &= \mathbf{207} \end{aligned}$$

Discussion To keep the circuit board surface from heating above 85°C, the finned surface having an overall effectiveness of 3 needs to have at least 207 fins. Number of fins on the circuit board may be reduced by using different fin material and geometry.

3-7 ■ BIOHEAT TRANSFER EQUATION

The study of heat transfer in biological systems is referred to as *bioheat transfer*. It is the study of heat transfer within the human body or external to the body. Bioheat transfer can be considered a subfield of biomedical engineering with its foundation in heat transfer engineering. Heat transfer within the human body, in particular in adverse environments, is an active area of research for the development of new medical treatments or devices to minimize the effects of the adverse conditions.

The transport of thermal energy in living tissues is a very complex process. It involves multiple mechanisms such as conduction, convection, radiation, evaporation, phase change, metabolic heat generation (heat generated by the body through the digestion of food, work, and exercise), and perfusion (exchange of thermal energy between flowing blood and the surrounding tissue). The addition of varied metabolic heat generation rates throughout the body is what typically complicates the study of bioheat transfer. Recall that the human body is a remarkably good insulator (see Topic of Special Interest: “Thermal Comfort”) and that the heat transfer properties can be altered based on the clothing worn or the conditions of the environment (e.g., wind speed, humidity, etc.). Also, some of the considerations based on blood perfusion will be discussed in Topic of Special Interest: “The Human Cardiovascular System as a Counter-Current Heat Exchanger” (Chap. 11).

A simple yet fairly accurate bioheat transfer equation or model was proposed by Harry Pennes in 1948 and published in Volume 1 of the *Journal of Applied Physiology*. For the development of his model, Pennes measured temperature distributions as a function of radial position in the forearms of nine human subjects. His data showed a temperature differential of 3 to 4°C between the skin and the interior of the arm. Pennes attributed this temperature difference to the effects of metabolic heat generation and perfusion in the arm. Based on his experimental results, Pennes proposed a model, now known as Pennes’ bioheat transfer equation, to describe the effects of metabolism and blood perfusion on the energy balance within living tissue. Pennes’ model was considered a major introductory effort in quantifying the heat transfer contribution of perfusion in tissue. His model is a modified version of the heat conduction equation introduced in Chap. 2 which accounts for the effects of metabolic heat generation and perfusion.

During the past decade, Pennes’ bioheat transfer model has been widely used but also criticized for various reasons. To gain a better understanding of one of the most influential articles ever published in the *Journal of Applied Physiology*, Pennes’ 1948 paper was republished for the 50th anniversary of the journal and was revisited in great depth by Wissler. Wissler reevaluated Pennes’ work by revisiting his original experimental data and analysis. Wissler in his concluding remarks states that “much of the criticism directed toward Pennes’ model is not justified and those who base their theoretical

model calculations on the Pennes' model can be somewhat more confident that their starting equations are valid."

In this section a simplified version of the bioheat transfer equation for the case of steady-state, constant properties, one-dimensional heat transfer in rectangular coordinates is introduced. Following Pennes' model, modifying Eq. 2–15 to account for the effects of metabolic heat generation and perfusion heat sources results in

$$\frac{d^2T}{dx^2} + \frac{\dot{e}_m + \dot{e}_p}{k} = 0 \quad (3-85)$$

where $\dot{e}_m$ and $\dot{e}_p$ are the *metabolic* and *perfusion* heat source terms (W/m³).

A simple expression for the perfusion term was proposed by Pennes. Pennes suggested that the rate of heat transfer from blood to tissue per unit volume of tissue ($\dot{e}_p$) is proportional to the perfusion rate $\dot{p}$ (volumetric blood flow rate per unit volume of tissue or 1/s) and the difference between the blood temperature entering small capillaries (on the order of 8 μ m in diameter) at an arterial inlet temperature of T_a and a perfusate of blood (consisting of mostly water and ions) exiting through the capillary wall at the local tissue temperature of T . It was difficult for Pennes to measure the temperature of the blood perfusate exiting the capillary, and he thus assumed that the temperature, T , would be the same as the temperature of blood exiting the capillary and entering the venous circulation. It is important to note that under normal physiological conditions blood can only leave the capillary through the venous end, but some perfusate of blood leaves through the wall. The perfusate that leaves the capillary is to help provide nutrients to the surrounding tissue. The expression proposed by Pennes for the exchange of thermal energy between flowing blood and the surrounding tissue (perfusion) is

$$\dot{e}_p = \dot{p} \rho_b c_b (T_a - T) \quad (\text{W/m}^3) \quad (3-86)$$

where ρ_b (kg/m³) is the density of blood and c_b (J/kg·K) is the specific heat of blood.

Substituting the Pennes' perfusion heat source term expression, Eq. 3–86, into Eq. 3–85 results in

$$\frac{d^2T}{dx^2} + \frac{\dot{e}_m + \dot{p} \rho_b c_b (T_a - T)}{k} = 0 \quad (3-87)$$

Following the procedure that led to Eq. 3–56 for extended surfaces and assuming constant $\dot{e}_m$, $\dot{p}$, ρ_b , c_b , and T_a , the differential Eq. 3–87 reduces to

$$\frac{d^2\theta}{dx^2} - B^2\theta = 0 \quad (3-88)$$

where $B^2 = \dot{p} \rho_b c_b / k$ has units of (1/m) and $\theta = T - T_a - \dot{e}_m / \dot{p} \rho_b c_b$ is the *temperature excess*. Equation 3–88 is in form identical to Eq. 3–56, and the solutions presented in the text for the different boundary conditions could be used. It should be noted that the density of blood, specific heat of blood, and arterial temperature are reasonably constant with respect to time and space. However, the metabolic heat source term is rarely constant, and the perfusion rate will have a small temporal variation in the capillaries. If the time duration of your problem is small, the variation in these terms would be negligible.

One of the applications of Eq. 3–88 is in a process called *thermoregulation*. Thermoregulation is the ability of an organism to regulate its body temperature within certain boundaries, even when the surrounding temperature is very different than the body temperature. Recall that when we discussed thermal comfort, we noted that most individuals live in areas that have an average temperature lower than both the core body temperature and the temperature that an average human finds comfortable. Thermoregulation in the human body is achieved by keeping a tight balance between heat gain and heat loss. Humans' temperature regulation system is similar to the operation of a home furnace. The human body regulates heat generation and preservation to maintain the internal body temperature or core temperature. Normal core temperature at rest varies between 36.5 and 37.5°C (97.7 and 99.5°F). However, the temperature at the extremities is not regulated as tightly as the core temperature, and it can vary significantly from the core temperature (imagine having a snowball fight without wearing gloves); under normal external operating temperatures it is in the range of 33–34°C. It is also important to note that the temperature of the extremities can fluctuate rather quickly depending on the activity level and the external environment.

A more realistic approach to the study of heat transfer within the human body is to solve the bioheat transfer equation in cylindrical coordinates as shown in the following equations. The one-dimensional steady-state bioheat transfer differential equation in cylindrical coordinates with constant properties is

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dT}{dr} \right) + \frac{\dot{e}_m + \dot{e}_p}{k} = 0 \quad (3-89)$$

Following the same procedure that led to the development of Eq. 3–88, the bioheat transfer equation in cylindrical coordinates in terms of excess temperature θ is

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{d\theta}{dr} \right) - B^2 \theta = 0 \quad (3-90)$$

Equation 3–90 is a *modified Bessel equation* of order zero, and its general solution is of the form

$$\theta(r) = C_1 I_0(Br) + C_2 K_0(Br) \quad (3-91)$$

where I_0 and K_0 are modified, zero-order Bessel functions of the first and second kinds, respectively. The values of I_0 and K_0 are given in Table 3–4.

One of the applications of Eq. 3–90 would be, for example, in the analysis of the steady-state heat transfer from a human forearm subjected to certain environmental conditions. One of the end-of-chapter problems will be on this subject. The following example demonstrates the use of Eq. 3–88 in calculating heat transfer between a human body and its surroundings.

EXAMPLE 3–14 Application of Bioheat Transfer Equation

We are interested in heat transfer analysis from a human body subjected to certain environmental conditions. For this purpose, consider a region of muscle with a skin/fat layer over it under steady-state conditions as shown in Fig. 3–51. For simplicity,

approximate this region as a one-dimensional plane wall with surface area A . The muscle thickness is L_m covered by a layer of skin/fat with a thickness L_{sf} . The metabolic heat generation rate ($\dot{e}_m$) and perfusion rate ($\dot{p}$) are both constant throughout the muscle. The blood density and specific heat are ρ_b and c_b , respectively. The core body temperature (T_c) and the arterial blood temperature (T_a) are both assumed to be the same and constant. The muscle and the skin/fat layer thermal conductivities are k_m and k_{sf} , respectively. The skin has an emissivity of ϵ and the body is subjected to an air environment with a temperature of T_∞ , a convection heat transfer coefficient of h_{conv} , and a radiation heat transfer coefficient of h_{rad} . Assuming blood properties and thermal conductivities are all constant, (a) develop an expression for the interface temperature (T_i) between the muscle and the outer skin/fat layer, and (b) determine the temperature T_i and the rate of heat loss from the body ($\dot{Q}_b$), and the skin temperature (T_s) for the following conditions:

$A = 1.8 \text{ m}^2$, $L_m = 0.03 \text{ m}$, $L_{sf} = 0.003 \text{ m}$, $\dot{e}_m = 700 \text{ W/m}^3$, $\dot{p} = 0.0005 \text{ l/s}$,
 $T_c = T_a = 37^\circ\text{C}$, $T_\infty = T_{surr} = 24^\circ\text{C}$, $\epsilon = 0.95$, $\rho_b = 1000 \text{ kg/m}^3$, $c_b = 3600 \text{ J/kg}\cdot\text{K}$,
 $k_m = 0.5 \text{ W/m}\cdot\text{K}$, $k_{sf} = 0.3 \text{ W/m}\cdot\text{K}$, $h_{conv} = 2 \text{ W/m}^2\cdot\text{K}$, $h_{rad} = 5.9 \text{ W/m}^2\cdot\text{K}$

SOLUTION A region of muscle with a skin/fat layer over it for a human body is subjected to certain environmental conditions. An expression for the interface temperature, the value of the interface temperature, the rate of heat loss from the body, and the skin temperature for a set of conditions are to be determined.

Assumptions 1 Muscle and skin/fat layer considered as a 1-D plain wall. 2 Steady-state conditions. 3 Blood properties, thermal conductivities, arterial temperature, core body temperature, metabolic heat generation rate, and perfusion rate are all constant. 4 Radiation exchange between the skin surface and the surroundings is between a small surface and a large enclosure at the air temperature. 5 Solar radiation is negligible.

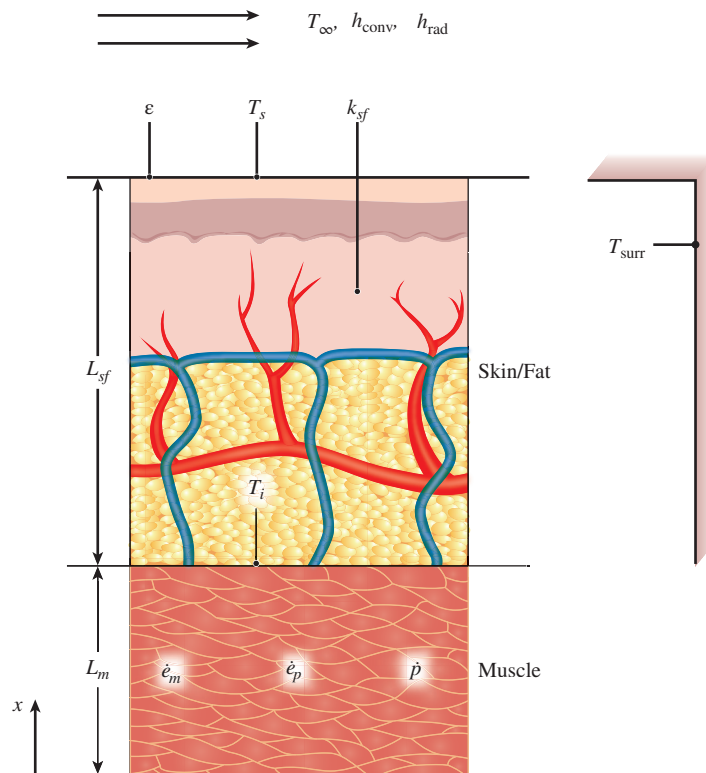


FIGURE 3–51
Schematic for Example 3–14.

Properties Muscle thermal conductivity $k_m = 0.5 \text{ W/m}\cdot\text{K}$, skin/fat layer thermal conductivity $k_{sf} = 0.3 \text{ W/m}\cdot\text{K}$, blood density $\rho_b = 1000 \text{ kg/m}^3$ and blood specific heat $c_b = 3600 \text{ J/kg}\cdot\text{K}$.

Analysis (a) Solve the bioheat transfer differential equation, Eq. 3–88, along with the appropriate boundary conditions to develop an expression for the interface temperature (T_i) between the muscle and the outer skin/fat layer. The bioheat differential equation is

$$\frac{d^2\theta}{dx^2} - B^2\theta = 0$$

where $B^2 = \dot{\rho}_b c_b / k$ has units of $(1/\text{m})$ and $\theta = T - T_a - \dot{e}_m / \dot{\rho}_b c_b$. The boundary conditions for the problem in terms of temperature excess θ are:

$$\theta(0) = T_c - T_a - \dot{e}_m / \dot{\rho}_b c_b = \theta_c \quad \text{and} \quad \theta(L_m) = T_i - T_a - \dot{e}_m / \dot{\rho}_b c_b = \theta_i$$

The solution to Eq. 3–88 with the two specified temperature boundary conditions θ_c and θ_i , is given by Eq. 3–67 developed for fins (case 3 – specified temperature). For our case Eq. 3–67 becomes

$$\frac{\theta}{\theta_c} = \frac{(\theta_i/\theta_c) \sinh Bx + \sinh B(L_m - x)}{\sinh BL_m} \quad (3-92)$$

Note that T_i that appears in θ_i in Eq. 3–92 is unknown. In order to find T_i , use Eq. 3–92 to calculate the rate at which heat leaves the muscle and enters the skin/fat layer at $x = L_m$ and equate it with the rate at which heat is transferred through the skin/fat layer and into the environment.

Using the Fourier's law of heat conduction, the rate of heat transfer that leaves the muscle at $x = L_m$ and enters the skin/fat layer is

$$\dot{Q}_{\text{specified temp.}} = -k_m A \left. \frac{dT}{dx} \right|_{x=L_m} = -k_m A \left. \frac{d\theta}{dx} \right|_{x=L_m} = -k_m A B \theta_c \frac{(\theta_i/\theta_c) \cosh BL_m - 1}{\sinh BL_m}$$

The rate at which heat is transferred through the skin/fat layer and into the environment is obtained by using the thermal resistance network concept (see Sec. 3–1). In this case the thermal resistance is a combined series-parallel arrangement. Heat is transferred through the skin/fat layer by conduction in series and is in parallel with heat transfer by convection and radiation. The total rate of heat transfer through the skin/fat layer and into the environment (the rate of heat loss from the body) is

$$\dot{Q}_b = \frac{T_i - T_\infty}{R_{\text{total}}}$$

where the total resistance is $R_{\text{total}} = R_{sf} + R_{\text{conv} - \text{rad}} = R_{sf} + \frac{R_{\text{conv}} R_{\text{rad}}}{R_{\text{conv}} + R_{\text{rad}}}$

and the individual resistances are $R_{sf} = \frac{L_{sf}}{k_{sf} A}$, $R_{\text{conv}} = \frac{1}{h_{\text{conv}} A}$ and $R_{\text{rad}} = \frac{1}{h_{\text{rad}} A}$

Equating the rate of heat transfer that leaves the muscle at $x = L_m$ and enters the skin/fat layer with the rate at which heat is transferred through the skin/fat layer and into the environment yields

$$-k_m A B \theta_c \frac{(\theta_i/\theta_c) \cosh BL_m - 1}{\sinh BL_m} = \frac{T_i - T_\infty}{R_{\text{total}}}$$

The above equation can be solved for T_i ; the final expression is

$$T_i = \frac{T_\infty \sinh BL_m + k_m A B R_{\text{total}} \left[\theta_c + \left(T_a + \frac{\dot{e}_m}{\dot{\rho}_b c_b} \right) \cosh BL_m \right]}{\sinh BL_m + k_m A B R_{\text{total}} \cosh BL_m}$$

(b) Using the data given in the problem statement and the expression for the interface temperature (T_i) between the muscle and the outer skin/fat layer, the interface temperature between the muscle and the outer skin/fat layer is

$$T_i = 34.8^\circ\text{C}$$

Using the calculated value of T_i and the equation for the total rate of heat transfer through the skin/fat layer and into the environment, the heat loss from the body is

$$\dot{Q}_b = 142 \text{ W}$$

The skin temperature (T_s) can be calculated by applying Fourier's law of heat conduction to the skin/fat layer

$$\dot{Q}_b = k_{sf} A \frac{T_i - T_s}{L_{sf}}$$

Solving the preceding equation for T_s ,

$$T_s = T_i - \frac{\dot{Q}_b L_{sf}}{k_{sf} A} \approx 34^\circ\text{C}$$

Discussion The skin temperature of 34°C is comfortable. However, if the environmental conditions change, our bodies will adjust. For example, if the temperatures of the air and the surroundings were lowered considerably, we would shiver. Shivering can increase the metabolic heat generation rate by up to six times the resting metabolic rate. If the air and the surroundings temperatures were increased considerably, we would sweat. Sweating will cause an increase in the perfusion rate near the skin surface, which causes an increase in the heat loss to the surroundings via evaporation.

3-8 ■ HEAT TRANSFER IN COMMON CONFIGURATIONS

So far, we have considered heat transfer in *simple* geometries such as large plane walls, long cylinders, and spheres. This is because heat transfer in such geometries can be approximated as *one-dimensional*, and simple analytical solutions can be obtained easily. But many problems encountered in practice are two- or three-dimensional and involve rather complicated geometries for which no simple solutions are available.

An important class of heat transfer problems for which simple solutions are obtained encompasses those involving two surfaces maintained at *constant* temperatures T_1 and T_2 . The steady rate of heat transfer between these two surfaces is expressed as

$$\dot{Q} = Sk(T_1 - T_2) \quad (3-93)$$

where S is the **conduction shape factor**, which has the dimension of *length*, and k is the thermal conductivity of the medium between the surfaces. The conduction shape factor depends on the *geometry* of the system only.

Conduction shape factors have been determined for a number of configurations encountered in practice and are given in Table 3-7 for some common cases. More comprehensive tables are available in the literature. Once the value of the shape factor is known for a specific geometry, the total steady heat transfer rate can be determined from Eq. 3-93 using the specified two

TABLE 3-7

Conduction shape factors S for several configurations for use in $\dot{Q} = kS(T_1 - T_2)$ to determine the steady rate of heat transfer through a medium of thermal conductivity k between the surfaces at temperatures T_1 and T_2

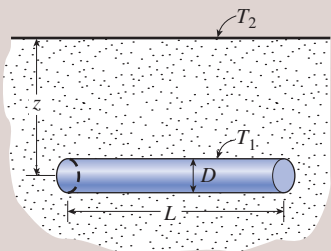
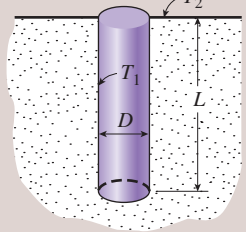
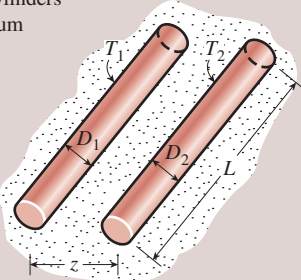
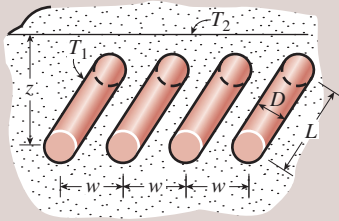
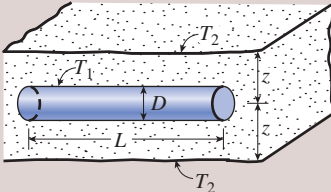
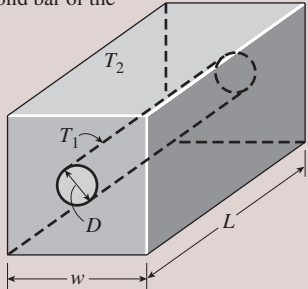
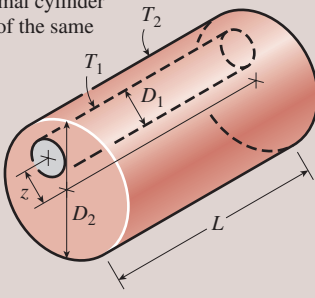
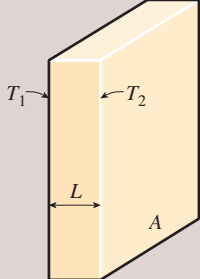
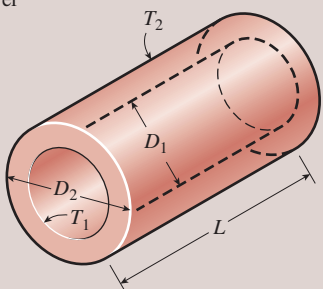
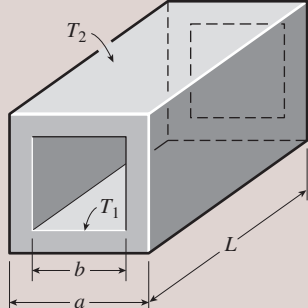
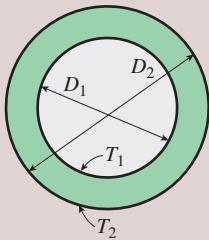
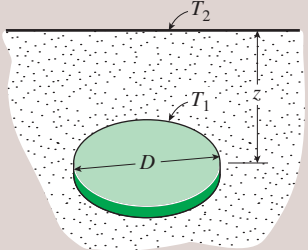
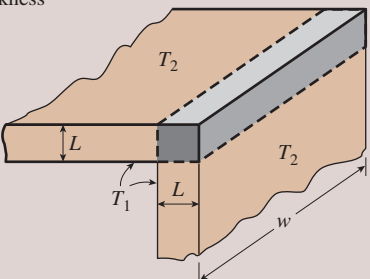
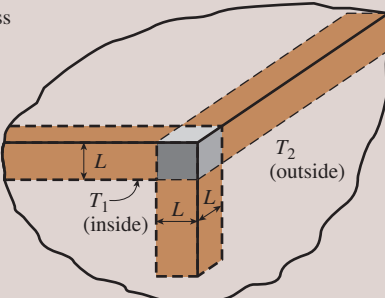
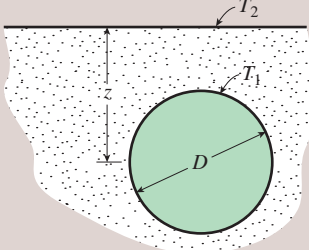
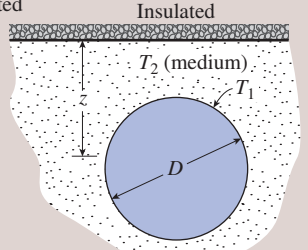
(1) Isothermal cylinder of length L buried in a semi-infinite medium ($L \gg D$ and $z > 1.5D$) $S = \frac{2\pi L}{\ln(4z/D)}$ 	(2) Vertical isothermal cylinder of length L buried in a semi-infinite medium ($L \gg D$) $S = \frac{2\pi L}{\ln(4L/D)}$ 
(3) Two parallel isothermal cylinders placed in an infinite medium ($L \gg D_1, D_2, z$) $S = \frac{2\pi L}{\cosh^{-1}\left(\frac{4z^2 - D_1^2 - D_2^2}{2D_1D_2}\right)}$ 	(4) A row of equally spaced parallel isothermal cylinders buried in a semi-infinite medium ($L \gg D, z$, and $w > 1.5D$) $S = \frac{2\pi L}{\ln\left(\frac{2w}{\pi D} \sinh \frac{2\pi z}{w}\right)}$ (per cylinder) 
(5) Circular isothermal cylinder of length L in the midplane of an infinite wall ($z > 0.5D$) $S = \frac{2\pi L}{\ln(8z/\pi D)}$ 	(6) Circular isothermal cylinder of length L at the center of a square solid bar of the same length $S = \frac{2\pi L}{\ln(1.08w/D)}$ 
(7) Eccentric circular isothermal cylinder of length L in a cylinder of the same length ($L > D_2$) $S = \frac{2\pi L}{\cosh^{-1}\left(\frac{D_1^2 + D_2^2 - 4z^2}{2D_1D_2}\right)}$ 	(8) Large plane wall $S = \frac{A}{L}$ 

TABLE 3-7 (CONTINUED)

(9) A long cylindrical layer $S = \frac{2\pi L}{\ln(D_2/D_1)}$ 	(10) A square flow passage (a) For $a/b > 1.4$, $S = \frac{2\pi L}{0.93 \ln(0.948 a/b)}$ (b) For $a/b < 1.4$, $S = \frac{2\pi L}{0.785 \ln(a/b)}$ 
(11) A spherical layer $S = \frac{2\pi D_1 D_2}{D_2 - D_1}$ 	(12) Disk buried parallel to the surface in a semi-infinite medium ($z \gg D$) $S = 4D$ ($S = 2D$ when $z = 0$) 
(13) The edge of two adjoining walls of equal thickness $S = 0.54 w$ 	(14) Corner of three walls of equal thickness $S = 0.15 L$ 
(15) Isothermal sphere buried in a semi-infinite medium $S = \frac{2\pi D}{1 - 0.25D/z}$ 	(16) Isothermal sphere buried in a semi-infinite medium at T_2 whose surface is insulated $S = \frac{2\pi D}{1 + 0.25D/z}$ 

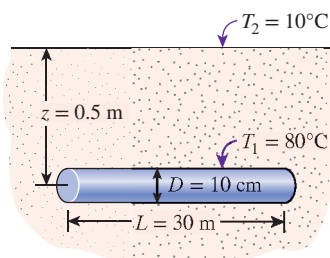


FIGURE 3-52

Schematic for Example 3-15.

constant temperatures of the two surfaces and the thermal conductivity of the medium between them. Note that conduction shape factors are applicable only when heat transfer between the two surfaces is by *conduction*. Therefore, they cannot be used when the medium between the surfaces is a liquid or gas, which involves natural or forced convection currents.

A comparison of Eqs. 3-4 and 3-93 reveals that the conduction shape factor S is related to the thermal resistance R by $R = 1/kS$ or $S = 1/kR$. Thus, these two quantities are the inverse of each other when the thermal conductivity of the medium is unity. The use of the conduction shape factors is illustrated with Examples 3-15 and 3-16.

It is well known that insulation reduces heat transfer and saves energy and money. Decisions on the right amount of insulation are based on a heat transfer analysis, followed by an economic analysis to determine the “monetary value” of energy loss. This is illustrated with Example 3-17.

EXAMPLE 3-15 Heat Loss from Buried Steam Pipes

A 30-m-long, 10-cm-diameter hot-water pipe of a district heating system is buried in the soil 50 cm below the ground surface, as shown in Fig. 3-52. The outer surface temperature of the pipe is 80°C. Taking the surface temperature of the earth to be 10°C and the thermal conductivity of the soil at that location to be 0.9 W/m·K, determine the rate of heat loss from the pipe.

SOLUTION The hot-water pipe of a district heating system is buried in the soil. The rate of heat loss from the pipe is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat transfer is two-dimensional (no change in the axial direction). 3 Thermal conductivity of the soil is constant.

Properties The thermal conductivity of the soil is given to be $k = 0.9$ W/m·K.

Analysis The shape factor for this configuration is given in Table 3-7 to be

$$S = \frac{2\pi L}{\ln(4z/D)}$$

since $z > 1.5D$, where z is the distance of the pipe from the ground surface, and D is the diameter of the pipe. Substituting,

$$S = \frac{2\pi \times (30 \text{ m})}{\ln(4 \times 0.5/0.1)} = 62.9 \text{ m}$$

Then the steady rate of heat transfer from the pipe becomes

$$\dot{Q} = Sk(T_1 - T_2) = (62.9 \text{ m})(0.9 \text{ W/m}\cdot\text{K})(80 - 10)^\circ\text{C} = \mathbf{3963 \text{ W}}$$

Discussion Note that this heat is conducted from the pipe surface to the surface of the earth through the soil and then transferred to the atmosphere by convection and radiation.

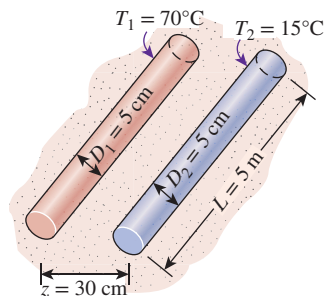


FIGURE 3-53

Schematic for Example 3-16.

EXAMPLE 3-16 Heat Transfer Between Hot- and Cold-Water Pipes

A 5-m-long section of hot- and cold-water pipes run parallel to each other in a thick concrete layer, as shown in Fig. 3-53. The diameters of both pipes are 5 cm, and the distance between the centerline of the pipes is 30 cm. The surface temperatures of the hot and cold pipes are 70°C and 15°C, respectively. Taking the thermal conductivity of the concrete to be $k = 0.75$ W/m·K, determine the rate of heat transfer between the pipes.

SOLUTION Hot- and cold-water pipes run parallel to each other in a thick concrete layer. The rate of heat transfer between the pipes is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat transfer is two-dimensional (no change in the axial direction). 3 Thermal conductivity of the concrete is constant.

Properties The thermal conductivity of concrete is given to be $k = 0.75 \text{ W/m}\cdot\text{K}$.

Analysis The shape factor for this configuration is given in Table 3–7 to be

$$S = \frac{2\pi L}{\cosh^{-1}\left(\frac{4z^2 - D_1^2 - D_2^2}{2D_1D_2}\right)}$$

where z is the distance between the centerlines of the pipes and L is their length. Substituting,

$$S = \frac{2\pi \times (5 \text{ m})}{\cosh^{-1}\left(\frac{4 \times 0.3^2 - 0.05^2 - 0.05^2}{2 \times 0.05 \times 0.05}\right)} = 6.34 \text{ m}$$

Then the steady rate of heat transfer between the pipes becomes

$$\dot{Q} = Sk(T_1 - T_2) = (6.34 \text{ m})(0.75 \text{ W/m}\cdot\text{K})(70 - 15^\circ) \text{ C} = \mathbf{262 \text{ W}}$$

Discussion We can reduce this heat loss by placing the hot- and cold-water pipes farther away from each other.

EXAMPLE 3–17 Cost of Heat Loss Through Walls in Winter

Consider an electrically heated house whose walls are 9 ft high and have an R -value of insulation of 13 (i.e., a thickness-to-thermal conductivity ratio of $L/k = 13 \text{ h}\cdot\text{ft}^2\cdot^\circ\text{F}/\text{Btu}$). Two of the walls of the house are 40 ft long and the others are 30 ft long. The house is maintained at 75°F at all times, while the temperature of the outdoors varies. Determine the amount of heat lost through the walls of the house on a certain day during which the average temperature of the outdoors is 45°F . Also, determine the cost of this heat loss to the homeowner if the unit cost of electricity is $\$0.075/\text{kWh}$. For combined convection and radiation heat transfer coefficients, use the ASHRAE (American Society of Heating, Refrigeration, and Air Conditioning Engineers) recommended values of $h_i = 1.46 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F}$ for the inner surface of the walls and $h_o = 6.0 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F}$ for the outer surface of the walls under 15 mph wind conditions in winter.

SOLUTION An electrically heated house with R -13 insulation is considered. The amount of heat lost through the walls and its cost are to be determined.

Assumptions 1 The indoor and outdoor air temperatures have remained at the given values for the entire day so that heat transfer through the walls is steady. 2 Heat transfer through the walls is one-dimensional since any significant temperature gradients in this case exist in the direction from the indoors to the outdoors. 3 The radiation effects are accounted for in the heat transfer coefficients.

Analysis This problem involves conduction through the wall and convection at its surfaces and can best be handled by making use of the thermal resistance concept and drawing the thermal resistance network, as shown in Fig. 3–54. The heat transfer area of the walls is

$$A = \text{Circumference} \times \text{Height} = (2 \times 30 \text{ ft} + 2 \times 40 \text{ ft})(9 \text{ ft}) = 1260 \text{ ft}^2$$

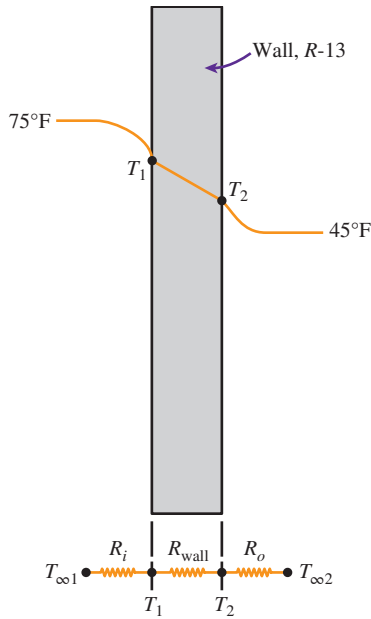


FIGURE 3-54

Schematic for Example 3-17.

Then the individual resistances are evaluated from their definitions to be

$$R_i = R_{\text{conv},i} = \frac{1}{h_i A} = \frac{1}{(1.46 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F})(1260 \text{ ft}^2)} = 0.00054 \text{ h} \cdot ^\circ\text{F/Btu}$$

$$R_{\text{wall}} = \frac{L}{kA} = \frac{R\text{-value}}{A} = \frac{13 \text{ h} \cdot \text{ft}^2 \cdot ^\circ\text{F/Btu}}{1260 \text{ ft}^2} = 0.01032 \text{ h} \cdot ^\circ\text{F/Btu}$$

$$R_o = R_{\text{conv},o} = \frac{1}{h_o A} = \frac{1}{(6.0 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F})(1260 \text{ ft}^2)} = 0.00013 \text{ h} \cdot ^\circ\text{F/Btu}$$

Noting that all three resistances are in series, the total resistance is

$$R_{\text{total}} = R_i + R_{\text{wall}} + R_o = 0.00054 + 0.01032 + 0.00013 = 0.01099 \text{ h} \cdot ^\circ\text{F/Btu}$$

Then the steady rate of heat transfer through the walls of the house becomes

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{\text{total}}} = \frac{(75 - 45)^\circ\text{F}}{0.01099 \text{ h} \cdot ^\circ\text{F/Btu}} = 2730 \text{ Btu/h}$$

Finally, the total amount of heat lost through the walls during a 24-h period and its cost to the homeowner are

$$Q = \dot{Q} \Delta t = (2730 \text{ Btu/h})(24\text{-h/day}) = \mathbf{65,514 \text{ Btu/day} = 19.2 \text{ kWh/day}}$$

since 1 kWh = 3412 Btu, and

$$\begin{aligned} \text{Heating cost} &= (\text{Energy lost})(\text{Cost of energy}) = (19.2 \text{ kWh/day})(\$0.075/\text{kWh}) \\ &= \mathbf{\$1.44/\text{day}} \end{aligned}$$

Discussion The heat losses through the walls of the house that day cost the homeowner \$1.44 worth of electricity. Most of this loss can be saved by insulation.

TOPIC OF SPECIAL INTEREST *

Heat Transfer Through Walls and Roofs

Under steady conditions, the rate of heat transfer through any section of a building wall or roof can be determined from

$$\dot{Q} = UA(T_i - T_o) = \frac{A(T_i - T_o)}{R} \quad (3-94)$$

where T_i and T_o are the indoor and outdoor air temperatures, A is the heat transfer area, U is the overall heat transfer coefficient (the U -factor), and

* This section can be skipped without a loss in continuity.

$R = 1/U$ is the overall unit thermal resistance (the R -value). Walls and roofs of buildings consist of various layers of materials, and the structure and operating conditions of the walls and the roofs may differ significantly from one building to another. Therefore, it is not practical to list the R -values (or U -factors) of different kinds of walls or roofs under different conditions. Instead, the overall R -value is determined from the thermal resistances of the individual components using the thermal resistance network. The overall thermal resistance of a structure can be determined most accurately in a lab by actually assembling the unit and testing it as a whole, but this approach is usually very time consuming and expensive. The analytical approach described here is fast and straightforward, and the results are usually in good agreement with the experimental values.

The unit thermal resistance of a plane layer of thickness L and thermal conductivity k can be determined from $R = L/k$. The thermal conductivity and other properties of common building materials are given in the appendix. The unit thermal resistances of various components used in building structures are listed in Table 3–8 for convenience.

Heat transfer through a wall or roof section is also affected by the convection and radiation heat transfer coefficients at the exposed surfaces. The effects of convection and radiation on the inner and outer surfaces of walls and roofs are usually combined into the *combined convection and radiation heat transfer coefficients* (also called *surface conductances*) h_i and h_o ,

TABLE 3–8Unit thermal resistance (the R -value) of common components used in buildings

R -value			R -value		
Component	$\text{m}^2 \cdot ^\circ\text{C}/\text{W}$	$\text{ft}^2 \cdot \text{h} \cdot ^\circ\text{F}/\text{Btu}$	Component	$\text{m}^2 \cdot ^\circ\text{C}/\text{W}$	$\text{ft}^2 \cdot \text{h} \cdot ^\circ\text{F}/\text{Btu}$
Outside surface (winter)	0.030	0.17	Wood stud, nominal 2 in $\times$ 6 in (5.5 in or 140 mm wide)	0.98	5.56
Outside surface (summer)	0.044	0.25	Clay tile, 100 mm (4 in)	0.18	1.01
Inside surface, still air	0.12	0.68	Acoustic tile	0.32	1.79
Plane airspace, vertical, ordinary surfaces ($\epsilon_{\text{eff}} = 0.82$):			Asphalt shingle roofing	0.077	0.44
13 mm ($\frac{1}{2}$ in)	0.16	0.90	Building paper	0.011	0.06
20 mm ($\frac{3}{4}$ in)	0.17	0.94	Concrete block, 100 mm (4 in):		
40 mm (1.5 in)	0.16	0.90	Lightweight	0.27	1.51
90 mm (3.5 in)	0.16	0.91	Heavyweight	0.13	0.71
Insulation, 25 mm (1 in) :			Plaster or gypsum board,		
Glass fiber	0.70	4.00	13 mm ($\frac{1}{2}$ in)	0.079	0.45
Mineral fiber batt	0.66	3.73	Wood fiberboard, 13 mm ($\frac{1}{2}$ in)	0.23	1.31
Urethane rigid foam	0.98	5.56	Plywood, 13 mm ($\frac{1}{2}$ in)	0.11	0.62
Stucco, 25 mm (1 in)	0.037	0.21	Concrete, 200 mm (8 in) :		
Face brick, 100 mm (4 in)	0.075	0.43	Lightweight	1.17	6.67
Common brick, 100 mm (4 in)	0.12	0.79	Heavyweight	0.12	0.67
Steel siding	0.00	0.00	Cement mortar, 13 mm ($\frac{1}{2}$ in)	0.018	0.10
Slag, 13 mm ($\frac{1}{2}$ in)	0.067	0.38	Wood bevel lapped siding,		
Wood, 25 mm (1 in)	0.22	1.25	13 mm $\times$ 200 mm		
Wood stud, nominal 2 in $\times$ 4 in (3.5 in or 90 mm wide)	0.63	3.58	($\frac{1}{2}$ in $\times$ 8 in)	0.14	0.81

TABLE 3-9

Combined convection and radiation heat transfer coefficients at window, wall, or roof surfaces (from ASHRAE *Handbook of Fundamentals*, Chap. 22, Table 1).

Position	Direction of Heat Flow	$h, \text{W/m}^2 \cdot \text{K}^*$		
		Surface Emittance, ϵ		
		0.90	0.20	0.05
Still air (both indoors and outdoors)				
Horiz.	Up $\uparrow$	9.26	5.17	4.32
Horiz.	Down $\downarrow$	6.13	2.10	1.25
45° slope	Up $\uparrow$	9.09	5.00	4.15
45° slope	Down $\downarrow$	7.50	3.41	2.56
Vertical	Horiz. $\rightarrow$	8.29	4.20	3.35
Moving air (any position, any direction)				
Winter condition				
(winds at 15 mph or 24 km/h)				
		34.0	—	—
Summer condition				
(winds at 7.5 mph or 12 km/h)				
		22.7	—	—

*Multiply by 0.176 to convert to $\text{Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F}$. Surface resistance can be obtained from $R = 1/h$.

respectively, whose values are given in Table 3-9 for ordinary surfaces ($\epsilon = 0.9$) and reflective surfaces ($\epsilon = 0.2$ or 0.05). Note that surfaces having a low emittance also have a low surface conductance due to the reduction in radiation heat transfer. The values in the table are based on a surface temperature of 21°C (72°F) and a surface-air temperature difference of 5.5°C (10°F). Also, the equivalent surface temperature of the environment is assumed to be equal to the ambient air temperature. Despite the convenience it offers, this assumption is not quite accurate because of the additional radiation heat loss from the surface to the clear sky. The effect of sky radiation can be accounted for approximately by taking the outside temperature to be the average of the outdoor air and sky temperatures.

The inner surface heat transfer coefficient h_i remains fairly constant throughout the year, but the value of h_o varies considerably because of its dependence on the orientation and wind speed, which can vary from less than 1 km/h in calm weather to over 40 km/h during storms. The commonly used values of h_i and h_o for peak load calculations are

$$h_i = 8.29 \text{ W/m}^2 \cdot \text{K} = 1.46 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F} \quad (\text{winter and summer})$$

$$h_o = \begin{cases} 34.0 \text{ W/m}^2 \cdot \text{K} = 6.0 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F} & (\text{winter}) \\ 22.7 \text{ W/m}^2 \cdot \text{K} = 4.0 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F} & (\text{summer}) \end{cases}$$

which correspond to design wind conditions of 24 km/h (15 mph) for winter and 12 km/h (7.5 mph) for summer. The corresponding surface thermal resistances (R -values) are determined from $R_i = 1/h_i$ and $R_o = 1/h_o$. The surface conductance values under still air conditions can be used for interior surfaces as well as exterior surfaces in calm weather.

Building components often involve *trapped airspaces* between various layers. Thermal resistances of such airspaces depend on the thickness of the layer, the temperature difference across the layer, the mean air temperature, the emissivity of each surface, the orientation of the air layer, and the direction of heat transfer. The emissivities of surfaces commonly encountered in buildings are given in Table 3-10. The **effective emissivity** of a plane-parallel airspace is given by

$$\frac{1}{\epsilon_{\text{effective}}} = \frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1 \quad (3-95)$$

where ϵ_1 and ϵ_2 are the emissivities of the surfaces of the airspace. Table 3-10 also lists the effective emissivities of airspaces for the cases where (1) the emissivity of one surface of the airspace is ϵ while the emissivity of the other surface is 0.9 (a building material) and (2) the emissivity of both surfaces is ϵ . Note that the effective emissivity of an airspace between building materials is $0.82/0.03 = 27$ times that of an airspace between surfaces covered with aluminum foil. For specified surface temperatures, radiation heat transfer through an airspace is proportional to effective emissivity, and thus the rate of radiation heat transfer in the ordinary surface case is 27 times that of the reflective surface case.

Table 3-11 lists the thermal resistances of 20-mm-, 40-mm-, and 90-mm- (0.75-in, 1.5-in, and 3.5-in) thick airspaces under various conditions. The

thermal resistance values in the table are applicable to airspaces of uniform thickness bounded by plane, smooth, parallel surfaces with no air leakage. Thermal resistances for other temperatures, emissivities, and airspaces can be obtained by interpolation and moderate extrapolation. Note that the presence of a low-emissivity surface reduces radiation heat transfer across an airspace and thus significantly increases the thermal resistance. The thermal effectiveness of a low-emissivity surface will decline, however, if the condition of the surface changes as a result of some effects such as condensation, surface oxidation, and dust accumulation.

The R -value of a wall or roof structure that involves layers of uniform thickness is determined easily by simply adding up the unit thermal resistances of the layers that are in series. But when a structure involves components such as wood studs and metal connectors, then the thermal resistance network involves parallel connections and possible two-dimensional effects. The overall R -value in this case can be determined by assuming (1) parallel heat flow paths through areas of different construction or (2) isothermal planes normal to the direction of heat transfer. The first approach usually overpredicts the overall thermal resistance, whereas the second approach usually underpredicts it. The parallel heat flow path approach is more suitable for wood frame walls and roofs, whereas the isothermal planes approach is more suitable for masonry or metal frame walls.

The thermal contact resistance between different components of building structures ranges between 0.01 and 0.1 m²·°C/W, which is negligible in most cases. However, it may be significant for metal building components such as steel framing members.

The construction of wood frame flat ceilings typically involves 2-in × 6-in joists on 400-mm (16-in) or 600-mm (24-in) centers. The fraction of framing is usually taken to be 0.10 for joists on 400-mm centers and 0.07 for joists on 600-mm centers.

Most buildings have a combination of a ceiling and a roof with an attic space in between, and the determination of the R -value of the roof–attic–ceiling combination depends on whether the attic is vented or not. For adequately ventilated attics, the attic air temperature is practically the same as the outdoor air temperature, and thus heat transfer through the roof is governed by the R -value of the ceiling only. However, heat is also transferred between the roof and the ceiling by radiation, and it needs to be considered (Fig. 3–55). The major function of the roof in this case is to serve as a radiation shield by blocking off solar radiation. Effectively ventilating the attic in summer should not lead one to believe that heat gain to the building through the attic is greatly reduced. This is because most of the heat transfer through the attic is by radiation.

Radiation heat transfer between the ceiling and the roof can be minimized by covering at least one side of the attic (the roof or the ceiling side) with a reflective material, called *radiant barrier*, such as aluminum foil or aluminum-coated paper. Tests on houses with R -19 attic floor insulation have shown that radiant barriers can reduce summer ceiling heat gains by 16 to 42 percent compared to an attic with the same insulation level and no radiant barrier. Considering that the ceiling heat gain represents about

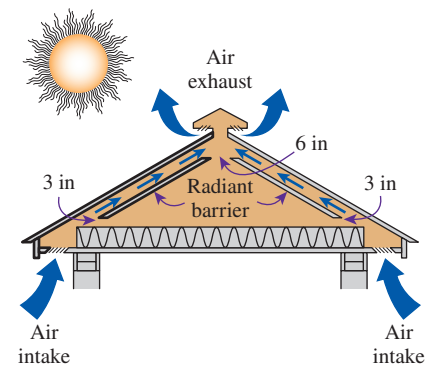
TABLE 3–10

Emissivities ϵ of various surfaces and the effective emissivity of airspaces

Surface	Effective Emissivity of Airspace		
	$\epsilon_1 = \epsilon$ $\epsilon_2 = 0.9$	$\epsilon_1 = \epsilon$ $\epsilon_2 = 0.9$	$\epsilon_1 = \epsilon$ $\epsilon_2 = \epsilon$
Aluminum foil, bright	0.05*	0.05	0.03
Aluminum sheet	0.12	0.12	0.06
Aluminum-coated paper, polished	0.20	0.20	0.11
Steel, galvanized, bright	0.25	0.24	0.15
Aluminum paint	0.50	0.47	0.35
Building materials:			
Wood, paper, masonry, nonmetallic			
paints	0.90	0.82	0.82
Ordinary glass	0.84	0.77	0.72

From ASHRAE Handbook of Fundamentals, Chap. 22, Table 3.

*Surface emissivity of aluminum foil increases to 0.30 with barely visible condensation, and to 0.70 with clearly visible condensation.

**FIGURE 3–55**

Ventilation paths for a naturally ventilated attic and the appropriate size of the flow areas around the radiant barrier for proper air circulation.

From DOE/CE-0335P, U.S. Dept. of Energy.

TABLE 3-11

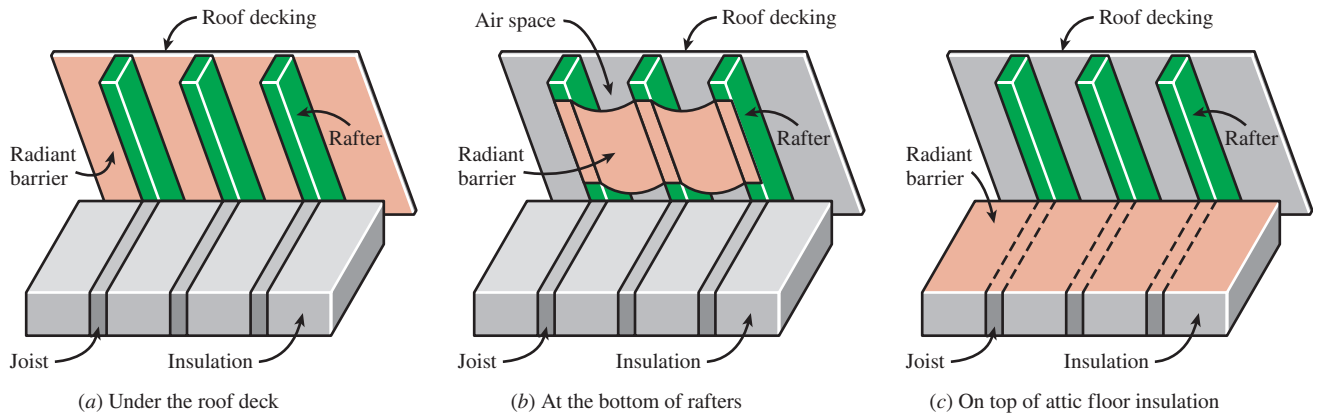
Unit thermal resistances (R -values) of well-sealed plane airspaces(a) SI units (in $\text{m}^2 \cdot ^\circ\text{C}/\text{W}$)

Position of Airspace	Direction of Heat Flow	Mean Temp., $^\circ\text{C}$	Temp. Diff., $^\circ\text{C}$	20-mm Airspace				40-mm Airspace				90-mm Airspace			
				Effective Emissivity, ϵ_{eff}				Effective Emissivity, ϵ_{eff}				Effective Emissivity, ϵ_{eff}			
				0.03	0.05	0.5	0.82	0.03	0.05	0.5	0.82	0.03	0.05	0.5	0.82
Horizontal	Up $\uparrow$	32.2	5.6	0.41	0.39	0.18	0.13	0.45	0.42	0.19	0.14	0.50	0.47	0.20	0.14
		10.0	16.7	0.30	0.29	0.17	0.14	0.33	0.32	0.18	0.14	0.27	0.35	0.19	0.15
		10.0	5.6	0.40	0.39	0.20	0.15	0.44	0.42	0.21	0.16	0.49	0.47	0.23	0.16
		-17.8	11.1	0.32	0.32	0.20	0.16	0.35	0.34	0.22	0.17	0.40	0.38	0.23	0.18
45° slope	Up $\uparrow$	32.2	5.6	0.52	0.49	0.20	0.14	0.51	0.48	0.20	0.14	0.56	0.52	0.21	0.14
		10.0	16.7	0.35	0.34	0.19	0.14	0.38	0.36	0.20	0.15	0.40	0.38	0.20	0.15
		10.0	5.6	0.51	0.48	0.23	0.17	0.51	0.48	0.23	0.17	0.55	0.52	0.24	0.17
		17.8	11.1	0.37	0.36	0.23	0.18	0.40	0.39	0.24	0.18	0.43	0.41	0.24	0.19
Vertical	Horizontal $\rightarrow$	32.2	5.6	0.62	0.57	0.21	0.15	0.70	0.64	0.22	0.15	0.65	0.60	0.22	0.15
		10.0	16.7	0.51	0.49	0.23	0.17	0.45	0.43	0.22	0.16	0.47	0.45	0.22	0.16
		10.0	5.6	0.65	0.61	0.25	0.18	0.67	0.62	0.26	0.18	0.64	0.60	0.25	0.18
		-17.8	11.1	0.55	0.53	0.28	0.21	0.49	0.47	0.26	0.20	0.51	0.49	0.27	0.20
45° slope	Down $\downarrow$	32.2	5.6	0.62	0.58	0.21	0.15	0.89	0.80	0.24	0.16	0.85	0.76	0.24	0.16
		10.0	16.7	0.60	0.57	0.24	0.17	0.63	0.59	0.25	0.18	0.62	0.58	0.25	0.18
		10.0	5.6	0.67	0.63	0.26	0.18	0.90	0.82	0.28	0.19	0.83	0.77	0.28	0.19
		-17.8	11.1	0.66	0.63	0.30	0.22	0.68	0.64	0.31	0.22	0.67	0.64	0.31	0.22
Horizontal	Down $\downarrow$	32.2	5.6	0.62	0.58	0.21	0.15	1.07	0.94	0.25	0.17	1.77	1.44	0.28	0.18
		10.0	16.7	0.66	0.62	0.25	0.18	1.10	0.99	0.30	0.20	1.69	1.44	0.33	0.21
		10.0	5.6	0.68	0.63	0.26	0.18	1.16	1.04	0.30	0.20	1.96	1.63	0.34	0.22
		-17.8	11.1	0.74	0.70	0.32	0.23	1.24	1.13	0.39	0.26	1.92	1.68	0.43	0.29

(b) English units (in $\text{h} \cdot \text{ft}^2 \cdot ^\circ\text{F}/\text{Btu}$)

Position of Airspace	Direction of Heat Flow	Mean Temp., $^\circ\text{F}$	Temp. Diff., $^\circ\text{F}$	0.75-in Airspace				1.5-in Airspace				3.5-in Airspace			
				Effective Emissivity, ϵ_{eff}				Effective Emissivity, ϵ_{eff}				Effective Emissivity, ϵ_{eff}			
				0.03	0.05	0.5	0.82	0.03	0.05	0.5	0.82	0.03	0.05	0.5	0.82
Horizontal	Up $\uparrow$	90	10	2.34	2.22	1.04	0.75	2.55	2.41	1.08	0.77	2.84	2.66	1.13	0.80
		50	30	1.71	1.66	0.99	0.77	1.87	1.81	1.04	0.80	2.09	2.01	1.10	0.84
		50	10	2.30	2.21	1.16	0.87	2.50	2.40	1.21	0.89	2.80	2.66	1.28	0.93
		0	20	1.83	1.79	1.16	0.93	2.01	1.95	1.23	0.97	2.25	2.18	1.32	1.03
45° slope	Up $\downarrow$	90	10	2.96	2.78	1.15	0.81	2.92	2.73	1.14	0.80	3.18	2.96	1.18	0.82
		50	30	1.99	1.92	1.08	0.82	2.14	2.06	1.12	0.84	2.26	2.17	1.15	0.86
		50	10	2.90	2.75	1.29	0.94	2.88	2.74	1.29	0.94	3.12	2.95	1.34	0.96
		0	20	2.13	2.07	1.28	1.00	2.30	2.23	1.34	1.04	2.42	2.35	1.38	1.06
Vertical	Horizontal $\rightarrow$	90	10	3.50	3.24	1.22	0.84	3.99	3.66	1.27	0.87	3.69	3.40	1.24	0.85
		50	30	2.91	2.77	1.30	0.94	2.58	2.46	1.23	0.90	2.67	2.55	1.25	0.91
		50	10	3.70	3.46	1.43	1.01	3.79	3.55	1.45	1.02	3.63	3.40	1.42	1.01
		0	20	3.14	3.02	1.58	1.18	2.76	2.66	1.48	1.12	2.88	2.78	1.51	1.14
45° slope	Down $\downarrow$	90	10	3.53	3.27	1.22	0.84	5.07	4.55	1.36	0.91	4.81	4.33	1.34	0.90
		50	30	3.43	3.23	1.39	0.99	3.58	3.36	1.42	1.00	3.51	3.30	1.40	1.00
		50	10	3.81	3.57	1.45	1.02	5.10	4.66	1.60	1.09	4.74	4.36	1.57	1.08
		0	20	3.75	3.57	1.72	1.26	3.85	3.66	1.74	1.27	3.81	3.63	1.74	1.27
Horizontal	Down $\downarrow$	90	10	3.55	3.29	1.22	0.85	6.09	5.35	1.43	0.94	10.07	8.19	1.57	1.00
		50	30	3.77	3.52	1.44	1.02	6.27	5.63	1.70	1.14	9.60	8.17	1.88	1.22
		50	10	3.84	3.59	1.45	1.02	6.61	5.90	1.73	1.15	11.15	9.27	1.93	1.24
		0	20	4.18	3.96	1.81	1.30	7.03	6.43	2.19	1.49	10.90	9.52	2.47	1.62

From ASHRAE Handbook of Fundamentals, Chap. 22, Table 2.

**FIGURE 3-56**

Three possible locations for an attic radiant barrier.

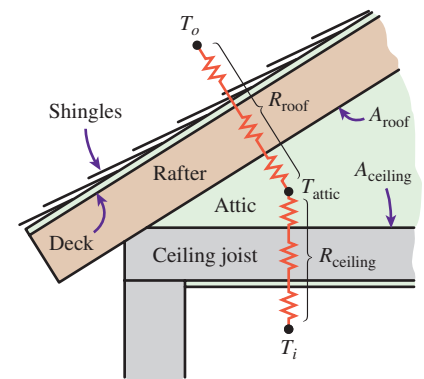
From DOE/CE-0335P, U.S. Dept. of Energy.

15 to 25 percent of the total cooling load of a house, radiant barriers will reduce the air conditioning costs by 2 to 10 percent. Radiant barriers also reduce the heat loss in winter through the ceiling, but tests have shown that the percentage reduction in heat losses is less. As a result, the percentage reduction in heating costs will be less than the reduction in the air-conditioning costs. Also, the values given are for new and undusted radiant barrier installations, and percentages will be lower for aged or dusty radiant barriers.

Some possible locations for attic radiant barriers are given in Fig. 3-56. In whole house tests on houses with R -19 attic floor insulation, radiant barriers have reduced the ceiling heat gain by an average of 35 percent when the radiant barrier is installed on the attic floor, and by 24 percent when it is attached to the bottom of roof rafters. Test cell tests also demonstrated that the best location for radiant barriers is the attic floor, provided that the attic is not used as a storage area and is kept clean.

For unvented attics, any heat transfer must occur through (1) the ceiling, (2) the attic space, and (3) the roof (Fig. 3-57). Therefore, the overall R -value of the roof–ceiling combination with an unvented attic depends on the combined effects of the R -value of the ceiling and the R -value of the roof as well as the thermal resistance of the attic space. The attic space can be treated as an air layer in the analysis. But a more practical way of accounting for its effect is to consider surface resistances on the roof and ceiling surfaces facing each other. In this case, the R -values of the ceiling and the roof are first determined separately (by using convection resistances for the still-air case for the attic surfaces). Then it can be shown that the overall R -value of the ceiling–roof combination per unit area of the ceiling can be expressed as

$$R = R_{\text{ceiling}} + R_{\text{roof}} \left(\frac{A_{\text{ceiling}}}{A_{\text{roof}}} \right) \quad (3-96)$$

**FIGURE 3-57**

Thermal resistance network for a pitched roof–attic–ceiling combination for the case of an unvented attic.

where A_{ceiling} and A_{roof} are the ceiling and roof areas, respectively. The area ratio is equal to 1 for flat roofs and is less than 1 for pitched roofs. For a 45° pitched roof, the area ratio is $A_{\text{ceiling}}/A_{\text{roof}} = 1/\sqrt{2} = 0.707$. Note that the pitched roof has a greater area for heat transfer than the flat ceiling, and the area ratio accounts for the reduction in the unit R -value of the roof when expressed per unit area of the ceiling. Also, the direction of heat flow is up in winter (heat loss through the roof) and down in summer (heat gain through the roof).

The R -value of a structure determined by analysis assumes that the materials used and the quality of workmanship meet the standards. Poor workmanship and substandard materials used during construction may result in R -values that deviate from predicted values. Therefore, some engineers use a safety factor in their designs based on experience in critical applications.

EXAMPLE 3–18 The R -Value of a Wood Frame Wall

Determine the overall unit thermal resistance (the R -value) and the overall heat transfer coefficient (the U -factor) of a wood frame wall that is built around $38\text{-mm} \times 90\text{-mm}$ (2×4 nominal) wood studs with a center-to-center distance of 400 mm. The 90-mm-wide cavity between the studs is filled with glass fiber insulation. The inside is finished with 13-mm gypsum wallboard and the outside with 13-mm wood fiberboard and 13-mm $\times$ 200-mm wood bevel lapped siding. The insulated cavity constitutes 75 percent of the heat transmission area while the studs, plates, and sills constitute 21 percent. The headers constitute 4 percent of the area, and they can be treated as studs.

Also, determine the rate of heat loss through the walls of a house whose perimeter is 50 m and wall height is 2.5 m in Las Vegas, Nevada, whose winter design temperature is -2°C . Take the indoor design temperature to be 22°C and assume 20 percent of the wall area is occupied by glazing.

SOLUTION The R -value and the U -factor of a wood frame wall as well as the rate of heat loss through such a wall in Las Vegas are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat transfer through the wall is one-dimensional. 3 Thermal properties of the wall and the heat transfer coefficients are constant.

Properties The R -values of different materials are given in Table 3–8.

Analysis The schematic of the wall as well as the different elements used in its construction are shown here. Heat transfer through the insulation and through the studs meets different resistances, and thus we need to analyze the thermal resistance for each path separately. Once the unit thermal resistances and the U -factors for the insulation and stud sections are available, the overall average thermal resistance for the entire wall can be determined from

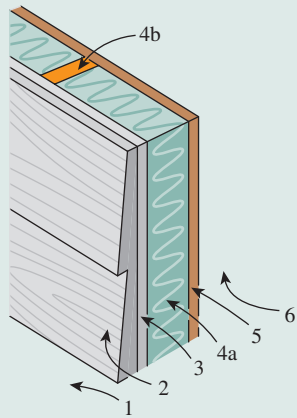
$$R_{\text{overall}} = 1/U_{\text{overall}}$$

where

$$U_{\text{overall}} = (U \times f_{\text{area}})_{\text{insulation}} + (U \times f_{\text{area}})_{\text{stud}}$$

and the value of the area fraction f_{area} is 0.75 for the insulation section and 0.25 for the stud section since the headers that constitute a small part of the wall are to be treated as studs. Using the available R -values from Table 3–8 and calculating others, the total R -values for each section can be determined in a systematic manner in the following table.

Schematic

 R -value, $\text{m}^2 \cdot ^\circ\text{C}/\text{W}$

Construction	R -value, $\text{m}^2 \cdot ^\circ\text{C}/\text{W}$	
	Between Studs	At Studs
1. Outside surface, 24 km/h wind	0.030	0.030
2. Wood bevel lapped siding	0.14	0.14
3. Wood fiberboard sheeting, 13 mm	0.23	0.23
4a. Glass fiber insulation, 90 mm	2.45	—
4b. Wood stud, 38 mm $\times$ 90 mm	—	0.63
5. Gypsum wallboard, 13 mm	0.079	0.079
6. Inside surface, still air	0.12	0.12

Total unit thermal resistance of each section,

$$R \text{ (in } \text{m}^2 \cdot ^\circ\text{C}/\text{W}) \quad \quad \quad 3.05 \quad \quad 1.23$$

The U -factor of each section, $U = 1/R$, in $\text{W}/\text{m}^2 \cdot \text{K}$ 0.328 0.813

Area fraction of each section, f_{area} 0.75 0.25

$$\text{Overall } U\text{-factor: } U = \sum f_{\text{area}, i} U_i = 0.75 \times 0.328 + 0.25 \times 0.813 \\ = \mathbf{0.449 \text{ W}/\text{m}^2 \cdot \text{K}}$$

$$\text{Overall unit thermal resistance: } R = 1/U = \mathbf{2.23 \text{ m}^2 \cdot ^\circ\text{C}/\text{W}}$$

We conclude that the overall unit thermal resistance of the wall is $2.23 \text{ m}^2 \cdot ^\circ\text{C}/\text{W}$, and this value accounts for the effects of the studs and headers. It corresponds to an R -value of $2.23 \times 5.68 = 12.7$ (or nearly R -13) in English units. Note that if there were no wood studs and headers in the wall, the overall thermal resistance would be $3.05 \text{ m}^2 \cdot ^\circ\text{C}/\text{W}$, which is 37 percent greater than $2.23 \text{ m}^2 \cdot ^\circ\text{C}/\text{W}$. Therefore, the wood studs and headers in this case serve as thermal bridges in wood frame walls, and their effect must be considered in the thermal analysis of buildings.

The perimeter of the building is 50 m, and the height of the walls is 2.5 m. Noting that glazing constitutes 20 percent of the walls, the total wall area is

$$A_{\text{wall}} = 0.80(\text{Perimeter})(\text{Height}) = 0.80(50 \text{ m})(2.5 \text{ m}) = 100 \text{ m}^2$$

Then the rate of heat loss through the walls under design conditions becomes

$$\begin{aligned} \dot{Q}_{\text{wall}} &= (UA)_{\text{wall}}(T_i - T_o) \\ &= (0.449 \text{ W}/\text{m}^2 \cdot \text{K})(100 \text{ m}^2)[22 - (-2)^\circ\text{C}] \\ &= \mathbf{1078 \text{ W}} \end{aligned}$$

Discussion Note that a 1-kW resistance heater in this house will make up almost all the heat lost through the walls, except through the doors and windows, when the outdoor air temperature drops to -2°C .

EXAMPLE 3–19 The R -Value of a Wall with Rigid Foam

The 13-mm-thick wood fiberboard sheathing of the wood stud wall discussed in the previous example is replaced by a 25-mm-thick rigid foam insulation. Determine the percent increase in the R -value of the wall as a result.

SOLUTION The overall R -value of the existing wall was determined in Example 3–18 to be $2.23 \text{ m}^2 \cdot ^\circ\text{C}/\text{W}$. Noting that the R -values of the fiberboard and the foam insulation are $0.23 \text{ m}^2 \cdot ^\circ\text{C}/\text{W}$ and $0.98 \text{ m}^2 \cdot ^\circ\text{C}/\text{W}$, respectively, and the added and removed thermal resistances are in series, the overall R -value of the wall after modification becomes

$$\begin{aligned} R_{\text{new}} &= R_{\text{old}} - R_{\text{removed}} + R_{\text{added}} \\ &= 2.23 - 0.23 + 0.98 \\ &= 2.98 \text{ m}^2 \cdot ^\circ\text{C}/\text{W} \end{aligned}$$

This represents an increase of $(2.98 - 2.23)/2.23 = 0.34$ or **34 percent** in the R -value of the wall. This example demonstrates how to evaluate the new R -value of a structure when some structural members are added or removed.

EXAMPLE 3–20 The R -Value of a Masonry Wall

Determine the overall unit thermal resistance (the R -value) and the overall heat transfer coefficient (the U -factor) of a masonry cavity wall that is built around 6-in-thick concrete blocks made of lightweight aggregate with three cores filled with perlite ($R = 4.2 \text{ h} \cdot \text{ft}^2 \cdot ^\circ\text{F}/\text{Btu}$). The outside is finished with 4-in face brick with $\frac{1}{2}$ -in cement mortar between the bricks and concrete blocks. The inside finish consists of $\frac{1}{2}$ -in gypsum wallboard separated from the concrete block by $\frac{3}{4}$ -in-thick (1-in $\times$ 3-in nominal) vertical furring ($R = 4.2 \text{ h} \cdot \text{ft}^2 \cdot ^\circ\text{F}/\text{Btu}$) whose center-to-center distance is 16 in. Both sides of the $\frac{1}{2}$ -in-thick airspace between the concrete block and the gypsum board are coated with reflective aluminum foil ($\varepsilon = 0.05$) so that the effective emissivity of the airspace is 0.03. For a mean temperature of 50°F and a temperature difference of 30°F , the R -value of the airspace is $2.91 \text{ h} \cdot \text{ft}^2 \cdot ^\circ\text{F}/\text{Btu}$. The reflective airspace constitutes 80 percent of the heat transmission area, while the vertical furring constitutes 20 percent.

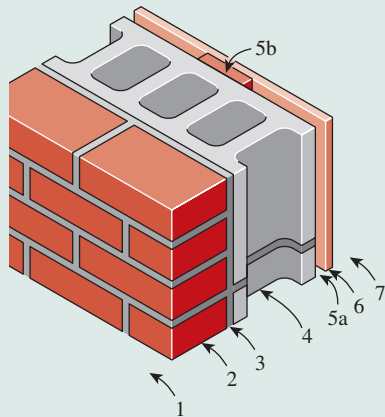
SOLUTION The R -value and the U -factor of a masonry cavity wall are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat transfer through the wall is one-dimensional. 3 Thermal properties of the wall and the heat transfer coefficients are constant.

Properties The R -values of different materials are given in Table 3–8.

Analysis The schematic of the wall as well as the different elements used in its construction are shown below. Following the approach described here and using the available R -values from Table 3–8, the overall R -value of the wall is determined in the following table.

Schematic



Construction	<i>R</i> -value, $\text{h}\cdot\text{ft}^2\cdot^\circ\text{F}/\text{Btu}$	
	Between Furring	At Furring
1. Outside surface, 15 mph wind	0.17	0.17
2. Face brick, 4 in	0.43	0.43
3. Cement mortar, 0.5 in	0.10	0.10
4. Concrete block, 6 in	4.20	4.20
5a. Reflective airspace, $\frac{3}{4}$ in	2.91	—
5b. Nominal 1 × 3 vertical furring	—	0.94
6. Gypsum wallboard, 0.5 in	0.45	0.45
7. Inside surface, still air	0.68	0.68

Total unit thermal resistance of each section, R	8.94	6.97
The U -factor of each section, $U = 1/R$, in $\text{Btu}/\text{h}\cdot\text{ft}^2\cdot^\circ\text{F}$	0.112	0.143
Area fraction of each section, f_{area}	0.80	0.20
Overall U -factor: $U = \sum f_{\text{area}, i} U_i = 0.80 \times 0.112 + 0.20 \times 0.143$ $= \mathbf{0.118 \text{ Btu}/\text{h}\cdot\text{ft}^2\cdot^\circ\text{F}}$		

Overall unit thermal resistance: $R = 1/U = \mathbf{8.46 \text{ h}\cdot\text{ft}^2\cdot^\circ\text{F}/\text{Btu}}$

Therefore, the overall unit thermal resistance of the wall is $8.46 \text{ h}\cdot\text{ft}^2\cdot^\circ\text{F}/\text{Btu}$ and the overall U -factor is $0.118 \text{ Btu}/\text{h}\cdot\text{ft}^2\cdot^\circ\text{F}$. These values account for the effects of the vertical furring.

EXAMPLE 3-21 The R -Value of a Pitched Roof

Determine the overall unit thermal resistance (the R -value) and the overall heat transfer coefficient (the U -factor) of a 45° pitched roof built around nominal 2-in $\times$ 4-in wood studs with a center-to-center distance of 16 in. The 3.5-in-wide airspace between the studs does not have any reflective surface and thus its effective emissivity is 0.84. For a mean temperature of 90°F and a temperature difference of 30°F , the R -value of the airspace is $0.86 \text{ h}\cdot\text{ft}^2\cdot^\circ\text{F}/\text{Btu}$. The lower part of the roof is finished with $\frac{1}{2}$ -in gypsum wallboard and the upper part with $\frac{5}{8}$ -in plywood, building paper, and asphalt shingle roofing. The airspace constitutes 75 percent of the heat transmission area, while the studs and headers constitute 25 percent.

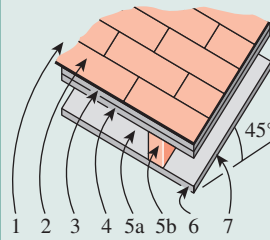
SOLUTION The R -value and the U -factor of a 45° pitched roof are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat transfer through the roof is one-dimensional. 3 Thermal properties of the roof and the heat transfer coefficients are constant.

Properties The R -values of different materials are given in Table 3–8.

Analysis The schematic of the pitched roof as well as the different elements used in its construction are shown below. Following the approach described above and using the available R -values from Table 3–8, the overall R -value of the roof can be determined in the table here.

Schematic



Construction	R -value, $\text{h}\cdot\text{ft}^2\cdot^\circ\text{F}/\text{Btu}$	
	Between Studs	At Studs
1. Outside surface, 15 mph wind	0.17	0.17
2. Asphalt shingle roofing	0.44	0.44
3. Building paper	0.06	0.06
4. Plywood deck, $\frac{5}{8}$ in	0.78	0.78
5a. Nonreflective airspace, 3.5 in	0.86	—
5b. Wood stud, 2 in $\times$ 4 in	—	3.58
6. Gypsum wallboard, 0.5 in	0.45	0.45
7. Inside surface, 45° slope, still air	0.63	0.63
Total unit thermal resistance of each section, R	3.39	6.11
The U -factor of each section, $U = 1/R$, in $\text{Btu}/\text{h}\cdot\text{ft}^2\cdot^\circ\text{F}$	0.292	0.163
Area fraction of each section, f_{area}	0.75	0.25
Overall U -factor: $U = \sum f_{\text{area}, i} U_i = 0.75 \times 0.292 + 0.25 \times 0.163$ $= 0.260 \text{ Btu}/\text{h}\cdot\text{ft}^2\cdot^\circ\text{F}$		
Overall unit thermal resistance:	$R = 1/U = 3.85 \text{ h}\cdot\text{ft}^2\cdot^\circ\text{F}/\text{Btu}$	

Therefore, the overall unit thermal resistance of this pitched roof is $3.85 \text{ h}\cdot\text{ft}^2\cdot^\circ\text{F}/\text{Btu}$, and the overall U -factor is $0.260 \text{ Btu}/\text{h}\cdot\text{ft}^2\cdot^\circ\text{F}$. Note that the wood studs offer much larger thermal resistance to heat flow than the airspace between the studs.

SUMMARY

One-dimensional heat transfer through a simple or composite body exposed to convection from both sides to media at temperatures $T_{\infty 1}$ and $T_{\infty 2}$ can be expressed as

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{\text{total}}}$$

where R_{total} is the total thermal resistance between the two media. For a plane wall exposed to convection on both sides, the total resistance is expressed as

$$R_{\text{total}} = R_{\text{conv}, 1} + R_{\text{wall}} + R_{\text{conv}, 2} = \frac{1}{h_1 A} + \frac{L}{kA} + \frac{1}{h_2 A}$$

This relation can be extended to plane walls that consist of two or more layers by adding an additional resistance for each

additional layer. The elementary thermal resistance relations can be expressed as follows:

Conduction resistance (plane wall): $R_{\text{wall}} = \frac{L}{kA}$

Conduction resistance (cylinder): $R_{\text{cyl}} = \frac{\ln(r_2/r_1)}{2\pi Lk}$

Conduction resistance (sphere): $R_{\text{sph}} = \frac{r_2 - r_1}{4\pi r_1 r_2 k}$

Convection resistance: $R_{\text{conv}} = \frac{1}{hA}$

Interface resistance: $R_{\text{interface}} = \frac{1}{h_c A} = \frac{R_c}{A}$

Radiation resistance: $R_{\text{rad}} = \frac{1}{h_{\text{rad}} A}$

where h_c is the thermal contact conductance, R_c is the thermal contact resistance, and the radiation heat transfer coefficient is defined as

$$h_{\text{rad}} = \varepsilon \sigma (T_s^2 + T_{\text{surr}}^2)(T_s + T_{\text{surr}})$$

Once the rate of heat transfer is available, the *temperature drop* across any layer can be determined from

$$\Delta T = \dot{Q}R$$

The thermal resistance concept can also be used to solve steady heat transfer problems involving parallel layers or combined series-parallel arrangements.

Adding insulation to a cylindrical pipe or a spherical shell increases the rate of heat transfer if the outer radius of the insulation is less than the *critical radius of insulation*, defined as

$$r_{\text{cr, cylinder}} = \frac{k_{\text{ins}}}{h}$$

$$r_{\text{cr, sphere}} = \frac{2k_{\text{ins}}}{h}$$

The effectiveness of an insulation is often given in terms of its *R-value*, the thermal resistance of the material for a unit surface area, expressed as

$$R\text{-value} = \frac{L}{k} \quad (\text{flat insulation})$$

where L is the thickness and k is the thermal conductivity of the material.

Finned surfaces are commonly used in practice to enhance heat transfer. Fins enhance heat transfer from a surface by exposing a larger surface area to convection. The temperature distribution along the fin is given by

$$\text{Very long fin:} \quad \frac{T(x) - T_{\infty}}{T_b - T_{\infty}} = e^{-x\sqrt{hp/kA_c}}$$

$$\text{Adiabatic fin tip:} \quad \frac{T(x) - T_{\infty}}{T_b - T_{\infty}} = \frac{\cosh m(L - x)}{\cosh mL}$$

Specified temperature at fin tip:

$$\frac{T(x) - T_{\infty}}{T_b - T_{\infty}} = \frac{[(T_L - T_{\infty})/(T_b - T_{\infty})]\sinh mx + \sinh m(L - x)}{\sinh mL}$$

Convection from fin tip:

$$\frac{T(x) - T_{\infty}}{T_b - T_{\infty}} = \frac{\cosh m(L - x) + (h/mk) \sinh m(L - x)}{\cosh mL + (h/mk) \sinh mL}$$

where $m = \sqrt{hp/kA_c}$, p is the perimeter, and A_c is the cross-sectional area of the fin. The rates of heat transfer for these cases are given to be

Very long fin:

$$\dot{Q}_{\text{long fin}} = -kA_c \left. \frac{dT}{dx} \right|_{x=0} = \sqrt{hp k A_c} (T_b - T_{\infty})$$

Adiabatic fin tip:

$$\dot{Q}_{\text{adiabatic tip}} = -kA_c \left. \frac{dT}{dx} \right|_{x=0} = \sqrt{hp k A_c} (T_b - T_{\infty}) \tanh mL$$

Specified temperature at fin tip:

$$\dot{Q}_{\text{specified temp.}} = \sqrt{hp k A_c} (T_b - T_{\infty}) \frac{\cosh mL - [(T_L - T_{\infty})/(T_b - T_{\infty})]}{\sinh mL}$$

Convection from the fin tip:

$$\dot{Q}_{\text{convection}} = \sqrt{hp k A_c} (T_b - T_{\infty}) \frac{\sinh mL + (h/mk) \cosh mL}{\cosh mL + (h/mk) \sinh mL}$$

Fins exposed to convection at their tips can be treated as fins with adiabatic tips by using the corrected length $L_c = L + A_c/p$ instead of the actual fin length.

The temperature of a fin drops along the fin, and thus the heat transfer from the fin is less because of the decreasing temperature difference toward the fin tip. To account for the effect of this decrease in temperature on heat transfer, we define *fin efficiency* as

$$\eta_{\text{fin}} = \frac{\dot{Q}_{\text{fin}}}{\dot{Q}_{\text{fin, max}}} = \frac{\text{Actual heat transfer rate from the fin}}{\text{Ideal heat transfer rate from the fin if the entire fin were at base temperature}}$$

When the fin efficiency is available, the rate of heat transfer from a fin can be determined from

$$\dot{Q}_{\text{fin}} = \eta_{\text{fin}} \dot{Q}_{\text{fin, max}} = \eta_{\text{fin}} h A_{\text{fin}} (T_b - T_{\infty})$$

The performance of the fins is judged on the basis of the enhancement in heat transfer relative to the no-fin case and is expressed in terms of the *fin effectiveness* ε_{fin} , defined as

$$\varepsilon_{\text{fin}} = \frac{\dot{Q}_{\text{fin}}}{\dot{Q}_{\text{no fin}}} = \frac{\dot{Q}_{\text{fin}}}{h A_b (T_b - T_{\infty})} = \frac{\text{Heat transfer rate from the fin of base area } A_b}{\text{Heat transfer rate from the surface of area } A_b}$$

Here, A_b is the cross-sectional area of the fin at the base, and $\dot{Q}_{\text{no fin}}$ represents the rate of heat transfer from this area if no fins are attached to the surface. The *overall effectiveness* for a finned surface is defined as the ratio of the total heat transfer from the finned surface to the heat transfer from the same surface if there were no fins,

$$\varepsilon_{\text{fin, overall}} = \frac{\dot{Q}_{\text{total, fin}}}{\dot{Q}_{\text{total, no fin}}} = \frac{h(A_{\text{unfin}} + \eta_{\text{fin}} A_{\text{fin}})(T_b - T_{\infty})}{h A_{\text{no fin}}(T_b - T_{\infty})}$$

Fin efficiency and fin effectiveness are related to each other by

$$\varepsilon_{\text{fin}} = \frac{A_{\text{fin}}}{A_b} \eta_{\text{fin}}$$

Certain multidimensional heat transfer problems involve two surfaces maintained at constant temperatures T_1 and T_2 . The steady rate of heat transfer between these two surfaces is expressed as

$$\dot{Q} = Sk(T_1 - T_2)$$

where S is the *conduction shape factor* that has the dimension of *length* and k is the thermal conductivity of the medium between the surfaces.

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PROBLEMS*

Steady Heat Conduction in Plane Walls

3-1C Consider heat conduction through a wall of thickness L and area A . Under what conditions will the temperature distributions in the wall be a straight line?

3-2C Consider heat conduction through a plane wall. Does the energy content of the wall change during steady heat conduction? How about during transient conduction? Explain.

3-3C What does the thermal resistance of a medium represent?

3-4C Can we define the convection resistance for a unit surface area as the inverse of the convection heat transfer coefficient?

*Problems designated by a "C" are concept questions, and students are encouraged to answer them all. Problems designated by an "E" are in English units, and the SI users can ignore them. Problems with the  icon are comprehensive in nature and are intended to be solved with appropriate software. Problems with the  icon are Prevention through Design problems. Problems with the  icon are Codes and Standards problems.

3-5C Consider steady heat transfer through the wall of a room in winter. The convection heat transfer coefficient at the outer surface of the wall is three times that of the inner surface as a result of the winds. On which surface of the wall do you think the temperature will be closer to the surrounding air temperature? Explain.

3-6C How is the combined heat transfer coefficient defined? What convenience does it offer in heat transfer calculations?

3-7C Why are the convection and the radiation resistances at a surface in parallel instead of being in series?

3-8C Consider steady one-dimensional heat transfer through a plane wall exposed to convection from both sides to environments at known temperatures $T_{\infty 1}$ and $T_{\infty 2}$ with known heat transfer coefficients h_1 and h_2 . Once the rate of heat transfer $\dot{Q}$ has been evaluated, explain how you would determine the temperature of each surface.

3-9C Someone comments that a microwave oven can be viewed as a conventional oven with zero convection resistance at the surface of the food. Is this an accurate statement?

3-10C Consider two cold canned drinks, one wrapped in a blanket and the other placed on a table in the same room. Which drink will warm up faster?

3-11C The bottom of a pan is made of a 4-mm-thick aluminum layer. In order to increase the rate of heat transfer through the bottom of the pan, someone proposes a design for the bottom that consists of a 3-mm-thick copper layer sandwiched between two 2-mm-thick aluminum layers. Will the new design conduct heat better? Explain. Assume perfect contact between the layers.

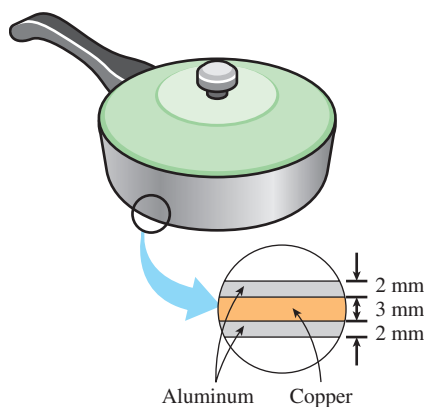


FIGURE P3-11C

3-12C Consider a surface of area A at which the convection and radiation heat transfer coefficients are h_{conv} and h_{rad} , respectively. Explain how you would determine (a) the single equivalent heat transfer coefficient, and (b) the equivalent thermal resistance. Assume the medium and the surrounding surfaces are at the same temperature.

3-13C How does the thermal resistance network associated with a single-layer plane wall differ from the one associated with a five-layer composite wall?

3-14C Consider steady one-dimensional heat transfer through a multilayer medium. If the rate of heat transfer $\dot{Q}$ is known, explain how you would determine the temperature drop across each layer.

3-15C Consider a window glass consisting of two 4-mm-thick glass sheets pressed tightly against each other. Compare the heat transfer rate through this window with that of one consisting of a single 8-mm-thick glass sheet under identical conditions.

3-16 Consider a 3-m-high, 6-m-wide, and 0.25-m-thick brick wall whose thermal conductivity is $k = 0.8 \text{ W/m}\cdot\text{K}$. On a certain day, the temperatures of the inner and the outer surfaces of the wall are measured to be 14°C and 5°C , respectively. Determine the rate of heat loss through the wall on that day.

3-17 Consider a person standing in a room at 20°C with an exposed surface area of 1.7 m^2 . The deep body temperature of the human body is 37°C , and the thermal conductivity of the human tissue near the skin is about $0.3 \text{ W/m}\cdot\text{K}$. The body is

losing heat at a rate of 150 W by natural convection and radiation to the surroundings. Taking the body temperature 0.5 cm beneath the skin to be 37°C , determine the skin temperature of the person. **Answer: 35.5°C**

3-18E Consider an electrically heated brick house ($k = 0.40 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$) whose walls are 9 ft high and 1 ft thick. Two of the walls of the house are 50 ft long and the others are 35 ft long. The house is maintained at 70°F at all times while the temperature of the outdoors varies. On a certain day, the temperature of the inner surface of the walls is measured to be at 55°F while the average temperature of the outer surface is observed to remain at 45°F during the day for 10 h and at 35°F at night for 14 h. Determine the amount of heat lost from the house that day. Also determine the cost of that heat loss to the homeowner for an electricity price of $\$0.09/\text{kWh}$.

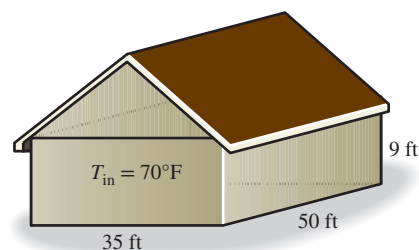


FIGURE P3-18E

3-19 A $12\text{-cm} \times 18\text{-cm}$ circuit board houses on its surface 100 closely spaced logic chips, each dissipating 0.06 W in an environment at 40°C . The heat transfer from the back surface of the board is negligible. If the heat transfer coefficient on the surface of the board is $10 \text{ W/m}^2\cdot\text{K}$, determine (a) the heat flux on the surface of the circuit board, in W/m^2 ; (b) the surface temperature of the chips; and (c) the thermal resistance between the surface of the circuit board and the cooling medium, in $^\circ\text{C/W}$.

3-20 Water is boiling in a 25-cm-diameter aluminum pan ($k = 237 \text{ W/m}\cdot\text{K}$) at 95°C . Heat is transferred steadily to the boiling water in the pan through its 0.5-cm-thick flat bottom at a rate of 800 W . If the inner surface temperature of the bottom of the pan is 108°C , determine (a) the boiling heat transfer coefficient on the inner surface of the pan and (b) the outer surface temperature of the bottom of the pan.

3-21 A cylindrical resistor element on a circuit board dissipates 0.15 W of power in an environment at 35°C . The resistor is 1.2 cm long and has a diameter of 0.3 cm . Assuming heat to be transferred uniformly from all surfaces, determine (a) the amount of heat this resistor dissipates during a 24-h period; (b) the heat flux on the surface of the resistor, in W/m^2 ; and (c) the surface temperature of the resistor for a combined convection and radiation heat transfer coefficient of $9 \text{ W/m}^2\cdot\text{K}$.

3–22 Consider a power transistor that dissipates 0.15 W of power in an environment at 30°C. The transistor is 0.4 cm long and has a diameter of 0.5 cm. Assuming heat to be transferred uniformly from all surfaces, determine (a) the amount of heat this resistor dissipates during a 24-h period, in kWh; (b) the heat flux on the surface of the transistor, in W/m²; and (c) the surface temperature of the resistor for a combined convection and radiation heat transfer coefficient of 18 W/m²·K.

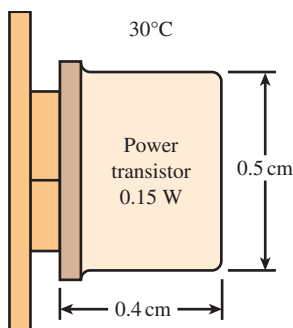


FIGURE P3–22

3–23 A 1.0 m × 1.5 m double-pane window consists of two 4-mm-thick layers of glass ($k = 0.78$ W/m·K) that are separated by a 5-mm air gap ($k_{\text{air}} = 0.025$ W/m·K). The heat flow through the air gap is assumed to be by conduction. The inside and outside air temperatures are 20°C and −20°C, respectively, and the inside and outside heat transfer coefficients are 40 and 20 W/m²·K. Determine (a) the daily rate of heat loss through the window in steady operation and (b) the temperature difference across the largest thermal resistance.

3–24 Consider a 1.5-m-high and 2.4-m-wide glass window whose thickness is 6 mm and thermal conductivity is $k = 0.78$ W/m·K. Determine the steady rate of heat transfer through this glass window and the temperature of its inner surface for a day during which the room is maintained at 24°C while the temperature of the outdoors is −5°C. Take the convection heat transfer coefficients on the inner and outer surfaces of the window to be $h_1 = 10$ W/m²·K and $h_2 = 25$ W/m²·K, and disregard any heat transfer by radiation.

3–25 Consider a 1.5-m-high and 2.4-m-wide double-pane window consisting of two 3-mm-thick layers of glass ($k = 0.78$ W/m·K) separated by a 12-mm-wide stagnant airspace ($k = 0.026$ W/m·K). Determine the steady rate of heat transfer through this double-pane window and the temperature of its inner surface for a day during which the room is maintained at 21°C while the temperature of the outdoors is −5°C. Take the convection heat transfer coefficients on the inner and outer surfaces of the window to be $h_1 = 10$ W/m²·K and $h_2 = 25$ W/m²·K, and disregard any heat transfer by radiation.

Answers: 154 W, 16.7°C

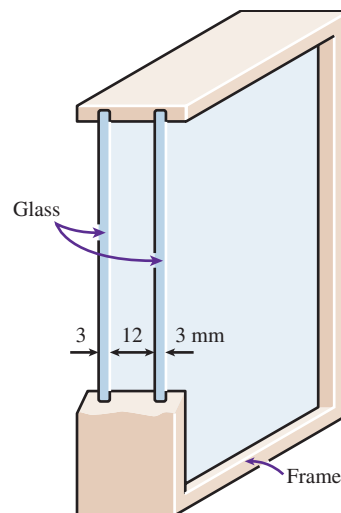


FIGURE P3–25

3–26 Repeat Prob. 3–25, assuming the space between the two glass layers is evacuated.

3–27  Reconsider Prob. 3–25. Using appropriate software, plot the rate of heat transfer through the window as a function of the width of airspace in the range of 2 mm to 20 mm, assuming pure conduction through the air. Discuss the results.

3–28E A wall is constructed of two layers of 0.6-in-thick sheetrock ($k = 0.10$ Btu/h·ft·°F), which is a plasterboard made of two layers of heavy paper separated by a layer of gypsum, placed 7 in apart. The space between the sheetrocks is filled with fiberglass insulation ($k = 0.020$ Btu/h·ft·°F). Determine (a) the thermal resistance of the wall and (b) its R -value of insulation in English units.

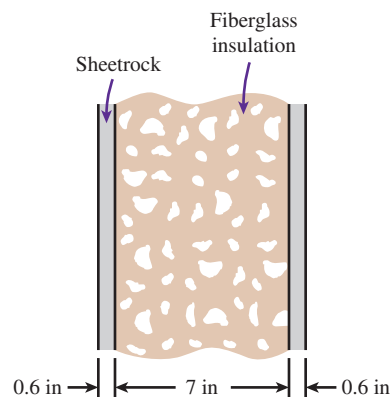


FIGURE P3–28E

3–29 To defog the rear window of an automobile, a very thin transparent heating element is attached to the inner surface of

the window. A uniform heat flux of 1300 W/m^2 is provided to the heating element for defogging a rear window with thickness of 5 mm . The interior temperature of the automobile is 22°C , and the convection heat transfer coefficient is $15 \text{ W/m}^2\cdot\text{K}$. The outside ambient temperature is -5°C , and the convection heat transfer coefficient is $100 \text{ W/m}^2\cdot\text{K}$. If the thermal conductivity of the window is $1.2 \text{ W/m}\cdot\text{K}$, determine the inner surface temperature of the window.

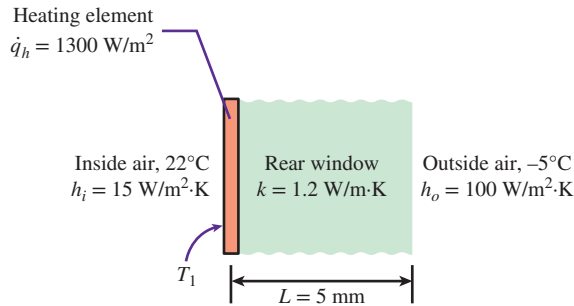


FIGURE P3-29

3-30 A transparent film is to be bonded onto the top surface of a solid plate inside a heated chamber. For the bond to cure properly, a temperature of 70°C is to be maintained at the bond, between the film and the solid plate. The transparent film has a thickness of 1 mm and thermal conductivity of $0.05 \text{ W/m}\cdot\text{K}$, while the solid plate is 13 mm thick and has a thermal conductivity of $1.2 \text{ W/m}\cdot\text{K}$. Inside the heated chamber, the convection heat transfer coefficient is $70 \text{ W/m}^2\cdot\text{K}$. If the bottom surface of the solid plate is maintained at 52°C , determine the temperature inside the heated chamber and the surface temperature of the transparent film. Assume thermal contact resistance is negligible.

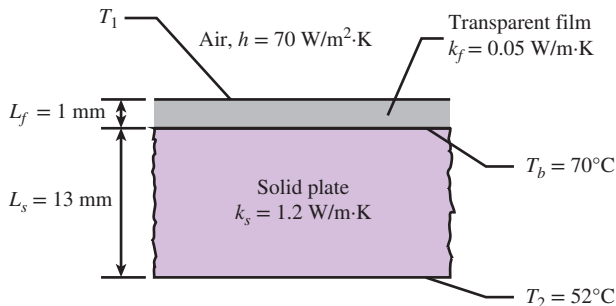


FIGURE P3-30

3-31 To defrost ice accumulated on the outer surface of an automobile windshield, warm air is blown over the inner surface of the windshield. Consider an automobile windshield with thickness of 5 mm and thermal conductivity of

$1.4 \text{ W/m}\cdot\text{K}$. The outside ambient temperature is -10°C and the convection heat transfer coefficient is $200 \text{ W/m}^2\cdot\text{K}$, while the ambient temperature inside the automobile is 25°C . Determine the value of the convection heat transfer coefficient for the warm air blowing over the inner surface of the windshield necessary to cause the accumulated ice to begin melting.

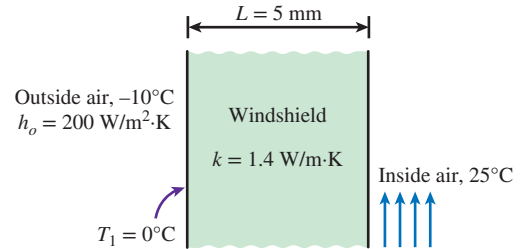


FIGURE P3-31

3-32 The roof of a house consists of a 15-cm -thick concrete slab ($k = 2 \text{ W/m}\cdot\text{K}$) that is 15 m wide and 20 m long. The convection heat transfer coefficients on the inner and outer surfaces of the roof are 5 and $12 \text{ W/m}^2\cdot\text{K}$, respectively. On a clear winter night, the ambient air is reported to be at 10°C , while the night sky temperature is 100 K . The house and the interior surfaces of the wall are maintained at a constant temperature of 20°C . The emissivity of both surfaces of the concrete roof is 0.9 . Considering both radiation and convection heat transfers, determine the rate of heat transfer through the roof, and the inner surface temperature of the roof.

If the house is heated by a furnace burning natural gas with an efficiency of 80 percent, and the price of natural gas is $\$1.20/\text{therm}$ ($1 \text{ therm} = 105,500 \text{ kJ}$ of energy content), determine the money lost through the roof that night during a 14-h period.

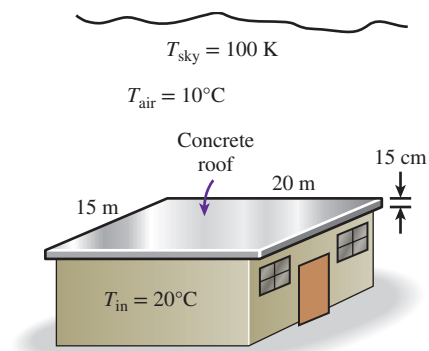


FIGURE P3-32

3-33 A $2\text{-m} \times 1.5\text{-m}$ section of wall of an industrial furnace burning natural gas is not insulated, and the temperature at the outer surface of this section is measured to be 110°C . The temperature of the furnace room is 32°C , and the combined

convection and radiation heat transfer coefficient at the surface of the outer furnace is $10 \text{ W/m}^2\cdot\text{K}$. It is proposed to insulate this section of the furnace wall with glass wool insulation ($k = 0.038 \text{ W/m}\cdot\text{K}$) in order to reduce the heat loss by 90 percent. Assuming the outer surface temperature of the metal section remains at about 110°C , determine the thickness of the insulation that needs to be used.

The furnace operates continuously and has an efficiency of 78 percent. The price of the natural gas is $\$1.10/\text{therm}$ (1 therm = 105,500 kJ of energy content). If the installation of the insulation will cost $\$250$ for materials and labor, determine how long it will take for the insulation to pay for itself from the energy it saves.

3–34 The wall of a refrigerator is constructed of fiberglass insulation ($k = 0.035 \text{ W/m}\cdot\text{K}$) sandwiched between two layers of 1-mm-thick sheet metal ($k = 15.1 \text{ W/m}\cdot\text{K}$). The refrigerated space is maintained at 2°C , and the average heat transfer coefficients at the inner and outer surfaces of the wall are $4 \text{ W/m}^2\cdot\text{K}$ and $9 \text{ W/m}^2\cdot\text{K}$, respectively. The kitchen temperature averages 24°C . It is observed that condensation occurs on the outer surfaces of the refrigerator when the temperature of the outer surface drops to 20°C . Determine the minimum thickness of fiberglass insulation that needs to be used in the wall in order to avoid condensation on the outer surfaces.

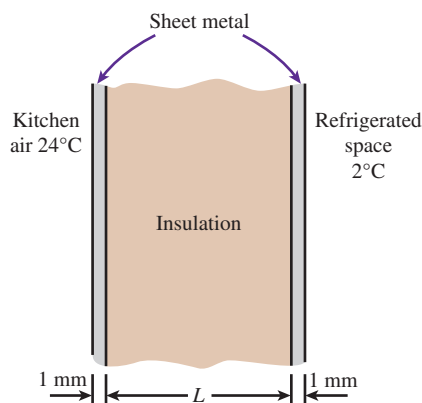


FIGURE P3–34

3–35  Reconsider Prob. 3–34. Using appropriate software, investigate the effects of the thermal conductivities of the insulation material and the sheet metal on the thickness of the insulation. Let the thermal conductivity vary from $0.02 \text{ W/m}\cdot\text{K}$ to $0.08 \text{ W/m}\cdot\text{K}$ for insulation and $10 \text{ W/m}\cdot\text{K}$ to $400 \text{ W/m}\cdot\text{K}$ for sheet metal. Plot the thickness of the insulation as the functions of the thermal conductivities of the insulation and the sheet metal, and discuss the results.

3–36 Heat is to be conducted along a circuit board that has a copper layer on one side. The circuit board is 15 cm long and 15 cm wide, and the thicknesses of the copper and epoxy layers are 0.1 mm and 1.2 mm, respectively. Disregarding heat transfer from side surfaces, determine the percentages of heat

conduction along the copper ($k = 386 \text{ W/m}\cdot\text{K}$) and epoxy ($k = 0.26 \text{ W/m}\cdot\text{K}$) layers. Also determine the effective thermal conductivity of the board. **Answers:** 0.8 percent, 99.2 percent, and $29.9 \text{ W/m}\cdot\text{K}$

3–37E A 0.05-in-thick copper plate ($k = 223 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$) is sandwiched between two 0.15-in-thick epoxy boards ($k = 0.15 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$) that are 7 in $\times$ 9 in in size. Determine the effective thermal conductivity of the board along its 9-in-long side. What fraction of the heat conducted along that side is conducted through copper?

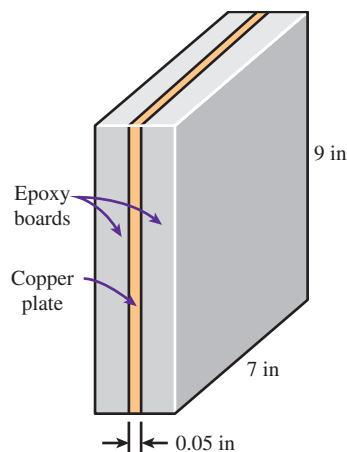


FIGURE P3–37E

3–38 Consider a house that has a 10-m $\times$ 20-m base and a 4-m-high wall. All four walls of the house have an R -value of $2.31 \text{ m}^2\cdot^\circ\text{C}/\text{W}$. The two 10-m $\times$ 4-m walls have no windows. The third wall has five windows made of 0.5-cm-thick glass ($k = 0.78 \text{ W/m}\cdot\text{K}$), 1.2 m $\times$ 1.8 m in size. The fourth wall has the same size and number of windows, but they are double-paned with a 1.5-cm-thick stagnant airspace ($k = 0.026 \text{ W/m}\cdot\text{K}$) enclosed between two 0.5-cm-thick glass layers. The thermostat in the house is set at 24°C , and the average temperature outside at that location is 8°C during the seven-month-long heating season. Disregarding any direct radiation gain or loss through the windows and taking the heat transfer coefficients at the inner and outer surfaces of the house to be 7 and $18 \text{ W/m}^2\cdot\text{K}$, respectively, determine the average rate of heat transfer through each wall.

If the house is electrically heated and the price of electricity is $\$0.08/\text{kWh}$, determine the amount of money this household will save per heating season by converting the single-pane windows to double-pane windows.

3–39E Consider a house whose walls are 12 ft high and 40 ft long. Two of the walls of the house have no windows, while each of the other two walls has four windows made of 0.25-in-thick glass ($k = 0.45 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$), 3 ft $\times$ 5 ft in size. The walls are certified to have an R -value of 19 (i.e., an L/k value of $19 \text{ h}\cdot\text{ft}^2\cdot^\circ\text{F}/\text{Btu}$). Disregarding any direct radiation gain or loss

through the windows and taking the heat transfer coefficients at the inner and outer surfaces of the house to be 2 and 4 Btu/h·ft²·°F, respectively, determine the ratio of the heat transfer through the walls with and without windows.

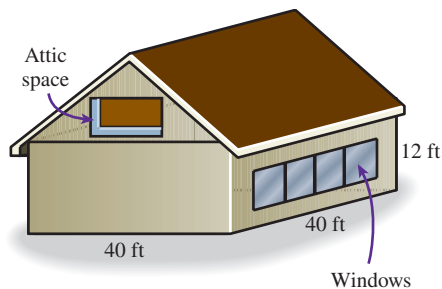


FIGURE P3-39E

3-40 **PtD** The outer surface of an engine is situated in a place where oil leakage can occur. When leaked oil comes in contact with a hot surface that has a temperature above its autoignition temperature, the oil can ignite spontaneously. Consider an engine cover that is made of a stainless steel plate with a thickness of 1 cm and a thermal conductivity of 14 W/m·K. The inner surface of the engine cover is exposed to hot air with a convection heat transfer coefficient of 7 W/m²·K at 333°C. The outer surface is exposed to an environment where the ambient air is 69°C with a convection heat transfer coefficient of 7 W/m²·K. To prevent fire hazard in the event of oil leak on the engine cover, a layer of thermal barrier coating (TBC) with a thermal conductivity of 1.1 W/m·K is applied on the engine cover outer surface. Would a TBC layer of 4 mm in thickness be sufficient to keep the engine cover surface below the autoignition temperature of 200°C to prevent a fire hazard?

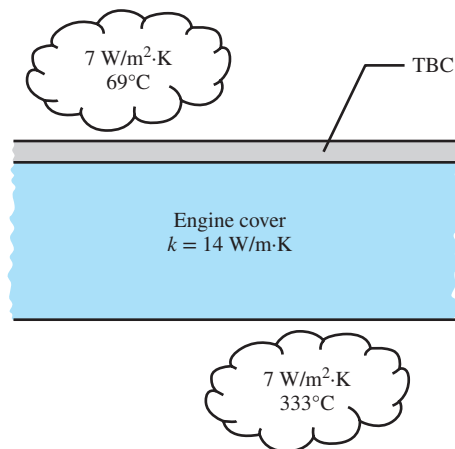


FIGURE P3-40

3-41 **PtD** Heat dissipated from a machine in operation can cause hot spots on its surface. Exposed hot spots can cause thermal burns when in contact with human skin

tissue and are considered to be hazards at the workplace. Consider a machine surface that is made of 5-mm-thick aluminum with a thermal conductivity of 237 W/m·K. During operation the machine dissipates about 300 W/m² of heat to the surroundings, and the inner aluminum surface is at 150°C. To protect machine operators from thermal burns, the machine surface can be covered with insulation. The aluminum/insulation interface has a thermal contact conductance of 3000 W/m²·K. What is the thickness required for the insulation layer with a thermal conductivity of 0.06 W/m·K in order to maintain the surface temperature at 45°C or lower?

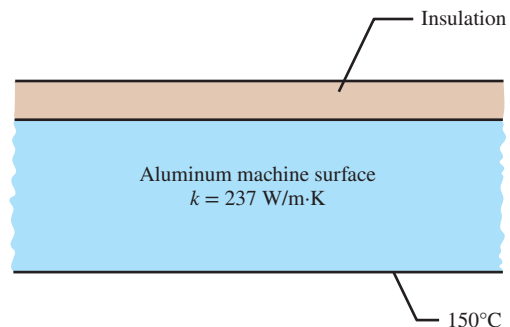


FIGURE P3-41

3-42 **C&S** A nonmetal plate ($k = 0.05$ W/m·K) is attached on the upper surface of an ASME SB-96 copper-silicon plate ($k = 36$ W/m·K). The nonmetal plate and the ASME SB-96 plate have thicknesses of 20 mm and 35 mm, respectively. The bottom surface of the ASME-SB-96 plate (surface 1) is subjected to a uniform heat flux of 200 W/m². The top nonmetal plate surface (surface 2) is exposed to convection at an air temperature of 15°C and a convection heat transfer coefficient of 10 W/m²·K. Copper-silicon alloys are not always suitable for applications where they are exposed to certain median and high temperatures. Therefore, the ASME Boiler and Pressure Vessel Code (ASME BPVC.IV-2015, HF-300) limits equipment constructed with ASME SB-96 plate to be operated at a temperature not exceeding 93°C. Determine the temperature at the bottom surface of the ASME SB-96 plate and the temperature at the interface. Would the use of the ASME SB-96 plate under these conditions be in compliance with the ASME Boiler and Pressure Vessel Code?

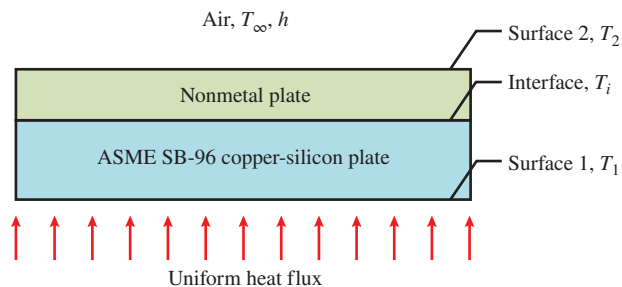


FIGURE P3-42

3-43 **C&S** An ASME SB-96 copper–silicon plate ($k = 36$ W/m·K) is attached on an ASTM A240 904L stainless steel plate ($k = 13$ W/m·K). Both plates are 25 mm thick. A series of ASTM B21 naval brass bolts are attached on the upper surface of the copper–silicon plate. The bottom surface of the stainless steel plate is exposed to convection with steam at 260°C and a convection heat transfer coefficient of 500 W/m²·K. The upper surface of the copper–silicon plate is exposed to convection with steam at 80°C and a convection heat transfer coefficient of 300 W/m²·K. According to the ASME Boiler and Pressure Vessel Code (ASME BPVC, IV-2015, HF-300), components constructed with ASME SB-96 plates should not be operated at temperatures above 93°C . According to the ASME Code for Process Piping (ASME B31.3-2014, Table A-2M), the maximum use temperature for the B21 bolts is 149°C , while the maximum use temperature for the ASTM A240 904L stainless steel plate is 260°C (ASME B31.3-2014, Table A-1M). Determine the temperatures at the upper surface of the stainless steel plate, at the bottom surface of the copper–silicon plate, and at the interface. Would the use of the copper–silicon plate, stainless steel plate, and ASTM B21 bolts be in compliance with the ASME Boiler and Pressure Vessel Code and the ASME Code for Process Piping?

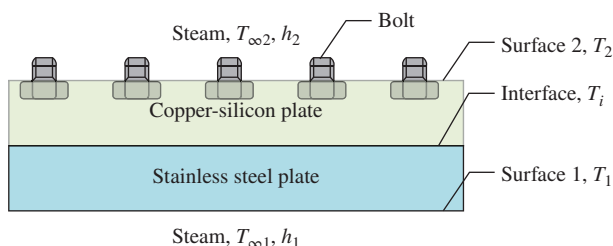


FIGURE P3-43

Thermal Contact Resistance

3-44C What is thermal contact resistance? How is it related to thermal contact conductance?

3-45C Will the thermal contact resistance be greater for smooth or rough plain surfaces?

3-46C Explain how the thermal contact resistance can be minimized.

3-47C A wall consists of two layers of insulation pressed against each other. Do we need to be concerned about the thermal contact resistance at the interface in a heat transfer analysis or can we just ignore it?

3-48C A plate consists of two thin metal layers pressed against each other. Do we need to be concerned about the thermal contact resistance at the interface in a heat transfer analysis or can we just ignore it?

3-49C Consider two surfaces pressed against each other. Now the air at the interface is evacuated. Will the thermal contact resistance at the interface increase or decrease as a result?

3-50 The thermal contact conductance at the interface of two 1-cm-thick aluminum plates is measured to be $11,000$ W/m²·K. Determine the thickness of the aluminum plate whose thermal resistance is equal to the thermal resistance of the interface between the plates.

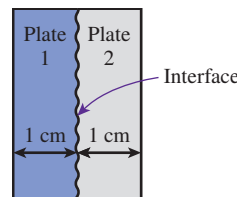


FIGURE P3-50

3-51 Two 5-cm-diameter, 15-cm-long aluminum bars ($k = 176$ W/m·K) with ground surfaces are pressed against each other with a pressure of 20 atm. The bars are enclosed in an insulation sleeve and, thus, heat transfer from the lateral surfaces is negligible. If the top and bottom surfaces of the two-bar system are maintained at temperatures of 150°C and 20°C , respectively, determine (a) the rate of heat transfer along the cylinders under steady conditions and (b) the temperature drop at the interface. **Answers:** (a) 142.4 W, (b) 6.4°C

3-52 A 1-mm-thick copper plate ($k = 401$ W/m·K) is sandwiched between two 7-mm-thick epoxy boards ($k = 0.26$ W/m·K) that are 15 cm $\times$ 20 cm in size. If the thermal contact conductance on both sides of the copper plate is estimated to be 6000 W/m²·K, determine the error involved in the total thermal resistance of the plate if the thermal contact conductances are ignored.

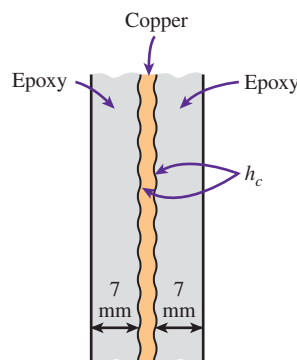


FIGURE P3-52

3-53 Two identical aluminum plates with thickness of 30 cm are pressed against each other at an average pressure of 1 atm. The interface, sandwiched between the two plates, is filled with glycerin. On the left outer surface, it is subjected to

a uniform heat flux of 7800 W/m^2 at a constant temperature of 50°C . On the right outer surface, the temperature is maintained constant at 30°C . Determine the thermal contact conductance of the glycerin at the interface if the thermal conductivity of the aluminum plates is $237 \text{ W/m}\cdot\text{K}$. Discuss whether the value of the thermal contact conductance is reasonable or not.

3-54 A two-layer wall is made of two metal plates, with surface roughness of about $25 \mu\text{m}$, pressed together at an average pressure of 10 MPa . The first layer is a stainless steel plate with a thickness of 5 mm and a thermal conductivity of $14 \text{ W/m}\cdot\text{K}$. The second layer is an aluminum plate with a thickness of 15 mm and a thermal conductivity of $237 \text{ W/m}\cdot\text{K}$. On the stainless steel side of the wall, the surface is subjected to a heat flux of 800 W/m^2 . On the aluminum side of the wall, the surface experiences convection heat transfer at an ambient temperature of 20°C , where the convection coefficient is $12 \text{ W/m}^2\cdot\text{K}$. Determine the surface temperature of the stainless steel plate.

3-55 An aluminum plate 25 mm thick ($k = 235 \text{ W/m}\cdot\text{K}$) is attached to a copper plate with thickness of 10 mm . The copper plate is heated electrically to dissipate a uniform heat flux of 5300 W/m^2 . The upper surface of the aluminum plate is exposed to convection heat transfer in a condition such that the convection heat transfer coefficient is $67 \text{ W/m}^2\cdot\text{K}$ and the surrounding room temperature is 20°C . Other surfaces of the two attached plates are insulated such that heat only dissipates through the upper surface of the aluminum plate. If the surface of the copper plate that is attached to the aluminum plate has a temperature of 100°C , determine the thermal contact conductance of the aluminum/copper interface.

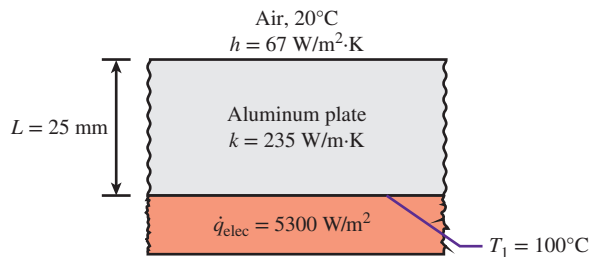


FIGURE P3-55

3-56 An aluminum plate and a stainless steel plate are pressed against each other at an average pressure of 20 MPa . Both plates have a surface roughness of $2 \mu\text{m}$. Determine the impact on the temperature drop at the interface if the surface roughness of the plates is increased tenfold.

3-57 A thin electronic component with a surface area of 950 cm^2 is cooled by having a heat sink attached on its top surface. The thermal contact conductance of the interface between the electronic component and the heat sink is $2000 \text{ W/m}^2\cdot\text{K}$.

According to the manufacturer, the heat sink has combined convection and radiation thermal resistance of 0.3 K/W . If the electronic component dissipates 45 W of heat through the heat sink in a surrounding temperature of 30°C , determine the temperature of the electronic component. Does the contact resistance at the interface of the electronic component and the heat sink play a significant role in the heat dissipation?

3-58 Consider an engine cover that is made with two layers of metal plates. The inner layer is stainless steel ($k_1 = 14 \text{ W/m}\cdot\text{K}$) with a thickness of 10 mm , and the outer layer is aluminum ($k_2 = 237 \text{ W/m}\cdot\text{K}$) with a thickness of 5 mm . Both metal plates have a surface roughness of about $23 \mu\text{m}$. The aluminum plate is attached on the stainless steel plate by screws that exert an average pressure of 20 MPa at the interface. The inside stainless steel surface of the cover is exposed to heat from the engine with a convection heat transfer coefficient of $10 \text{ W/m}^2\cdot\text{K}$ at an ambient temperature of 150°C . The outside aluminum surface is exposed to a convection heat transfer coefficient of $25 \text{ W/m}^2\cdot\text{K}$ at an ambient temperature of 40°C . Determine the heat flux through the engine cover.

3-59 Inconel® refers to a class of nickel-chromium-based superalloys that are used in high-temperature applications, such as gas turbine blades. For further improvement in the performance of gas turbine engines, the outer blade surface is coated with ceramic-based thermal barrier coating (TBC). A flat Inconel® plate with a thickness of 12 mm is coated with a layer of TBC with a thickness of $300 \mu\text{m}$ on its surface. At the interface between the Inconel® and the TBC, the thermal contact conductance is $3500 \text{ W/m}^2\cdot\text{K}$. The thermal conductivities of the Inconel® and the TBC are $25 \text{ W/m}\cdot\text{K}$ and $1.5 \text{ W/m}\cdot\text{K}$, respectively. The plate is in a surrounding of hot combustion gasses at 1500°C , and the convection heat transfer coefficient is $750 \text{ W/m}^2\cdot\text{K}$. Determine the temperature at the mid-plane of the Inconel® plate if the outer surface temperature is 1200°C .

3-60  A composite wall is made of stainless steel ($k = 13 \text{ W/m}\cdot\text{K}$, 30 mm thick), copper-silicon ($k = 36 \text{ W/m}\cdot\text{K}$, 30 mm thick), and nonmetal ($k = 0.1 \text{ W/m}\cdot\text{K}$, 15 mm thick) plates. The copper-silicon plate is sandwiched between the stainless steel plate at the bottom and the nonmetal plate at the top. A series of ASTM B21 naval brass bolts are bolted to the nonmetal plate, and the upper surface of the plate is exposed to convection heat transfer with air at 20°C and $h = 20 \text{ W/m}^2\cdot\text{K}$. At the bottom surface, the stainless steel plate is subjected to a uniform heat flux of 2000 W/m^2 . Thermal contact resistances exist at the plate interfaces. The thermal contact conductance between the stainless steel and copper-silicon plates is $20,000 \text{ W/m}^2\cdot\text{K}$; and the thermal contact conductance between the copper-silicon and nonmetal plates is $10,000 \text{ W/m}^2\cdot\text{K}$. The ASME Code for Process Piping (ASME B31.3-2014, Table A-2M) limits the maximum use temperature for the ASTM B21 bolts to 149°C . Determine the total thermal resistance of the

wall for an area of 1 m^2 , between T_1 and T_∞ . Would the ASTM B21 bolts in the nonmetal plate comply with the ASME code?

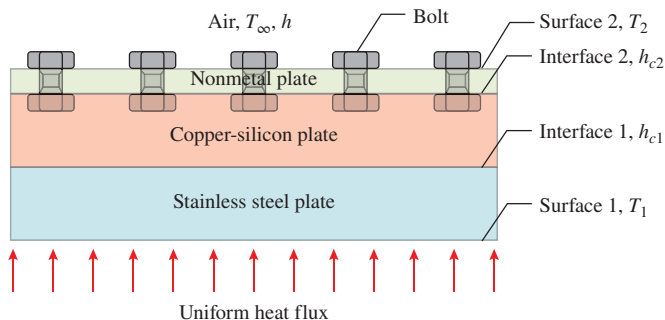


FIGURE P3-60

3-61 **c&s** A nonmetal plate ($k = 3 \text{ W/m}\cdot\text{K}$) is attached on an ASTM A240 904L stainless steel plate ($k = 13 \text{ W/m}\cdot\text{K}$). The nonmetal plate and the stainless steel plate have thicknesses of 10 mm and 30 mm, respectively. The nonmetal plate is bolted to the stainless steel plate with a series of ASTM B211 6061 aluminum alloy bolts. The upper surface of the nonmetal plate is exposed to convection heat transfer with air at 20°C and $h = 50 \text{ W/m}^2\cdot\text{K}$. The bottom surface of the stainless steel plate is exposed to convection with steam at 260°C and $h = 500 \text{ W/m}^2\cdot\text{K}$. The thermal contact conductance between the stainless steel and nonmetal plates is $10,000 \text{ W/m}^2\cdot\text{K}$. According to the ASME Code for Process Piping (ASME B31.3-2014, Table A-2M), the maximum use temperature for the B211 6061 bolts is 204°C , while the maximum use temperature for the ASTM A240 904L plate is 260°C (ASME B31.3-2014, Table A-1M). Would the use of the ASTM A240 904L plate and ASTM B211 6061 bolts be in compliance with the ASME Code for Process Piping?

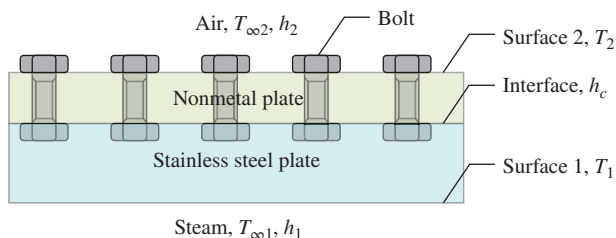


FIGURE P3-61

Generalized Thermal Resistance Networks

3-62C What are the two approaches used in the development of the thermal resistance network for two-dimensional problems?

3-63C The thermal resistance networks can also be used approximately for multidimensional problems. For what kind

of multidimensional problems will the thermal resistance approach give adequate results?

3-64C When plotting the thermal resistance network associated with a heat transfer problem, explain when two resistances are in series and when they are in parallel.

3-65 A 10-cm-thick wall is to be constructed with 2.5-m-long wood studs ($k = 0.11 \text{ W/m}\cdot\text{K}$) that have a cross section of $10 \text{ cm} \times 10 \text{ cm}$. At some point the builder ran out of those studs and started using pairs of 2.5-m-long wood studs that have a cross section of $5 \text{ cm} \times 10 \text{ cm}$ nailed to each other instead. The manganese steel nails ($k = 50 \text{ W/m}\cdot\text{K}$) are 10 cm long and have a diameter of 0.4 cm. A total of 50 nails are used to connect the two studs, which are mounted to the wall such that the nails cross the wall. The temperature difference between the inner and outer surfaces of the wall is 8°C . Assuming the thermal contact resistance between the two layers to be negligible, determine the rate of heat transfer (a) through a solid stud and (b) through a stud pair of equal length and width nailed to each other. (c) Also determine the effective conductivity of the nailed stud pair.

3-66E Consider a 10-in $\times$ 12-in epoxy glass laminate ($k = 0.10 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$) whose thickness is 0.05 in. In order to reduce the thermal resistance across its thickness, cylindrical copper fillings ($k = 223 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$) of 0.02 in diameter are to be planted throughout the board, with a center-to-center distance of 0.06 in. Determine the new value of the thermal resistance of the epoxy board for heat conduction across its thickness as a result of this modification. *Answer: $0.000256 \text{ h}\cdot^\circ\text{F/Btu}$*

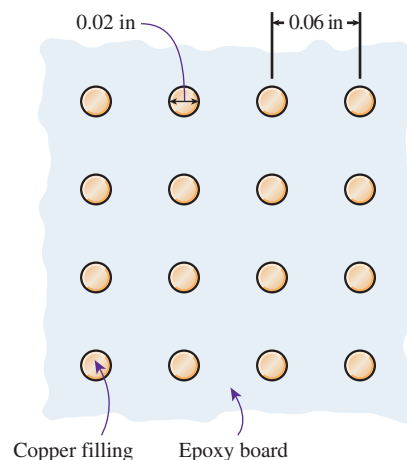


FIGURE P3-66E

3-67 Clothing made of several thin layers of fabric with trapped air in between, often called ski clothing, is commonly used in cold climates because it is light, fashionable, and a very effective thermal insulator. So it is no surprise that such clothing has largely replaced thick and heavy old-fashioned coats.

Consider a jacket made of five layers of 0.15-mm-thick synthetic fabric ($k = 0.13 \text{ W/m}\cdot\text{K}$) with 1.5-mm-thick air-space ($k = 0.026 \text{ W/m}\cdot\text{K}$) between the layers. Assuming the inner surface temperature of the jacket to be 25°C and the surface area to be 1.25 m^2 , determine the rate of heat loss through the jacket when the temperature of the outdoors is 0°C and the heat transfer coefficient at the outer surface is $25 \text{ W/m}^2\cdot\text{K}$.

What would your response be if the jacket is made of a single layer of 0.75-mm-thick synthetic fabric? What should be the thickness of a wool fabric ($k = 0.035 \text{ W/m}\cdot\text{K}$) if the person is to achieve the same level of thermal comfort wearing a thick wool coat instead of a five-layer ski jacket?

3–68 A 5-m-wide, 4-m-high, and 40-m-long kiln used to cure concrete pipes is made of 20-cm-thick concrete walls and ceiling ($k = 0.9 \text{ W/m}\cdot\text{K}$). The kiln is maintained at 40°C by injecting hot steam into it. The two ends of the kiln, $4 \text{ m} \times 5 \text{ m}$ in size, are made of a 3-mm-thick sheet metal covered with 2-cm-thick Styrofoam ($k = 0.033 \text{ W/m}\cdot\text{K}$). The convection heat transfer coefficients on the inner and the outer surfaces of the kiln are $3000 \text{ W/m}^2\cdot\text{K}$ and $25 \text{ W/m}^2\cdot\text{K}$, respectively. Disregarding any heat loss through the floor, determine the rate of heat loss from the kiln when the ambient air is at -4°C .

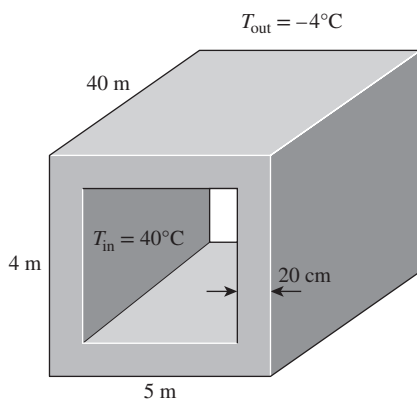


FIGURE P3–68

3–69  Reconsider Prob. 3–68. Using appropriate software, investigate the effects of the thickness of the wall and the convection heat transfer coefficient on the outer surface on the rate of heat loss from the kiln. Let the thickness vary from 10 cm to 30 cm and the convection heat transfer coefficient from $5 \text{ W/m}^2\cdot\text{K}$ to $50 \text{ W/m}^2\cdot\text{K}$. Plot the rate of heat transfer as functions of wall thickness and the convection heat transfer coefficient, and discuss the results.

3–70 A typical section of a building wall is shown in Fig. P3–70. This section extends in and out of the page and is repeated in the vertical direction. The wall support members are made of steel ($k = 50 \text{ W/m}\cdot\text{K}$). The support members are $8 \text{ cm}(t_{23}) \times 0.5 \text{ cm}(L_B)$. The remainder of the inner wall space

is filled with insulation ($k = 0.03 \text{ W/m}\cdot\text{K}$) and measures $8 \text{ cm}(t_{23}) \times 60 \text{ cm}(L_B)$. The inner wall is made of gypsum board ($k = 0.5 \text{ W/m}\cdot\text{K}$) that is 1 cm thick (t_{12}) and the outer wall is made of brick ($k = 1.0 \text{ W/m}\cdot\text{K}$) that is 10 cm thick (t_{34}). What is the average heat flux through this wall when $T_1 = 20^\circ\text{C}$ and $T_4 = 35^\circ\text{C}$?

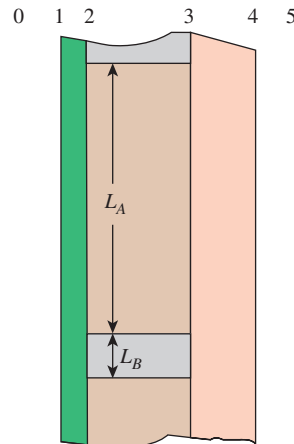


FIGURE P3–70

3–71 A 4-m-high and 6-m-wide wall consists of a long $15\text{-cm} \times 25\text{-cm}$ cross section of horizontal bricks ($k = 0.72 \text{ W/m}\cdot\text{K}$) separated by 3-cm-thick plaster layers ($k = 0.22 \text{ W/m}\cdot\text{K}$). There are also 2-cm-thick plaster layers on each side of the wall, and a 2-cm-thick rigid foam ($k = 0.026 \text{ W/m}\cdot\text{K}$) on the inner side of the wall. The indoor and the outdoor temperatures are 22°C and -4°C , and the convection heat transfer coefficients on the inner and the outer sides are $h_1 = 10 \text{ W/m}^2\cdot\text{K}$ and $h_2 = 20 \text{ W/m}^2\cdot\text{K}$, respectively. Assuming one-dimensional heat transfer and disregarding radiation, determine the rate of heat transfer through the wall.

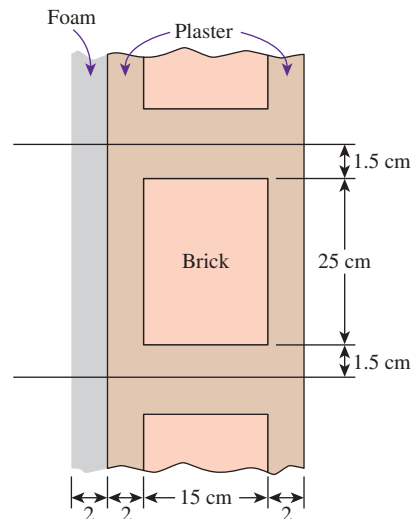


FIGURE P3–71

3-72  Reconsider Prob. 3-71. Using appropriate software, plot the rate of heat transfer through the wall as a function of the thickness of the rigid foam in the range of 1 cm to 10 cm. Discuss the results.

3-73 A 12-m-long and 5-m-high wall is constructed of two layers of 1-cm-thick sheetrock ($k = 0.17 \text{ W/m}\cdot\text{K}$) spaced 16 cm by wood studs ($k = 0.11 \text{ W/m}\cdot\text{K}$) whose cross section is $16 \text{ cm} \times 5 \text{ cm}$. The studs are placed vertically 60 cm apart, and the space between them is filled with fiberglass insulation ($k = 0.034 \text{ W/m}\cdot\text{K}$). The house is maintained at 20°C and the ambient temperature outside is -9°C . Taking the heat transfer coefficients at the inner and outer surfaces of the house to be 8.3 and $34 \text{ W/m}^2\cdot\text{K}$, respectively, determine (a) the thermal resistance of the wall considering a representative section of it and (b) the rate of heat transfer through the wall.

3-74E A 10-in-thick, 30-ft-long, and 10-ft-high wall is to be constructed using 9-in-long solid bricks ($k = 0.40 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$) of cross section $7 \text{ in} \times 7 \text{ in}$, or identical-size bricks with nine square air holes ($k = 0.015 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$) that are 9 in long and have a cross section of $1.5 \text{ in} \times 1.5 \text{ in}$. There is a 0.5-in-thick plaster layer ($k = 0.10 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$) between two adjacent bricks on all four sides and on both sides of the wall. The house is maintained at 80°F and the ambient temperature outside is 35°F . Taking the heat transfer coefficients at the inner and outer surfaces of the wall to be 1.5 and $6 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F}$, respectively, determine the rate of heat transfer through the wall constructed of (a) solid bricks and (b) bricks with air holes.

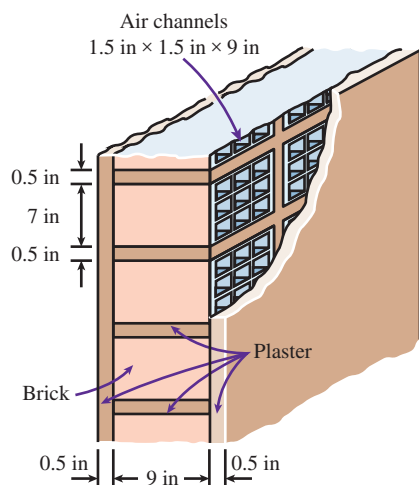


FIGURE P3-74E

3-75 Consider a 5-m-high, 8-m-long, and 0.22-m-thick wall whose representative cross section is as given in the figure. The thermal conductivities of various materials used, in $\text{W/m}\cdot\text{K}$, are $k_A = k_F = 2$, $k_B = 8$, $k_C = 20$, $k_D = 15$, and $k_E = 35$. The left and right surfaces of the wall are maintained at uniform temperatures of 300°C and 100°C , respectively. Assuming heat transfer through the wall to be one-dimensional, determine

(a) the rate of heat transfer through the wall; (b) the temperature at the point where the sections B, D, and E meet; and (c) the temperature drop across the section F. Disregard any contact resistances at the interfaces.

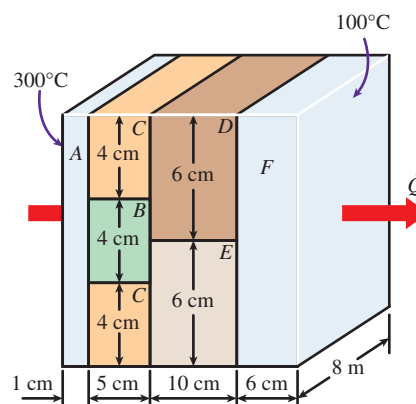


FIGURE P3-75

3-76 In an experiment to measure convection heat transfer coefficients, a very thin metal foil of very low emissivity (e.g., highly polished copper) is attached on the surface of a slab of material with very low thermal conductivity. The other surface of the metal foil is exposed to convection heat transfer by flowing fluid over the foil surface. This setup diminishes heat conduction through the slab and radiation on the metal foil surface, while heat convection plays the prominent role. The slab to which the metal foil is attached has a thickness of 25 mm and a thermal conductivity of $0.023 \text{ W/m}\cdot\text{K}$. In a condition where the surrounding room temperature is 20°C , the metal foil is heated electrically with a uniform heat flux of 5000 W/m^2 . If the bottom surface of the slab is 20°C and the metal foil has an emissivity of 0.02, determine (a) the convection heat transfer coefficient if air is flowing over the metal foil and the surface temperature of the foil is 150°C ; and (b) the convection heat transfer coefficient if water is flowing over the metal foil and the surface temperature of the foil is 30°C .

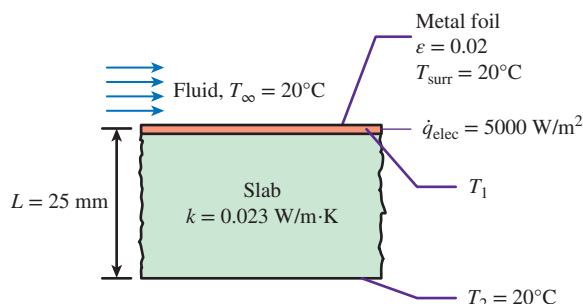


FIGURE P3-76

3-77 **C&S** A stainless steel plate ($k = 13 \text{ W/m}\cdot\text{K}$, 1 cm thick) is attached on an ASME SB-96 copper-silicon plate ($k = 36 \text{ W/m}\cdot\text{K}$, 3 cm thick) to form a plane wall. The bottom surface of the ASME SB-96 plate (surface 1) is subjected to a uniform heat flux of 750 W/m^2 . The top surface of the stainless steel plate (surface 2) is exposed to convection heat transfer with air, at $T_\infty = 20^\circ\text{C}$ and $h = 5 \text{ W/m}^2\cdot\text{K}$, and thermal radiation with the surroundings at $T_{\text{surr}} = 20^\circ\text{C}$. The stainless steel surface has an emissivity of 0.3. The thermal contact conductance at the interface of stainless steel and copper-silicon plates is $5000 \text{ W/m}^2\cdot\text{K}$. The ASME Boiler and Pressure Vessel Code (ASME BPVC.IV-2015, HF-300) limits equipment constructed with ASME SB-96 plate to be operated at a temperature not exceeding 93°C . Determine the total thermal resistance of the wall for an area of 1 m^2 , between T_1 and T_∞ . Would the use of the ASME SB-96 plate under these conditions be in compliance with the ASME Boiler and Pressure Vessel Code?

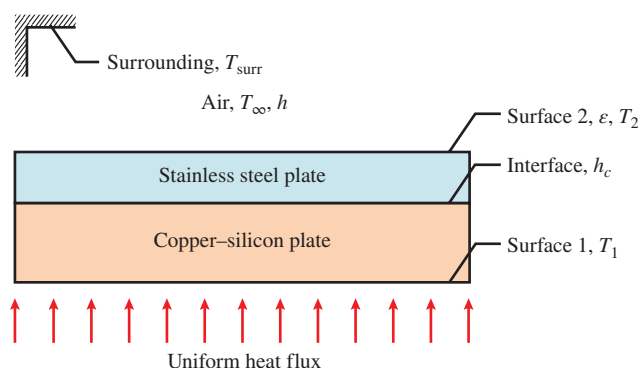


FIGURE P3-77

Heat Conduction in Cylinders and Spheres

3-78C What is an infinitely long cylinder? When is it proper to treat an actual cylinder as being infinitely long, and when is it not?

3-79C Can the thermal resistance concept be used for a solid cylinder or sphere in steady operation? Explain.

3-80C Consider a short cylinder whose top and bottom surfaces are insulated. The cylinder is initially at a uniform temperature T_i and is subjected to convection from its side surface to a medium at temperature T_∞ , with a heat transfer coefficient of h . Is the heat transfer in this short cylinder one- or two-dimensional? Explain.

3-81 Steam at 280°C flows in a stainless steel pipe ($k = 15 \text{ W/m}\cdot\text{K}$) whose inner and outer diameters are 5 cm and 5.5 cm, respectively. The pipe is covered with 3-cm-thick glass wool insulation ($k = 0.038 \text{ W/m}\cdot\text{K}$). Heat is lost to the surroundings at 5°C by natural convection and radiation, with a combined natural convection and radiation heat transfer

coefficient of $22 \text{ W/m}^2\cdot\text{K}$. Taking the heat transfer coefficient inside the pipe to be $80 \text{ W/m}^2\cdot\text{K}$, determine the rate of heat loss from the steam per unit length of the pipe. Also determine the temperature drops across the pipe shell and the insulation.

3-82 **C&S** Reconsider Prob. 3-81. Using appropriate software, investigate the effect of the thickness of the insulation on the rate of heat loss from the steam and the temperature drop across the insulation layer. Let the insulation thickness vary from 1 cm to 10 cm. Plot the rate of heat loss and the temperature drop as a function of insulation thickness, and discuss the results.

3-83 A 50-m-long section of a steam pipe whose outer diameter is 10 cm passes through an open space at 15°C . The average temperature of the outer surface of the pipe is measured to be 150°C . If the combined heat transfer coefficient on the outer surface of the pipe is $20 \text{ W/m}^2\cdot\text{K}$, determine (a) the rate of heat loss from the steam pipe; (b) the annual cost of this energy lost if steam is generated in a natural gas furnace that has an efficiency of 75 percent and the price of natural gas is $\$0.52/\text{therm}$ (1 therm = 105,500 kJ); and (c) the thickness of fiberglass insulation ($k = 0.035 \text{ W/m}\cdot\text{K}$) needed in order to save 90 percent of the heat lost. Assume the pipe temperature remains constant at 150°C .

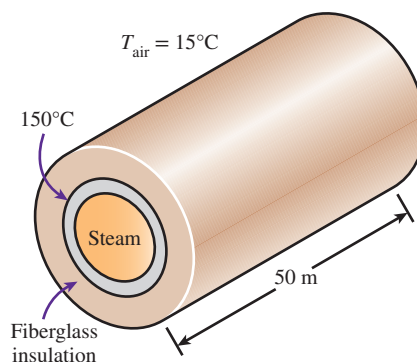


FIGURE P3-83

3-84 Superheated steam at an average temperature 200°C is transported through a steel pipe ($k = 50 \text{ W/m}\cdot\text{K}$, $D_o = 8.0 \text{ cm}$, $D_i = 6.0 \text{ cm}$, and $L = 20.0 \text{ m}$). The pipe is insulated with a 4-cm-thick layer of gypsum plaster ($k = 0.5 \text{ W/m}\cdot\text{K}$). The insulated pipe is placed horizontally inside a warehouse where the average air temperature is 10°C . The steam and the air heat transfer coefficients are estimated to be 800 and $200 \text{ W/m}^2\cdot\text{K}$, respectively. Calculate (a) the daily rate of heat transfer from the superheated steam, and (b) the temperature on the outside surface of the gypsum plaster insulation.

3-85 **C&S** Hot liquid is flowing in a steel pipe with an inner diameter of $D_1 = 22 \text{ mm}$ and an outer diameter of $D_2 = 27 \text{ mm}$. The inner surface of the pipe is coated with a thin fluorinated ethylene propylene (FEP) lining. The thermal conductivity of the pipe wall is $15 \text{ W/m}\cdot\text{K}$. The pipe outer surface is

subjected to a uniform flux of 1200 W/m^2 for a length of 1 m. The hot liquid flowing inside the pipe has a mean temperature of 180°C and a convection heat transfer coefficient of $50 \text{ W/m}^2\cdot\text{K}$. The interface between the FEP lining and the steel surface has a thermal contact conductance of $1500 \text{ W/m}^2\cdot\text{K}$. Determine the temperatures at the lining and at the pipe outer surface for the pipe length subjected to the uniform heat flux. What is the total thermal resistance between the two temperatures? The ASME Code for Process Piping (ASME B31.3-2014, A323) recommends a maximum temperature for FEP lining to be 204°C . Does the FEP lining comply with the recommendation of the code?

3-86  Hot water flows in a 1-m-long section of a pipe that is made of acrylonitrile butadiene styrene (ABS) thermoplastic. The ABS pipe section has an inner diameter of $D_1 = 22 \text{ mm}$ and an outer diameter of $D_2 = 27 \text{ mm}$. The thermal conductivity of the ABS pipe wall is $0.1 \text{ W/m}\cdot\text{K}$. The outer pipe surface is exposed to convection heat transfer with air at 20°C and $h = 10 \text{ W/m}^2\cdot\text{K}$. The water flowing inside the pipe has a convection heat transfer coefficient of $50 \text{ W/m}^2\cdot\text{K}$. According to the ASME Code for Process Piping (ASME B31.3-2014, Table B-1), the maximum recommended temperature for ABS pipe is 80°C . Determine the maximum temperature of the water flowing in the pipe, such that the ABS pipe is operating at the recommended temperature or lower. What is the temperature at the outer pipe surface when the water is at maximum temperature?

3-87E Steam exiting the turbine of a steam power plant at 100°F is to be condensed in a large condenser by cooling water flowing through copper pipes ($k = 223 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$) of inner diameter 0.4 in and outer diameter 0.6 in at an average temperature of 70°F . The heat of vaporization of water at 100°F is 1037 Btu/lbm . The heat transfer coefficients are $2400 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F}$ on the steam side and $35 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F}$ on the water side. Determine the length of the tube required to condense steam at a rate of 250 lbm/h . *Answer: 2380 ft*

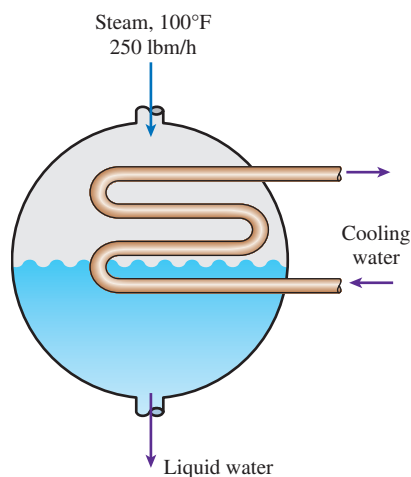


FIGURE P3-87E

3-88E Repeat Prob. 3-87E, assuming that a 0.01-in-thick layer of mineral deposit ($k = 0.5 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$) has formed on the inner surface of the pipe.

3-89E  Reconsider Prob. 3-87E. Using appropriate software, investigate the effects of the thermal conductivity of the pipe material and the outer diameter of the pipe on the length of the tube required. Let the thermal conductivity vary from $10 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$ to $400 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$ and the outer diameter from 0.5 in to 1.0 in. Plot the length of the tube as functions of pipe conductivity and the outer pipe diameter, and discuss the results.

3-90 A 2.2-mm-diameter and 14-m-long electric wire is tightly wrapped with a 1-mm-thick plastic cover whose thermal conductivity is $k = 0.15 \text{ W/m}\cdot\text{K}$. Electrical measurements indicate that a current of 13 A passes through the wire, and there is a voltage drop of 8 V along the wire. If the insulated wire is exposed to a medium at $T_\infty = 30^\circ\text{C}$ with a heat transfer coefficient of $h = 24 \text{ W/m}^2\cdot\text{K}$, determine the temperature at the interface of the wire and the plastic cover in steady operation. Also determine if doubling the thickness of the plastic cover will increase or decrease this interface temperature.

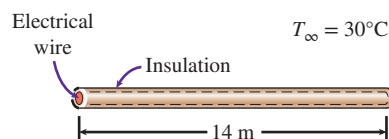


FIGURE P3-90

3-91 Consider a 1.5-m-high electric hot-water heater that has a diameter of 40 cm and maintains the hot water at 60°C . The tank is located in a small room whose average temperature is 27°C , and the heat transfer coefficients on the inner and outer surfaces of the heater are 50 and $12 \text{ W/m}^2\cdot\text{K}$, respectively. The tank is placed in another 46-cm-diameter sheet metal tank of negligible thickness, and the space between the two tanks is filled with foam insulation ($k = 0.03 \text{ W/m}\cdot\text{K}$). The thermal resistances of the water tank and the outer thin sheet metal shell are very small and can be neglected. The price of electricity is $\$0.08/\text{kWh}$, and the homeowner pays $\$280$ a year for water heating. Determine the fraction of the hot-water energy cost of this household that is due to the heat loss from the tank.

Hot-water tank insulation kits consisting of 3-cm-thick fiberglass insulation ($k = 0.035 \text{ W/m}\cdot\text{K}$) large enough to wrap the entire tank are available in the market for about $\$30$. If such an insulation is installed on this water tank by the homeowner himself, how long will it take for this additional insulation to pay for itself? *Answers: 15.2 percent, 21 months*

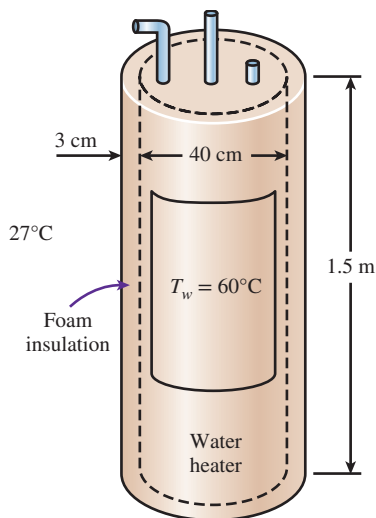


FIGURE P3-91

3-92 Chilled water enters a thin-shelled 4-cm-diameter, 200-m-long pipe at 7°C at a rate of 0.98 kg/s and leaves at 8°C. The pipe is exposed to ambient air at 30°C with a heat transfer coefficient of 9 W/m²·K. If the pipe is to be insulated with glass wool insulation ($k = 0.05$ W/m·K) in order to decrease the temperature rise of water to 0.25°C, determine the required thickness of the insulation.

3-93E Steam at 450°F is flowing through a steel pipe ($k = 8.7$ Btu/h·ft·°F) whose inner and outer diameters are 3.5 in and 4.0 in, respectively, in an environment at 55°F. The pipe is insulated with 2-in-thick fiberglass insulation ($k = 0.020$ Btu/h·ft·°F). If the heat transfer coefficients on the inside and the outside of the pipe are 30 and 5 Btu/h·ft²·°F, respectively, determine the rate of heat loss from the steam per foot length of the pipe. What is the error involved in neglecting the thermal resistance of the steel pipe in calculations?

3-94 Hot water at an average temperature of 90°C is flowing through a 15-m section of a cast iron pipe ($k = 52$ W/m·K) whose inner and outer diameters are 4 cm and 4.6 cm, respectively. The outer surface of the pipe, whose emissivity is 0.7, is exposed to the cold air at 10°C in the basement, with a heat transfer coefficient of 15 W/m²·K. The heat transfer coefficient at the inner surface of the pipe is 120 W/m²·K. Taking the walls of the basement to be at 10°C also, determine the rate of heat loss from the hot water. Also, determine the average velocity of the water in the pipe if the temperature of the water drops by 3°C as it passes through the basement.

3-95  Liquid flows in a metal pipe with an inner diameter of $D_1 = 22$ mm and an outer diameter of $D_2 = 32$ mm. The thermal conductivity of the pipe wall is 12 W/m·K. The inner surface of the pipe is coated with a thin polyvinylidene chloride (PVDC) lining. Along a length of 1 m,

the pipe outer surface is exposed to convection heat transfer with hot gas, at $T_\infty = 100^\circ\text{C}$ and $h = 5$ W/m²·K, and thermal radiation with a surrounding at $T_{\text{surr}} = 100^\circ\text{C}$. The emissivity at the pipe outer surface is 0.3. The liquid flowing inside the pipe has a convection heat transfer coefficient of 50 W/m²·K. If the outer surface of the pipe is at 85°C, determine the temperature at the PVDC lining and the temperature of the liquid. The ASME Code for Process Piping (ASME B31.3-2014, A323) recommends a maximum temperature for PVDC lining to be 79°C. Does the PVDC lining comply with the recommendation of the code?

3-96 In a pharmaceutical plant, a copper pipe ($k_c = 400$ W/m·K) with inner diameter of 20 mm and wall thickness of 2.5 mm is used for carrying liquid oxygen to a storage tank. The liquid oxygen flowing in the pipe has an average temperature of -200°C and a convection heat transfer coefficient of 120 W/m²·K. The condition surrounding the pipe has an ambient air temperature of 20°C and a combined heat transfer coefficient of 20 W/m²·K. If the dew point is 10°C, determine the thickness of the insulation ($k_i = 0.05$ W/m·K) around the copper pipe to avoid condensation on the outer surface. Assume thermal contact resistance is negligible.

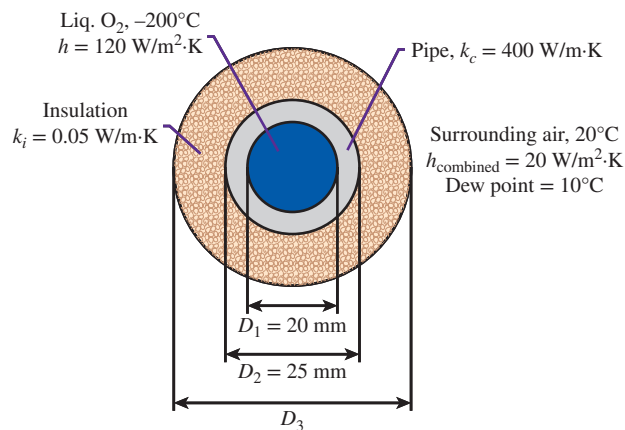


FIGURE P3-96

3-97  Liquid hydrogen is flowing through an insulated pipe ($k = 23$ W/m·K, $D_i = 3$ cm, $D_o = 4$ cm, and $L = 20$ m). The pipe is situated in a chemical plant, where the average air temperature is 40°C. The convection heat transfer coefficients of the liquid hydrogen and the ambient air are 200 W/m²·K and 50 W/m²·K, respectively. If the outer surface temperature of the insulated pipe is 5°C, determine the thickness of the pipe insulation ($k = 0.6$ W/m·K) in order to keep the liquid hydrogen flowing at an average temperature of -300°C .

3-98  Exposure to high concentrations of gaseous ammonia can cause lung damage. To prevent gaseous ammonia from leaking out, ammonia is transported in

its liquid state through a pipe ($k = 25 \text{ W/m}\cdot\text{K}$, $D_i = 2.5 \text{ cm}$, $D_o = 4 \text{ cm}$, and $L = 10 \text{ m}$). Since liquid ammonia has a normal boiling point of -33.3°C , the pipe needs to be properly insulated to prevent the surrounding heat from causing the ammonia to boil. The pipe is situated in a laboratory, where the average ambient air temperature is 20°C . The convection heat transfer coefficients of the liquid ammonia and the ambient air are $100 \text{ W/m}^2\cdot\text{K}$ and $20 \text{ W/m}^2\cdot\text{K}$, respectively. Determine the insulation thickness for the pipe using a material with $k = 0.75 \text{ W/m}\cdot\text{K}$ to keep the liquid ammonia flowing at an average temperature of -35°C , while maintaining the insulated pipe outer surface temperature at 10°C .

3-99  A mixture of chemicals is flowing in a pipe ($k = 14 \text{ W/m}\cdot\text{K}$, $D_i = 2.5 \text{ cm}$, $D_o = 3 \text{ cm}$, and $L = 10 \text{ m}$). During the transport, the mixture undergoes an exothermic reaction having an average temperature of 135°C and a convection heat transfer coefficient of $150 \text{ W/m}^2\cdot\text{K}$. To prevent any incident of thermal burn, the pipe needs to be insulated. However, due to the vicinity of the pipe, there is only enough room to fit a 2.5-cm-thick layer of insulation over the pipe. The pipe is situated in a plant where the average ambient air temperature is 20°C and the convection heat transfer coefficient is $25 \text{ W/m}^2\cdot\text{K}$. Determine the insulation for the pipe such that the thermal conductivity of the insulation is sufficient to maintain the outside surface temperature at 45°C or lower.

3-100  Ice slurry is being transported in a pipe ($k = 15 \text{ W/m}\cdot\text{K}$, $D_i = 2.5 \text{ cm}$, $D_o = 3 \text{ cm}$, and $L = 5 \text{ m}$) with an inner surface temperature of 0°C . The ambient condition surrounding the pipe has a temperature of 20°C , a convection heat transfer coefficient of $10 \text{ W/m}^2\cdot\text{K}$, and a dew point of 10°C . If the outer surface temperature of the pipe drops below the dew point, condensation can occur on the surface. Since this pipe is located in a vicinity of high-voltage devices, water droplets from the condensation can create an electrical hazard. To prevent an electrical accident, the pipe surface needs to be insulated. Determine the insulation thickness for the pipe using a material with $k = 0.95 \text{ W/m}\cdot\text{K}$ to prevent the outer surface temperature from dropping below the dew point.

3-101 An 8-m-internal-diameter spherical tank made of 1.5-cm-thick stainless steel ($k = 15 \text{ W/m}\cdot\text{K}$) is used to store iced water at 0°C . The tank is located in a room whose temperature is 25°C . The walls of the room are also at 25°C . The outer surface of the tank is black (emissivity $\varepsilon = 1$), and heat transfer between the outer surface of the tank and the surroundings is by natural convection and radiation. The convection heat transfer coefficients at the inner and the outer surfaces of the tank are $80 \text{ W/m}^2\cdot\text{K}$ and $10 \text{ W/m}^2\cdot\text{K}$, respectively. Determine (a) the rate of heat transfer to the iced water in the tank and (b) the amount of ice at 0°C that melts during a 24-h period. The heat of fusion of water at atmospheric pressure is $h_{if} = 333.7 \text{ kJ/kg}$.

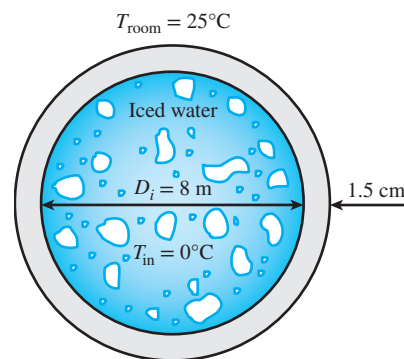


FIGURE P3-101

3-102 The boiling temperature of nitrogen at atmospheric pressure at sea level (1 atm pressure) is -196°C . Therefore, nitrogen is commonly used in low-temperature scientific studies since the temperature of liquid nitrogen in a tank open to the atmosphere will remain constant at -196°C until it is depleted. Any heat transfer to the tank will result in the evaporation of some liquid nitrogen, which has a heat of vaporization of 198 kJ/kg and a density of 810 kg/m^3 at 1 atm.

Consider a 3-m-diameter spherical tank that is initially filled with liquid nitrogen at 1 atm and -196°C . The tank is exposed to ambient air at 15°C , with a combined convection and radiation heat transfer coefficient of $35 \text{ W/m}^2\cdot\text{K}$. The temperature of the thin-shelled spherical tank is observed to be almost the same as the temperature of the nitrogen inside. Determine the rate of evaporation of the liquid nitrogen in the tank as a result of the heat transfer from the ambient air if the tank is (a) not insulated, (b) insulated with 5-cm-thick fiberglass insulation ($k = 0.035 \text{ W/m}\cdot\text{K}$), and (c) insulated with 2-cm-thick superinsulation which has an effective thermal conductivity of $0.00005 \text{ W/m}\cdot\text{K}$.

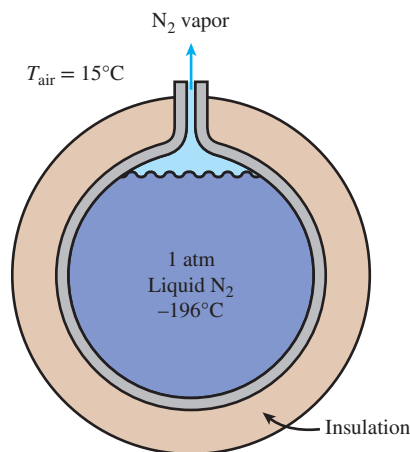


FIGURE P3-102

3-103 Repeat Prob. 3-102 for liquid oxygen, which has a boiling temperature of -183°C , a heat of vaporization of 213 kJ/kg , and a density of 1140 kg/m^3 at 1 atm pressure.

Critical Radius of Insulation

3-104C What is the critical radius of insulation? How is it defined for a cylindrical layer?

3-105C Consider an insulated pipe exposed to the atmosphere. Will the critical radius of insulation be greater on calm days or on windy days? Why?

3-106C A pipe is insulated to reduce the heat loss from it. However, measurements indicate that the rate of heat loss has increased instead of decreasing. Can the measurements be right?

3-107C Consider a pipe at a constant temperature whose radius is greater than the critical radius of insulation. Someone claims that the rate of heat loss from the pipe has increased when some insulation is added to the pipe. Is this claim valid?

3-108C A pipe is insulated such that the outer radius of the insulation is less than the critical radius. Now the insulation is taken off. Will the rate of heat transfer from the pipe increase or decrease for the same pipe surface temperature?

3-109E A 0.083-in-diameter electrical wire at 90°F is covered by 0.02-in-thick plastic insulation ($k = 0.075\text{ Btu/h}\cdot\text{ft}\cdot^{\circ}\text{F}$). The wire is exposed to a medium at 50°F , with a combined convection and radiation heat transfer coefficient of $2.5\text{ Btu/h}\cdot\text{ft}^2\cdot^{\circ}\text{F}$. Determine if the plastic insulation on the wire will increase or decrease heat transfer from the wire. *Answer: It helps*

3-110E Repeat Prob. 3-109E, assuming a thermal contact resistance of $0.01\text{ h}\cdot\text{ft}^2\cdot^{\circ}\text{F/Btu}$ at the interface of the wire and the insulation.

3-111 A 4-mm-diameter spherical ball at 50°C is covered by a 1-mm-thick plastic insulation ($k = 0.13\text{ W/m}\cdot\text{K}$). The ball is exposed to a medium at 15°C , with a combined convection and radiation heat transfer coefficient of $20\text{ W/m}^2\cdot\text{K}$. Determine if the plastic insulation on the ball will help or hurt heat transfer from the ball.

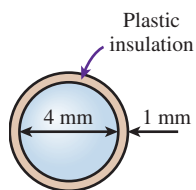


FIGURE P3-111

3-112  Reconsider Prob. 3-111. Using appropriate software, plot the rate of heat transfer from the ball as a function of the plastic insulation thickness in the range of 0.5 mm to 20 mm. Discuss the results.

Heat Transfer from Finned Surfaces

3-113C Hot air is to be cooled as it is forced to flow through the tubes exposed to atmospheric air. Fins are to be added in order to enhance heat transfer. Would you recommend attaching the fins inside or outside the tubes? Why? When would you recommend attaching fins both inside and outside the tubes?

3-114C What is the reason for the widespread use of fins on surfaces?

3-115C What is the difference between the fin effectiveness and the fin efficiency?

3-116C The fins attached to a surface are determined to have an effectiveness of 0.9. Do you think the rate of heat transfer from the surface has increased or decreased as a result of the addition of these fins?

3-117C Explain how the fins enhance heat transfer from a surface. Also, explain how the addition of fins may actually decrease heat transfer from a surface.

3-118C How does the overall effectiveness of a finned surface differ from the effectiveness of a single fin?

3-119C Hot water is to be cooled as it flows through the tubes exposed to atmospheric air. Fins are to be attached in order to enhance heat transfer. Would you recommend attaching the fins inside or outside the tubes? Why?

3-120C Consider two finned surfaces that are identical except that the fins on the first surface are formed by casting or extrusion, whereas they are attached to the second surface afterwards by welding or tight fitting. For which case do you think the fins will provide greater enhancement in heat transfer? Explain.

3-121C The heat transfer surface area of a fin is equal to the sum of all surfaces of the fin exposed to the surrounding medium, including the surface area of the fin tip. Under what conditions can we neglect heat transfer from the fin tip?

3-122C Does the (a) efficiency and (b) effectiveness of a fin increase or decrease as the fin length is increased?

3-123C Two pin fins are identical, except that the diameter of one of them is twice the diameter of the other. For which fin is the (a) fin effectiveness and (b) fin efficiency higher? Explain.

3-124C Two plate fins of constant rectangular cross section are identical, except that the thickness of one of them is twice the thickness of the other. For which fin is the (a) fin effectiveness and (b) fin efficiency higher? Explain.

3-125C Two finned surfaces are identical, except that the convection heat transfer coefficient of one of them is twice that of the other. For which finned surface is the (a) fin effectiveness and (b) fin efficiency higher? Explain.

3-126 Obtain a relation for the fin efficiency for a fin of constant cross-sectional area A_c , perimeter p , length L , and

thermal conductivity k exposed to convection to a medium at T_∞ with a heat transfer coefficient h . Assume the fins are sufficiently long so that the temperature of the fin at the tip is nearly T_∞ . Take the temperature of the fin at the base to be T_b and neglect heat transfer from the fin tips. Simplify the relation for (a) a circular fin of diameter D and (b) rectangular fins of thickness t .

3-127 A 4-mm-diameter and 10-cm-long aluminum fin ($k = 237 \text{ W/m}\cdot\text{K}$) is attached to a surface. If the heat transfer coefficient is $12 \text{ W/m}^2\cdot\text{K}$, determine the percent error in the rate of heat transfer from the fin when the infinitely long fin assumption is used instead of the adiabatic fin tip assumption.

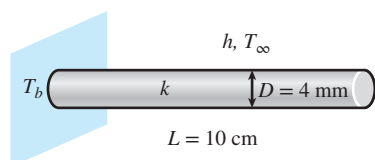


FIGURE P3-127

3-128 Consider a very long rectangular fin attached to a flat surface such that the temperature at the end of the fin is essentially that of the surrounding air, i.e., 20°C . Its width is 5.0 cm; thickness is 1.0 mm; thermal conductivity is $200 \text{ W/m}\cdot\text{K}$; and base temperature is 40°C . The heat transfer coefficient is $20 \text{ W/m}^2\cdot\text{K}$. Estimate the fin temperature at a distance of 5.0 cm from the base and the rate of heat loss from the entire fin.

3-129 Consider a very long, slender rod. One end of the rod is attached to a base surface maintained at T_b , while the surface of the rod is exposed to an air temperature of 400°C . Thermocouples imbedded in the rod at locations 25 and 120 mm from the base surface register temperatures of 325°C and 375°C , respectively. (a) Calculate the rod base temperature ($^\circ\text{C}$). (b) Determine the rod length (mm) for the case where the ratio of the heat transfer from a finite length fin to the heat transfer from a very long fin under the same conditions is 99 percent.

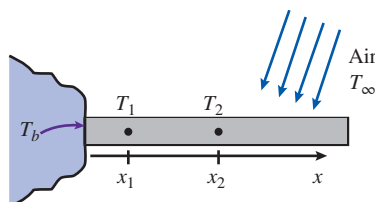


FIGURE P3-129

3-130 Two very long, slender rods of the same diameter and length are given. One rod (Rod 1) is made of aluminum and has a thermal conductivity $k_1 = 200 \text{ W/m}\cdot\text{K}$, but the thermal conductivity of Rod 2, k_2 , is not known. To determine the thermal conductivity of Rod 2, both rods at one end are thermally attached to a metal surface which is maintained at a constant

temperature T_b . Both rods are losing heat by convection, with a convection heat transfer coefficient h into the ambient air at T_∞ . The surface temperature of each rod is measured at various distances from the hot base surface. The measurements reveal that the temperature of the aluminum rod (Rod 1) at $x_1 = 40 \text{ cm}$ from the base is the same as that of the rod of unknown thermal conductivity (Rod 2) at $x_2 = 20 \text{ cm}$ from the base. Determine the thermal conductivity k_2 of the second rod ($\text{W/m}\cdot\text{K}$).

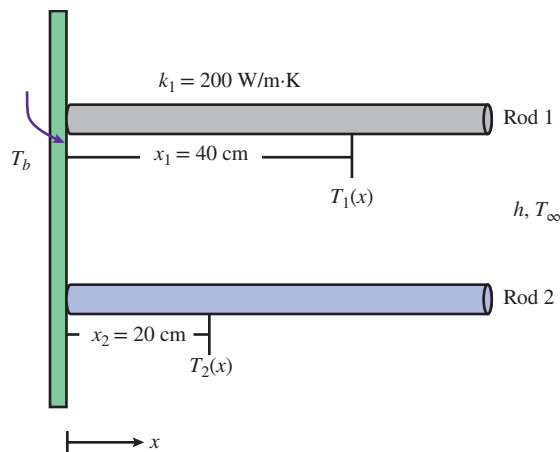


FIGURE P3-130

3-131 A turbine blade made of a metal alloy ($k = 17 \text{ W/m}\cdot\text{K}$) has a length of 5.3 cm, a perimeter of 11 cm, and a cross-sectional area of 5.13 cm^2 . The turbine blade is exposed to hot gas from the combustion chamber at 973°C with a convection heat transfer coefficient of $538 \text{ W/m}^2\cdot\text{K}$. The base of the turbine blade maintains a constant temperature of 450°C , and the tip is adiabatic. Determine the heat transfer rate to the turbine blade and the temperature at the tip.

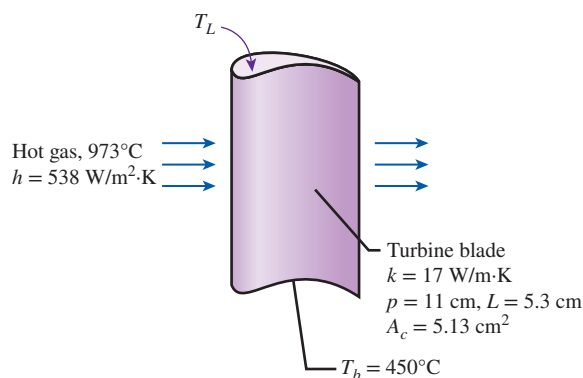


FIGURE P3-131

3-132  An ASTM B209 5154 aluminum alloy plate is connected to an insulation plate by long metal bolts 4.8 mm in diameter. The portion of the bolts exposed to convection heat transfer with air is 5 cm long. The thermal

conductivity of the bolt is known to be 15 W/m·K. The ambient condition of the air is at 20°C with a convection heat transfer coefficient of 20 W/m²·K. According to the ASME Code for Process Piping (ASME B31.3-2014, Table A-1M), the maximum use temperature for the ASTM B209 5154 aluminum alloy plate is 65°C. If the temperature of the bolt at mid-length (2.5 cm from the upper surface of the aluminum plate) is 50°C, determine the temperature T_b at the upper surface of the aluminum plate. What is the rate of heat loss from each bolt to convection? Is the use of the ASTM B209 5154 plate in compliance with the ASME Code for Process Piping?

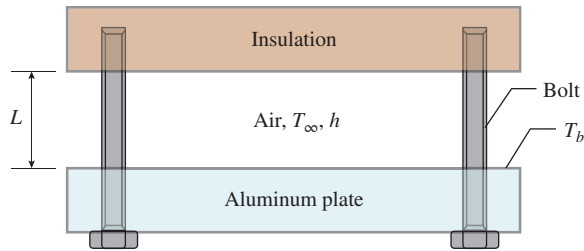


FIGURE P3-132

3-133E Consider a stainless steel spoon ($k = 8.7$ Btu/h·ft²·°F) partially immersed in boiling water at 200°F in a kitchen at 75°F. The handle of the spoon has a cross section of 0.08 in × 0.5 in and extends 7 in in the air from the free surface of the water. If the heat transfer coefficient at the exposed surfaces of the spoon handle is 3 Btu/h·ft²·°F, determine the temperature difference across the exposed surface of the spoon handle. State your assumptions. **Answer:** 124.6°F

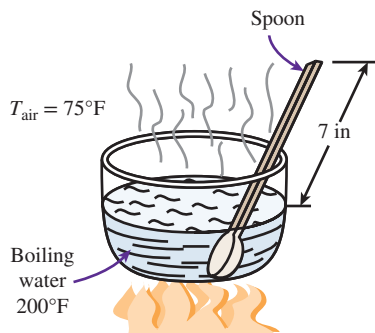


FIGURE P3-133E

3-134E  Reconsider Prob. 3-133E. Using appropriate software, investigate the effects of the thermal conductivity of the spoon material and the length of its extension in the air on the temperature difference across the exposed surface of the spoon handle. Let the thermal conductivity vary from 5 Btu/h·ft²·°F to 225 Btu/h·ft²·°F and the length from 5 in to 12 in. Plot the temperature difference as the function of thermal conductivity and length, and discuss the results.

3-135 A DC motor delivers mechanical power to a rotating stainless steel shaft ($k = 15.1$ W/m·K) with a length of 25 cm and a diameter of 25 mm. In a surrounding with ambient air temperature of 20°C and convection heat transfer coefficient of 25 W/m²·K, the surface area of the motor housing that is exposed to the ambient air is 0.075 m². The motor uses 300 W of electrical power and delivers 55 percent of it as mechanical power to rotate the stainless steel shaft. If the tip of the stainless steel shaft has a temperature of 22°C, determine the surface temperature of the motor housing. Assume the base temperature of the shaft is equal to the surface temperature of the motor housing.

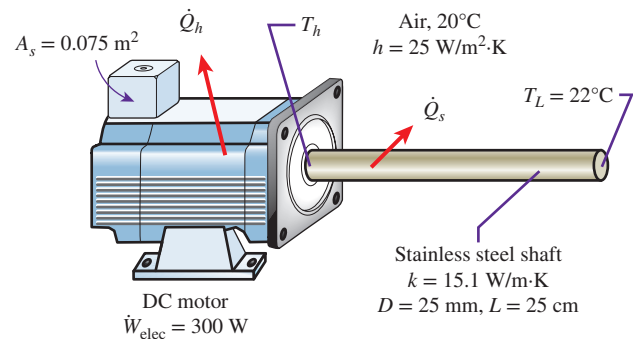


FIGURE P3-135

3-136  A stainless steel plate is connected to a copper plate by long ASTM B98 copper–silicon bolts of 9.5 mm in diameter. The portion of the bolts exposed to convection heat transfer with air is 5 cm long. The air temperature for convection is at 20°C with a convection heat transfer coefficient of 5 W/m²·K. The thermal conductivity of the bolt is known to be 36 W/m·K. The copper plate has a uniform temperature of 70°C. If each bolt is estimated to have a rate of heat loss to convection of 5 W, determine the temperature T_b at the upper surface of the stainless steel plate. According to the ASME Code for Process Piping (ASME B31.3-2014, Table A-2M), the maximum use temperature for the ASTM B98 copper–silicon bolt is 149°C. Does the use of the ASTM B98 bolts in this condition comply with the ASME code?

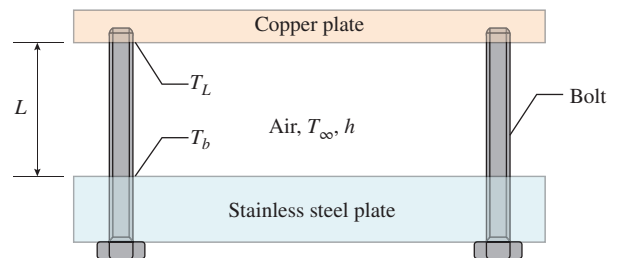


FIGURE P3-136

3–137 A plane wall with surface temperature of 350°C is attached with straight rectangular fins ($k = 235 \text{ W/m}\cdot\text{K}$). The fins are exposed to an ambient air condition of 25°C , and the convection heat transfer coefficient is $154 \text{ W/m}^2\cdot\text{K}$. Each fin has a length of 50 mm , a base 5 mm thick, and a width of 100 mm . Determine the efficiency, heat transfer rate, and effectiveness of each fin, using (a) Table 3–3 and (b) Figure 3–43.

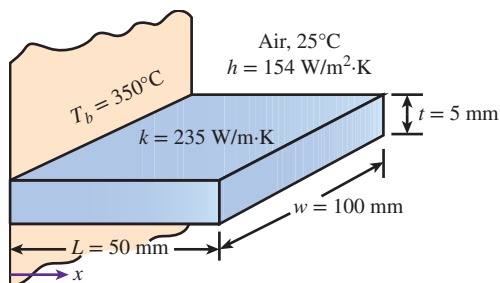


FIGURE P3–137

3–138 Two 4-m-long and 0.4-cm-thick cast iron ($k = 52 \text{ W/m}\cdot\text{K}$) steam pipes of outer diameter 10 cm are connected to each other through two 1-cm-thick flanges of outer diameter 18 cm. The steam flows inside the pipe at an average temperature of 200°C with a heat transfer coefficient of $180 \text{ W/m}^2\cdot\text{K}$. The outer surface of the pipe is exposed to an ambient at 12°C , with a heat transfer coefficient of $25 \text{ W/m}^2\cdot\text{K}$. (a) Disregarding the flanges, determine the average outer surface temperature of the pipe. (b) Using this temperature for the base of the flange and treating the flanges as the fins, determine the fin efficiency and the rate of heat transfer from the flanges. (c) What length of pipe is the flange section equivalent to for heat transfer purposes?

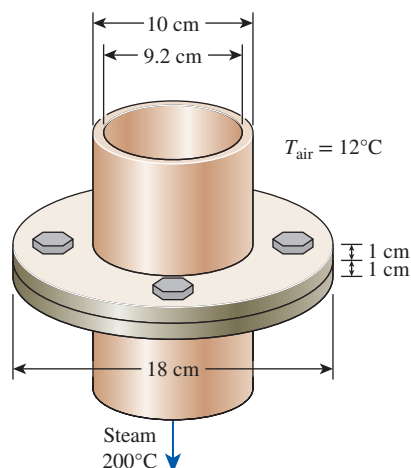


FIGURE P3–138

3–139 Pipes with inner and outer diameters of 50 mm and 60 mm, respectively, are used for transporting superheated vapor in a manufacturing plant. The pipes with thermal conductivity of

$16 \text{ W/m}\cdot\text{K}$ are connected together by flanges with combined thickness of 20 mm and outer diameter of 90 mm. Air condition surrounding the pipes has a temperature of 25°C and a convection heat transfer coefficient of $10 \text{ W/m}^2\cdot\text{K}$. If the inner surface temperature of the pipe is maintained at a constant temperature of 150°C , determine the temperature at the base of the flange and the rate of heat loss through the flange.

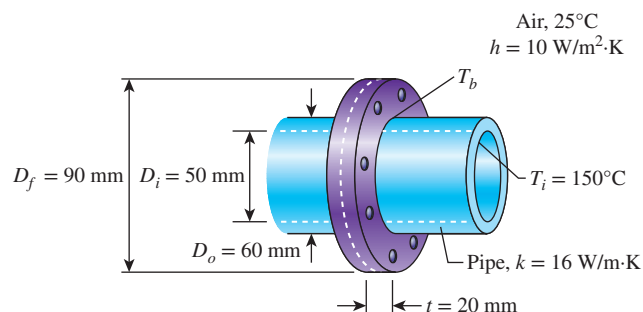


FIGURE P3–139

3–140 Steam in a heating system flows through tubes whose outer diameter is 5 cm and whose walls are maintained at a temperature of 130°C . Circular aluminum alloy 2024-T6 fins ($k = 186 \text{ W/m}\cdot\text{K}$) of outer diameter 6 cm and constant thickness 1 mm are attached to the tube. The space between the fins is 3 mm, and thus there are 250 fins per meter length of the tube. Heat is transferred to the surrounding air at $T_{\infty} = 25^{\circ}\text{C}$, with a heat transfer coefficient of $40 \text{ W/m}^2\cdot\text{K}$. Determine the increase in heat transfer from the tube per meter of its length as a result of adding fins. **Answer: 1788 W**

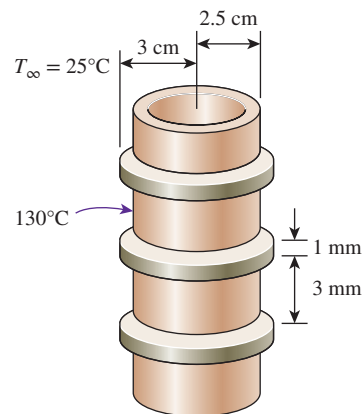


FIGURE P3–140

3–141 A 0.4-cm-thick, 12-cm-high, and 18-cm-long circuit board houses 80 closely spaced logic chips on one side, each dissipating 0.04 W . The board is impregnated with copper fillings and has an effective thermal conductivity of $30 \text{ W/m}\cdot\text{K}$. All the heat generated in the chips is conducted across the circuit board and is dissipated from the back side of the board to a medium at 40°C , with a heat transfer coefficient of

52 W/m²·K. (a) Determine the temperatures on the two sides of the circuit board. (b) Now a 0.2-cm-thick, 12-cm-high, and 18-cm-long aluminum plate ($k = 237$ W/m·K) with 864 2-cm-long aluminum pin fins of diameter 0.25 cm is attached to the back side of the circuit board with a 0.02-cm-thick epoxy adhesive ($k = 1.8$ W/m·K). Determine the new temperatures on the two sides of the circuit board.

3-142 A hot surface at 100°C is to be cooled by attaching 3-cm-long, 0.25-cm-diameter aluminum pin fins ($k = 237$ W/m·K) to it, with a center-to-center distance of 0.6 cm. The temperature of the surrounding medium is 30°C, and the heat transfer coefficient on the surfaces is 35 W/m²·K. Determine the rate of heat transfer from the surface for a 1-m × 1-m section of the plate. Also determine the overall effectiveness of the fins.

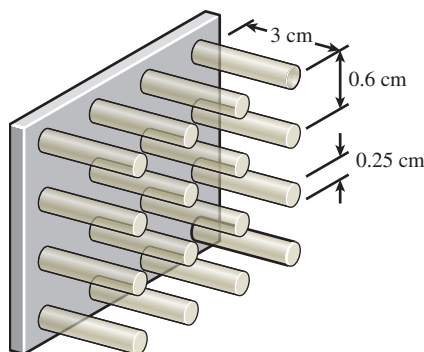


FIGURE P3-142

3-143  Reconsider Prob. 3-142. Using appropriate software, investigate the effect of the center-to-center distance of the fins on the rate of heat transfer from the surface and the overall effectiveness of the fins. Let the center-to-center distance vary from 0.4 cm to 2.0 cm. Plot the rate of heat transfer and the overall effectiveness as a function of the center-to-center distance, and discuss the results.

3-144 Circular cooling fins of diameter $D = 1$ mm and length $L = 30$ mm, made of copper ($k = 380$ W/m·K), are used to enhance heat transfer from a surface that is maintained at temperature $T_{s1} = 132^\circ\text{C}$. Each rod has one end attached to this surface ($x = 0$), while the opposite end ($x = L$) is joined to a second surface, which is maintained at $T_{s2} = 0^\circ\text{C}$. The air flowing between the surfaces and the rods is also at $T_\infty = 0^\circ\text{C}$, and the convection coefficient is $h = 100$ W/m²·K.

- Express the function $\theta(x) = T(x) - T_\infty$ along a fin, and calculate the temperature at $x = L/2$.
- Determine the rate of heat transferred from the hot surface through each fin and the fin effectiveness. Is the use of fins justified? Why?
- What is the total rate of heat transfer from a 10-cm by 10-cm section of the wall, which has 625 uniformly distributed fins? Assume the same convection coefficient for the fin and for the unfinned wall surface.

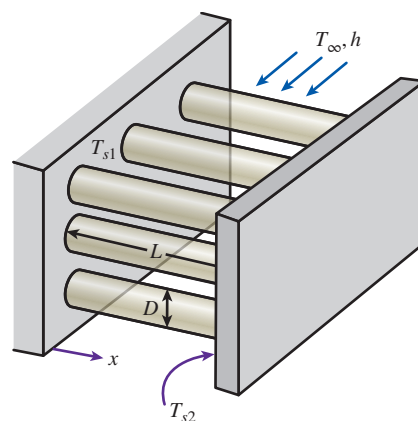


FIGURE P3-144

3-145 A 40-W power transistor is to be cooled by attaching it to one of the commercially available heat sinks shown in Table 3-6. Select a heat sink that will allow the case temperature of the transistor not to exceed 90°C in the ambient air at 20°C.

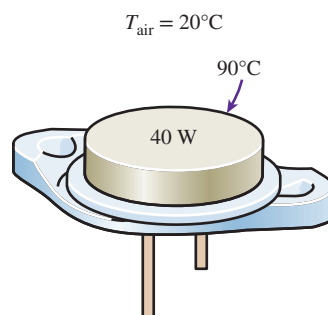


FIGURE P3-145

3-146 A 60-W power transistor is to be cooled by attaching it to one of the commercially available heat sinks shown in Table 3-6. Select a heat sink that will allow the case temperature of the transistor not to exceed 90°C in the ambient air at 30°C.

Bioheat Transfer Equation

3-147 The human body is adaptable to extreme climatic conditions and keeps the body core and skin temperatures within the comfort zone by regulating the metabolic heat generation rate. For example, in extreme cold conditions, the human body will maintain the body temperature by increasing metabolic heat generation, while in very hot conditions, the body will sweat and release heat. To understand this effect of ambient conditions on the human body, repeat Example 3-14 in the text and consider a case where climatic conditions change from -20°C and 20°C . For this change in ambient air temperature, calculate the metabolic heat generation rate required with

skin/fat thicknesses of 0.0075, 0.005, and 0.0025 m to maintain the skin temperature at 34°C. Assume that in spite of the change in ambient air temperature, the perspiration rate remains constant at 0.0005 s⁻¹. Plot a graph of metabolic heat generation rate against the ambient temperature with temperature increments of 5°C.

3-148 Consider the conditions of Example 3-14 in the text for two different environments of air and water with convective heat transfer coefficient of 2 W/m²·K and 20 W/m²·K, respectively. The ambient (T_∞) and surrounding temperature (T_{surr}) for both air and water may be assumed to be 15°C. What would be the metabolic heat generation rate required to maintain the skin temperature at 34°C?

3-149 Consider the conditions of Example 3-14 in the text except that the ambient air is at a temperature of 30°C. A person with skin/fat layer thickness of 0.003 m is doing vigorous exercise which raises the metabolic heat generation rate from 700 to 7000 W/m³ over a period of time. Calculate the perspiration rate required in L/s so as to maintain the skin temperature at 34°C. Assume that perspiration properties are the same as those of liquid water at the average surface skin temperature of 35.5°C.

3-150 We are interested in steady-state heat transfer analysis from a human forearm subjected to certain environmental conditions. For this purpose, consider the forearm to be made up of muscle with thickness r_m with a skin/fat layer of thickness t_{sf} over it, as shown in Figure P3-150. For simplicity, approximate the forearm as a one-dimensional cylinder and ignore the presence of bones. The metabolic heat generation rate ($\dot{e}_m$) and perfusion rate ($\dot{p}$) are both constant throughout the muscle. The blood density and specific heat are ρ_b and c_b , respectively. The core body temperature (T_c) and the arterial blood temperature (T_a) are both assumed to be the same and constant. The muscle and the skin/fat layer thermal conductivities are k_m and k_{sf} , respectively. The skin has an emissivity of ϵ , and the forearm is subjected to an air environment with a temperature of T_∞ , a convection heat transfer coefficient of h_{conv} , and a radiation heat transfer coefficient of h_{rad} . Assuming blood properties and thermal conductivities are all constant, (a) write the bioheat transfer equation in radial coordinates. The boundary conditions for the forearm are specified constant temperature at the outer surface of the muscle (T_i) and temperature symmetry at the centerline of the forearm. (b) Solve the differential equation and apply the boundary conditions to develop an expression for the temperature distribution in the forearm. (c) Determine the temperature at the outer surface of the muscle (T_i) and the maximum temperature in the forearm (T_{max}) for the following conditions:

$$\begin{aligned} r_m &= 0.05 \text{ m}, t_{sf} = 0.003 \text{ m}, \dot{e}_m = 700 \text{ W/m}^3, \dot{p} = 0.0005 \text{ l/s}, \\ T_a &= 37^\circ\text{C}, T_\infty = T_{\text{surr}} = 24^\circ\text{C}, \epsilon = 0.95, \\ \rho_b &= 1000 \text{ kg/m}^3, c_b = 3600 \text{ J/kg}\cdot\text{K}, k_m = 0.5 \text{ W/m}\cdot\text{K}, \\ k_{sf} &= 0.3 \text{ W/m}\cdot\text{K}, h_{\text{conv}} = 2 \text{ W/m}^2\cdot\text{K}, h_{\text{rad}} = 5.9 \text{ W/m}^2\cdot\text{K} \end{aligned}$$

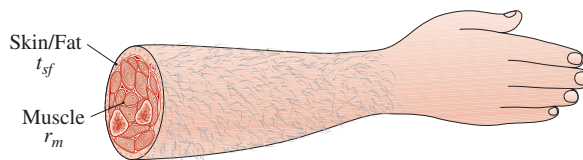


FIGURE P3-150

Heat Transfer in Common Configurations

3-151C What is a conduction shape factor? How is it related to the thermal resistance?

3-152C What is the value of conduction shape factors in engineering?

3-153 A 12-m-long and 8-cm-diameter hot-water pipe of a district heating system is buried in the soil 80 cm below the ground surface. The outer surface temperature of the pipe is 60°C. Taking the surface temperature of the earth to be 2°C and the thermal conductivity of the soil at that location to be 0.9 W/m·K, determine the rate of heat loss from the pipe.

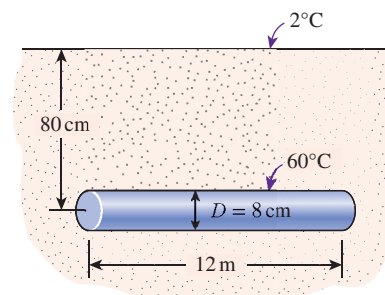


FIGURE P3-153

3-154 A thin-walled cylindrical container is placed horizontally on snow-covered ground. The container is 1.5 m long and has a diameter of 10 cm. The container contains chemicals undergoing an exothermic reaction and generating heat at 900 W/m³. After a severe winter storm, the container is covered with approximately 30 cm of fresh snow. At the surface, the snow has a temperature of -5°C. Determine the surface temperature of the container. Discuss whether or not the snow around the container will begin to melt.

3-155 Hot water at an average temperature of 53°C and an average velocity of 0.4 m/s is flowing through a 5-m section of a thin-walled hot-water pipe that has an outer diameter of 2.5 cm. The pipe passes through the center of a 14-cm-thick wall filled with fiberglass insulation ($k = 0.035 \text{ W/m}\cdot\text{K}$). If the surfaces of the wall are at 18°C, determine (a) the rate of heat transfer from the pipe to the air in the rooms and (b) the temperature drop of the hot water as it flows through this 5-m-long section of the wall. **Answers:** 19.6 W, 0.024°C

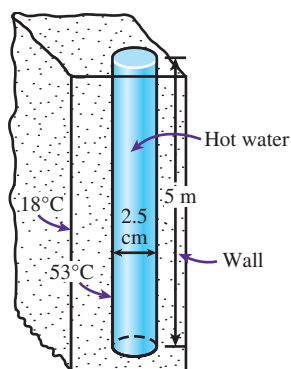


FIGURE P3-155

3-156 Hot- and cold-water pipes 12 m long run parallel to each other in a thick concrete layer. The diameters of both pipes are 6 cm, and the distance between the centerlines of the pipes is 40 cm. The surface temperatures of the hot and cold pipes are 60°C and 15°C, respectively. Taking the thermal conductivity of the concrete to be $k = 0.75 \text{ W/m}\cdot\text{K}$, determine the rate of heat transfer between the pipes. *Answer: 492 W*

3-157  Reconsider Prob. 3-156. Using appropriate software, plot the rate of heat transfer between the pipes as a function of the distance between the centerlines of the pipes in the range of 10 cm to 1.0 m. Discuss the results.

3-158E A row of 3-ft-long and 1-in-diameter used uranium fuel rods that are still radioactive are buried in the ground parallel to each other with a center-to-center distance of 8 in at a depth of 15 ft from the ground surface at a location where the thermal conductivity of the soil is $0.6 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$. If the surface temperatures of the rods and the ground are 350°F and 60°F, respectively, determine the rate of heat transfer from the fuel rods to the atmosphere through the soil.

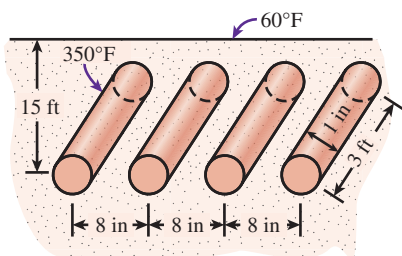


FIGURE P3-158E

3-159 Hot water at an average temperature of 80°C and an average velocity of 1.5 m/s is flowing through a 25-m section of a pipe that has an outer diameter of 5 cm. The pipe extends 2 m in the ambient air above the ground, dips into the ground ($k = 1.5 \text{ W/m}\cdot\text{K}$) vertically for 3 m, and continues horizontally at this depth for 20 m more before it enters the next building.

The first section of the pipe is exposed to the ambient air at 5°C, with a heat transfer coefficient of $22 \text{ W/m}^2\cdot\text{K}$. If the surface of the ground is covered with snow at -3°C , determine (a) the total rate of heat loss from the hot water and (b) the temperature drop of the hot water as it flows through this 25-m-long section of the pipe.

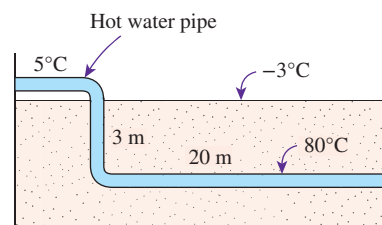


FIGURE P3-159

3-160 Hot water at an average temperature of 90°C passes through a row of eight parallel pipes that are 4 m long and have an outer diameter of 3 cm, located vertically in the middle of a concrete wall ($k = 0.75 \text{ W/m}\cdot\text{K}$) that is 4 m high, 8 m long, and 15 cm thick. If the surfaces of the concrete walls are exposed to a medium at 32°C, with a heat transfer coefficient of $12 \text{ W/m}^2\cdot\text{K}$, determine the rate of heat loss from the hot water and the surface temperature of the wall.

3-161 Two flow passages with different cross-sectional shapes, one circular another square, are each centered in a square solid bar of the same dimension and thermal conductivity. Both configurations have the same length, T_1 , and T_2 . Determine which configuration has the higher rate of heat transfer through the square solid bar for (a) $a = 1.2b$ and (b) $a = 2b$.

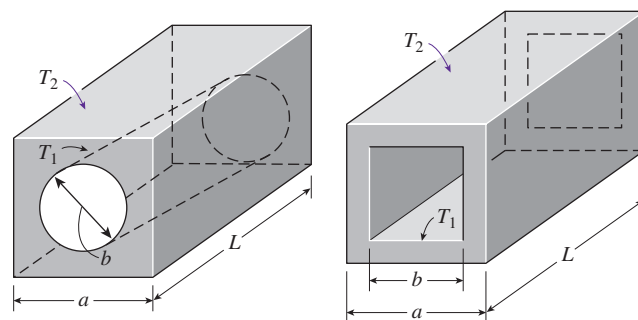


FIGURE P3-161

3-162 Consider a tube for transporting steam that is not centered properly in a cylindrical insulation material ($k = 0.73 \text{ W/m}\cdot\text{K}$). The tube diameter is $D_1 = 20 \text{ cm}$ and the insulation diameter is $D_2 = 40 \text{ cm}$. The distance between the center of the tube and the center of the insulation is $z = 5 \text{ mm}$. If the surface of the tube maintains a temperature of 100°C and the outer surface temperature of the insulation is constant at 30°C ,

determine the rate of heat transfer per unit length of the tube through the insulation.

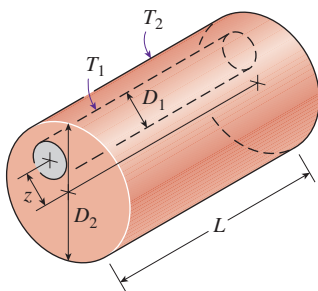


FIGURE P3-162

3-163 Consider two tubes of the same diameter ($D_1 = 20$ cm), length, and surface temperature (T_1). One tube is properly centered in a cylindrical insulation of $D_2 = 40$ cm; the other tube is placed in eccentric with the same diameter cylindrical insulation, where the distance between the center of the tube and the center of the insulation is $z = 5$ mm. If both configurations have the same outer surface temperature T_2 , determine which configuration has the higher rate of heat transfer through the insulation.

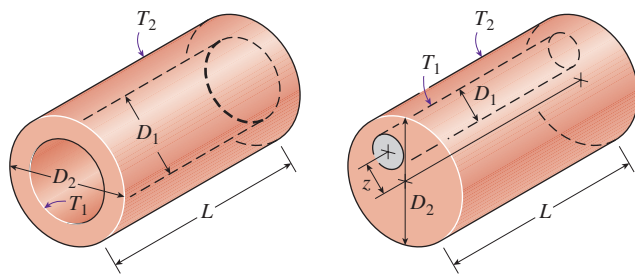


FIGURE P3-163

3-164 Consider a 25-m-long thick-walled concrete duct ($k = 0.75$ W/m·K) of square cross section. The outer dimensions of the duct are 20 cm $\times$ 20 cm, and the thickness of the duct wall is 2 cm. If the inner and outer surfaces of the duct are at 100°C and 30°C, respectively, determine the rate of heat transfer through the walls of the duct. *Answer: 47.1 kW*

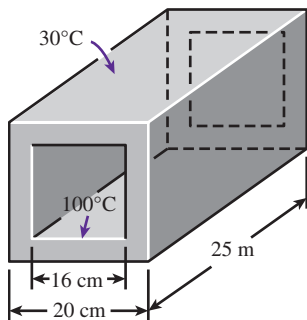


FIGURE P3-164

3-165 Consider a house with a flat roof whose outer dimensions are 12 m $\times$ 12 m. The outer walls of the house are 6 m high. The walls and the roof of the house are made of 20-cm-thick concrete ($k = 0.75$ W/m·K). The temperatures of the inner and outer surfaces of the house are 15°C and 3°C, respectively. Accounting for the effects of the edges of adjoining surfaces, determine the rate of heat loss from the house through its walls and the roof. What is the error involved in ignoring the effects of the edges and corners and treating the roof as a 12-m $\times$ 12-m surface and the walls as 6-m $\times$ 12-m surfaces for simplicity?

3-166 A 3-m-diameter spherical tank containing some radioactive material is buried in the ground ($k = 1.4$ W/m·K). The distance between the top surface of the tank and the ground surface is 4 m. If the surface temperatures of the tank and the ground are 140°C and 15°C, respectively, determine the rate of heat transfer from the tank.

3-167 Radioactive material, stored in a spherical vessel of diameter $D = 3.5$ m, is buried underground at a depth of 10 m. The radioactive material inside the vessel releases heat at a rate of 1000 W/m³. The surface temperature of the vessel is constant at 480°C, and the thermal conductivity of the ground is 2 W/m·K. When it snows in the winter, will the area directly above the spherical vessel be covered with snow?

Special Topics: Heat Transfer Through the Walls and Roofs

3-168C What is the R -value of a wall? How does it differ from the unit thermal resistance of the wall? How is it related to the U -factor?

3-169C What is effective emissivity for a plane-parallel airspace? How is it determined? How is radiation heat transfer through the airspace determined when the effective emissivity is known?

3-170C The unit thermal resistances (R -values) of both 40-mm and 90-mm vertical airspaces are given in Table 3-9 to be 0.22 m²·C/W, which implies that more than doubling the thickness of airspace in a wall has no effect on heat transfer through the wall. Do you think this is a typing error? Explain.

3-171C What is a radiant barrier? What kinds of materials are suitable for use as radiant barriers? Is it worthwhile to use radiant barriers in the attics of homes?

3-172C Consider a house whose attic space is ventilated effectively so that the air temperature in the attic is the same as the ambient air temperature at all times. Will the roof still have any effect on heat transfer through the ceiling? Explain.

3-173 Determine the summer R -value and the U -factor of a wood frame wall that is built around 38-mm $\times$ 140-mm wood studs with a center-to-center distance of 400 mm. The 140-mm-wide cavity between the studs is filled with mineral fiber batt insulation. The inside is finished with 13-mm gypsum wallboard and the outside with 13-mm wood fiberboard and 13-mm $\times$ 200-mm wood bevel lapped siding. The insulated cavity constitutes 80 percent of the heat transmission area, while the studs, headers, plates, and sills constitute 20 percent. *Answers: 3.213 m²·K/W, 0.311 W/m²·K*

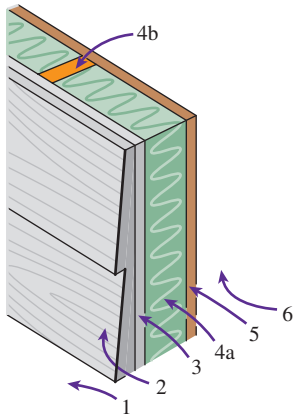


FIGURE P3-173

3-174 The 13-mm-thick wood fiberboard sheathing of the wood stud wall in Prob. 3-173 is replaced by a 25-mm-thick rigid foam insulation. Determine the percent increase in the R -value of the wall as a result.

3-175 The overall heat transfer coefficient (the U -value) of a wall under winter design conditions is $U = 2.25 \text{ W/m}^2\cdot\text{K}$. Now a layer of 100-mm face brick is added to the outside, leaving a 20-mm airspace between the wall and the bricks. Determine the new U -value of the wall. Also, determine the rate of heat transfer through a 3-m-high, 7-m-long section of the wall after modification when the indoor and outdoor temperatures are 22°C and -25°C , respectively.

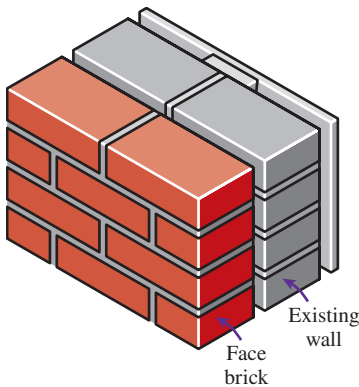


FIGURE P3-175

3-176 Consider a flat ceiling that is built around $38\text{-mm} \times 90\text{-mm}$ wood studs with a center-to-center distance of 400 mm. The lower part of the ceiling is finished with 13-mm gypsum wallboard, while the upper part consists of a wood subfloor ($R = 0.166 \text{ m}^2\cdot^\circ\text{C/W}$), a 13-mm plywood layer, a layer of felt ($R = 0.011 \text{ m}^2\cdot^\circ\text{C/W}$), and linoleum ($R = 0.009 \text{ m}^2\cdot^\circ\text{C/W}$). Both sides of the ceiling are exposed to still air. The airspace constitutes 82 percent of the heat transmission area, while the studs and headers constitute 18 percent. Determine the winter R -value and the U -factor of the ceiling assuming the 90-mm-wide airspace

between the studs (a) does not have any reflective surface, (b) has a reflective surface with $\epsilon = 0.05$ on one side, and (c) has reflective surfaces with $\epsilon = 0.05$ on both sides. Assume a mean temperature of 10°C and a temperature difference of 5.6°C for the airspace.

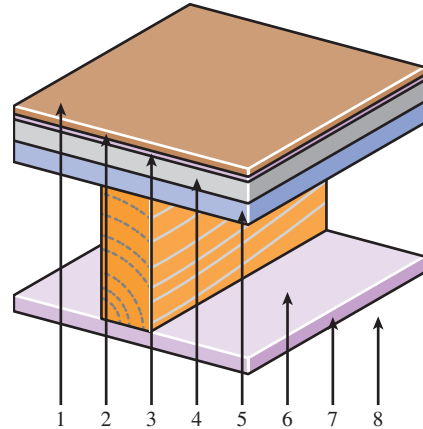


FIGURE P3-176

3-177 Determine the winter R -value and the U -factor of a masonry cavity wall that consists of 100-mm common bricks, a 90-mm airspace, 100-mm concrete blocks made of lightweight aggregate, 20-mm airspace, and 13-mm gypsum wallboard separated from the concrete block by 20-mm-thick (1-in $\times$ 3-in nominal) vertical furring whose center-to-center distance is 400 mm. Neither side of the two airspaces is coated with any reflective films. When determining the R -value of the airspaces, the temperature difference across them can be taken to be 16.7°C with a mean air temperature of 10°C . The airspace constitutes 84 percent of the heat transmission area, while the vertical furring and similar structures constitute 16 percent. **Answers:** $1.02 \text{ m}^2\cdot^\circ\text{C/W}$, $0.978 \text{ W/m}^2\cdot\text{K}$

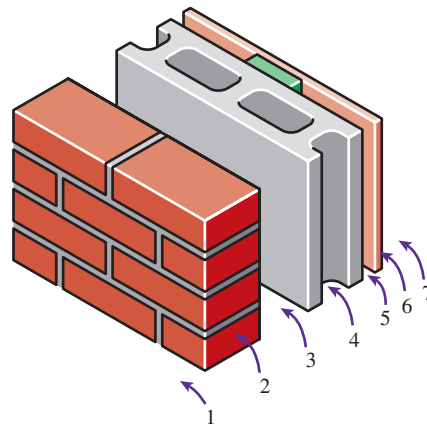


FIGURE P3-177

3-178 Repeat Prob. 3-177 assuming one side of both airspaces is coated with a reflective film of $\epsilon = 0.05$.

3-179 Determine the winter R -value and the U -factor of a masonry wall that consists of the following layers: 100-mm

face bricks, 100-mm common bricks, 25-mm urethane rigid foam insulation, and 13-mm gypsum wallboard. **Answers:** $1.404 \text{ m}^2\cdot^\circ\text{C}/\text{W}$, $0.712 \text{ W}/\text{m}^2\cdot\text{K}$

3-180 The overall heat transfer coefficient (the U -value) of a wall under winter design conditions is $U = 1.40 \text{ W}/\text{m}^2\cdot\text{K}$. Determine the U -value of the wall under summer design conditions.

3-181E Determine the winter R -value and the U -factor of a masonry cavity wall that is built around 4-in-thick concrete blocks made of lightweight aggregate. The outside is finished with 4-in face brick with $\frac{1}{2}$ -in cement mortar between the bricks and concrete blocks. The inside finish consists of $\frac{1}{2}$ -in gypsum wallboard separated from the concrete block by $\frac{3}{4}$ -in-thick (1-in by 3-in nominal) vertical furring whose center-to-center distance is 16 in. Neither side of the $\frac{3}{4}$ -in-thick airspace between the concrete block and the gypsum board is coated with any reflective film. When determining the R -value of the airspace, the temperature difference across it can be taken to be 30°F with a mean air temperature of 50°F . The airspace constitutes 80 percent of the heat transmission area, while the vertical furring and similar structures constitute 20 percent.

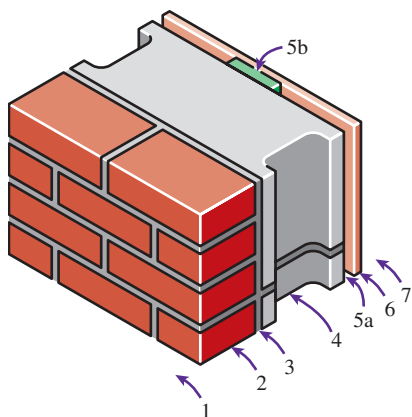


FIGURE P3-181E

3-182 Determine the summer and winter R -values, in $\text{m}^2\cdot^\circ\text{C}/\text{W}$, of a masonry wall that consists of 100-mm face bricks, 13 mm of cement mortar, 100-mm lightweight concrete block, 40-mm airspace, and 20-mm plasterboard. **Answers:** 0.809 and $0.795 \text{ m}^2\cdot^\circ\text{C}/\text{W}$

3-183E The overall heat transfer coefficient of a wall is determined to be $U = 0.075 \text{ Btu}/\text{h}\cdot\text{ft}^2\cdot^\circ\text{F}$ under the conditions of still air inside and winds of 7.5 mph outside. What will the U -factor be when the wind velocity outside is doubled? **Answer:** $0.0755 \text{ Btu}/\text{h}\cdot\text{ft}^2\cdot^\circ\text{F}$

3-184 Two homes are identical, except that the walls of one house consist of 200-mm lightweight concrete blocks, 20-mm airspace, and 20-mm plasterboard, while the walls of the other house involve the standard R -2.4 $\text{m}^2\cdot^\circ\text{C}/\text{W}$ frame wall construction. Which house do you think is more energy efficient?

3-185 Determine the R -value of a ceiling that consists of a layer of 19-mm acoustical tiles whose top surface is covered with a highly reflective aluminum foil for winter conditions. Assume still air below and above the tiles.

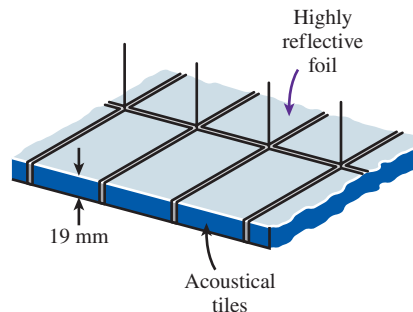


FIGURE P3-185

Review Problems

3-186 Consider two identical people each generating 60 W of metabolic heat steadily while doing sedentary work and dissipating it by convection and perspiration. The first person is wearing clothes made of 1-mm-thick leather ($k = 0.159 \text{ W}/\text{m}\cdot\text{K}$) that covers half of the body, while the second one is wearing clothes made of 1-mm-thick synthetic fabric ($k = 0.13 \text{ W}/\text{m}\cdot\text{K}$) that covers the body completely. The ambient air is at 30°C , the heat transfer coefficient at the outer surface is $15 \text{ W}/\text{m}^2\cdot\text{K}$, and the inner surface temperature of the clothes can be taken to be 32°C . Treating the body of each person as a 25-cm-diameter, 1.7-m-long cylinder, determine the fractions of heat lost from each person by perspiration.

3-187 Cold conditioned air at 12°C is flowing inside a 1.5-cm-thick square aluminum ($k = 237 \text{ W}/\text{m}\cdot\text{K}$) duct of inner cross section $22 \text{ cm} \times 22 \text{ cm}$ at a mass flow rate of $0.8 \text{ kg}/\text{s}$. The duct is exposed to air at 33°C with a combined convection-radiation heat transfer coefficient of $13 \text{ W}/\text{m}^2\cdot\text{K}$. The convection heat transfer coefficient at the inner surface is $75 \text{ W}/\text{m}^2\cdot\text{K}$. If the air temperature in the duct should not increase by more than 1°C , determine the maximum length of the duct.

3-188 Hot water is flowing at an average velocity of $1.5 \text{ m}/\text{s}$ through a cast iron pipe ($k = 52 \text{ W}/\text{m}\cdot\text{K}$) whose inner and outer diameters are 3 cm and 3.5 cm, respectively. The pipe passes through a 15-m-long section of a basement whose temperature is 15°C . If the temperature of the water drops from 70°C to 67°C as it passes through the basement and the heat transfer coefficient on the inner surface of the pipe is $400 \text{ W}/\text{m}^2\cdot\text{K}$, determine the combined convection and radiation heat transfer coefficient at the outer surface of the pipe. **Answer:** $272.5 \text{ W}/\text{m}^2\cdot\text{K}$

3-189 The plumbing system of a house involves a 0.5-m section of a plastic pipe ($k = 0.16 \text{ W}/\text{m}\cdot\text{K}$) of inner diameter 2 cm and outer diameter 2.4 cm exposed to the ambient air. During a cold and windy night, the ambient air temperature remains at about -5°C for a period of 14 h. The combined convection and radiation heat transfer coefficient on the outer surface of the

pipe is estimated to be $40 \text{ W/m}^2\cdot\text{K}$, and the heat of fusion of water is 333.7 kJ/kg . Assuming the pipe to contain stationary water initially at 0°C , determine if the water in that section of the pipe will completely freeze that night.

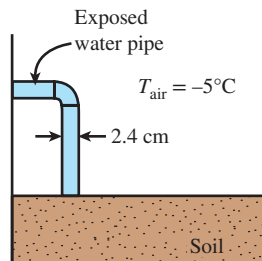


FIGURE P3–189

3–190 A cylindrical nuclear fuel rod of 15 mm in diameter is encased in a concentric hollow ceramic cylinder with inner diameter of 35 mm and outer diameter of 110 mm. This created an air gap between the fuel rod and the hollow ceramic cylinder with a convection heat transfer coefficient of $10 \text{ W/m}^2\cdot\text{K}$. The hollow ceramic cylinder has a thermal conductivity of $0.07 \text{ W/m}\cdot\text{K}$, and its outer surface maintains a constant temperature of 30°C . If the fuel rod generates heat at a rate of 1 MW/m^3 , determine the temperature at the surface of the fuel rod. *Answer: 1026°C*

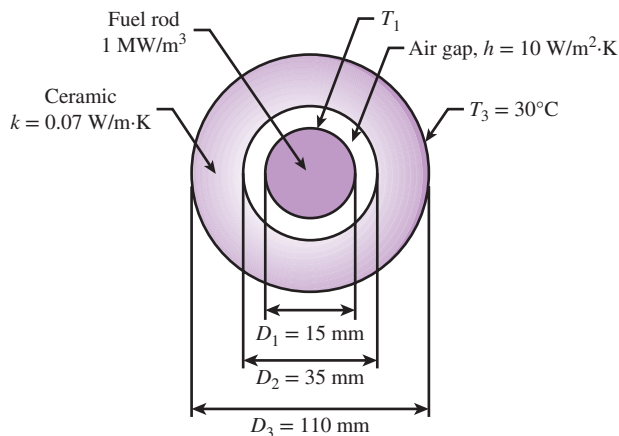


FIGURE P3–190

3–191 Steam at 260°C is flowing inside a steel pipe ($k = 61 \text{ W/m}\cdot\text{K}$) whose inner and outer diameters are 10 cm and 12 cm, respectively, in an environment at 20°C . The heat transfer coefficients inside and outside the pipe are $120 \text{ W/m}^2\cdot\text{K}$ and $14 \text{ W/m}^2\cdot\text{K}$, respectively. Determine (a) the thickness of the insulation ($k = 0.038 \text{ W/m}\cdot\text{K}$) needed to reduce the heat loss by 95 percent and (b) the thickness of the insulation needed to reduce the exposed surface temperature of insulated pipe to 40°C for safety reasons.

3–192 A spherical vessel, 3.0 m in diameter (and negligible wall thickness), is used for storing a fluid at a temperature of 0°C . The vessel is covered with a 5.0-cm-thick layer of an insulation ($k = 0.20 \text{ W/m}\cdot\text{K}$). The surrounding air is at 22°C . The inside and outside heat transfer coefficients are 40 and $10 \text{ W/m}^2\cdot\text{K}$, respectively. Calculate (a) all thermal resistances, in K/W , (b) the steady rate of heat transfer, and (c) the temperature difference across the insulation layer.

3–193 One wall of a refrigerated warehouse is 10.0 m high and 5.0 m wide. The wall is made of three layers: 1.0-cm-thick aluminum ($k = 200 \text{ W/m}\cdot\text{K}$), 8.0-cm-thick fiberglass ($k = 0.038 \text{ W/m}\cdot\text{K}$), and 3.0-cm-thick gypsum board ($k = 0.48 \text{ W/m}\cdot\text{K}$). The warehouse inside and outside temperatures are -10°C and 20°C , respectively, and the average value of both inside and outside heat transfer coefficients is $40 \text{ W/m}^2\cdot\text{K}$.

- Calculate the rate of heat transfer across the warehouse wall in steady operation.
- Suppose that 400 metal bolts ($k = 43 \text{ W/m}\cdot\text{K}$), each 2.0 cm in diameter and 12.0 cm long, are used to fasten (i.e., hold together) the three wall layers. Calculate the rate of heat transfer for the “bolted” wall.
- What is the percent change in the rate of heat transfer across the wall due to metal bolts?

3–194 A 4-m-high and 6-m-long wall is constructed of two large 0.8-cm-thick steel plates ($k = 15 \text{ W/m}\cdot\text{K}$) separated by 1-cm-thick and 22-cm-wide steel bars placed 99 cm apart. The remaining space between the steel plates is filled with fiberglass insulation ($k = 0.035 \text{ W/m}\cdot\text{K}$). If the temperature difference between the inner and the outer surfaces of the walls is 22°C , determine the rate of heat transfer through the wall. Can we ignore the steel bars between the plates in heat transfer analysis since they occupy only 1 percent of the heat transfer surface area?

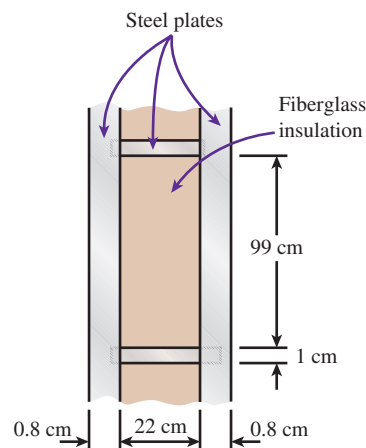


FIGURE P3–194

3–195 A typical section of a building wall is shown in Fig. P3–195. This section extends in and out of the page and is repeated in the vertical direction. The wall support members are made of steel ($k = 50 \text{ W/m}\cdot\text{K}$). The support members

are 8 cm (t_{23}) $\times$ 0.5 cm (L_B). The remainder of the inner wall space is filled with insulation ($k = 0.03$ W/m·K) and measures 8 cm (t_{23}) $\times$ 60 cm (L_B). The inner wall is made of gypsum board ($k = 0.5$ W/m·K) that is 1 cm thick (t_{12}), and the outer wall is made of brick ($k = 1.0$ W/m·K) that is 10 cm thick (t_{34}). What is the temperature on the interior brick surface, 3, when $T_1 = 20^\circ\text{C}$ and $T_4 = 35^\circ\text{C}$?

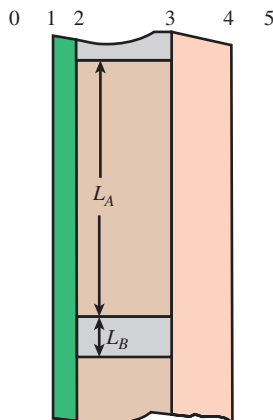


FIGURE P3-195

3-196 A 12-cm-long bar with a square cross section, as shown in Fig. P3-196, consists of a 1-cm-thick copper layer ($k = 380$ W/m·K) and a 1-cm-thick epoxy composite layer ($k = 0.4$ W/m·K). Calculate the rate of heat transfer under a thermal driving force of 50°C when the direction of steady one-dimensional heat transfer is (a) from front to back (i.e., along its length), (b) from left to right, and (c) from top to bottom.

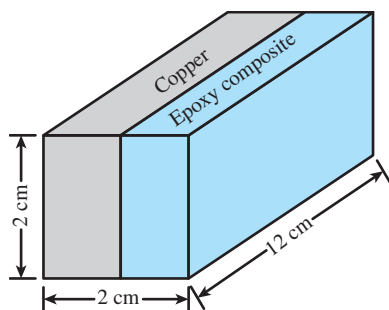


FIGURE P3-196

3-197 Circular fins of uniform cross section, with diameter of 10 mm and length of 50 mm, are attached to a wall with surface temperature of 350°C . The fins are made of material with thermal conductivity of 240 W/m·K, they are exposed to an ambient air condition of 25°C , and the convection heat transfer coefficient is 250 W/m²·K. Determine the heat transfer rate and plot the temperature variation of a single fin for the following boundary conditions:

- Infinitely long fin
- Adiabatic fin tip
- Fin with tip temperature of 250°C
- Convection from the fin tip

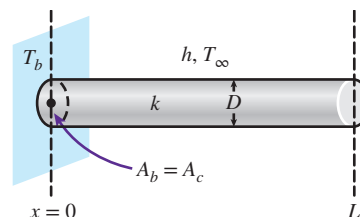


FIGURE P3-197

3-198 A total of 10 rectangular aluminum fins ($k = 203$ W/m·K) are placed on the outside flat surface of an electronic device. Each fin is 100 mm wide, 20 mm high, and 4 mm thick. The fins are located parallel to each other at a center-to-center distance of 8 mm. The temperature at the outside surface of the electronic device is 72°C . The air is at 20°C , and the heat transfer coefficient is 80 W/m²·K. Determine (a) the rate of heat loss from the electronic device to the surrounding air and (b) the fin effectiveness.

3-199 A plane wall with surface temperature of 300°C is attached with straight aluminum triangular fins ($k = 236$ W/m·K). The fins are exposed to an ambient air condition of 25°C , and the convection heat transfer coefficient is 25 W/m²·K. Each fin has a length of 55 mm, a base 4 mm thick, and a width of 110 mm. Using Table 3-3, determine the efficiency, heat transfer rate, and effectiveness of each fin.

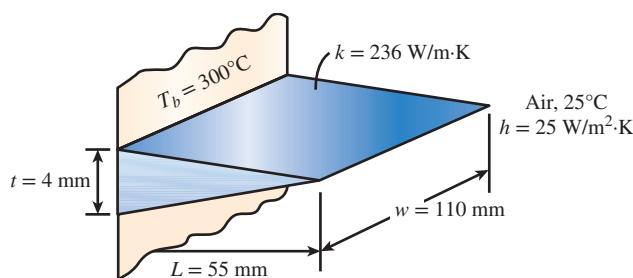


FIGURE P3-199

3-200 A plane wall surface at 200°C is to be cooled with aluminum pin fins of parabolic profile with blunt tips. Each fin has a length of 25 mm and a base diameter of 4 mm. The fins are exposed to ambient air at 25°C , and the heat transfer coefficient is 45 W/m²·K. If the thermal conductivity of the fins is 230 W/m·K, determine the heat transfer rate from a single fin and the increase in the rate of heat transfer per m² surface area as a result of attaching fins. Assume there are 100 fins per m² surface area.

3-201 Steam in a heating system flows through tubes whose outer diameter is 3 cm and whose walls are maintained at a temperature of 120°C . Circular aluminum alloy fins ($k = 180$ W/m·K) of outer diameter 6 cm and constant thickness $t = 2$ mm are attached to the tube, as shown in Fig. P3-201. The space between the fins is 3 mm, and thus there are 200 fins per meter length of the tube. Heat is transferred to the surrounding air at 25°C , with a combined heat transfer coefficient of 60 W/m²·K. Determine the increase in heat transfer from the tube per meter of its length as a result of adding fins.

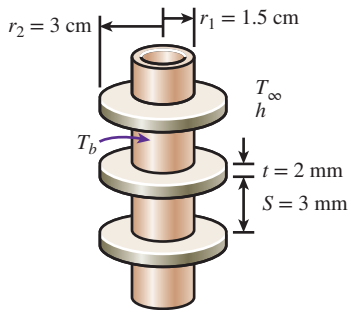


FIGURE P3-201

3-202 A 0.2-cm-thick, 10-cm-high, and 15-cm-long circuit board houses electronic components on one side that dissipate a total of 15 W of heat uniformly. The board is impregnated with conducting metal fillings and has an effective thermal conductivity of 12 W/m·K. All the heat generated in the components is conducted across the circuit board and is dissipated from the back side of the board to a medium at 37°C, with a heat transfer coefficient of 45 W/m²·K. (a) Determine the surface temperatures on the two sides of the circuit board. (b) Now a 0.1-cm-thick, 10-cm-high, and 15-cm-long aluminum plate ($k = 237$ W/m·K) with 20 0.2-cm-thick, 2-cm-long, and 15-cm-wide aluminum fins of rectangular profile are attached to the back side of the circuit board with a 0.03-cm-thick epoxy adhesive ($k = 1.8$ W/m·K). Determine the new temperatures on the two sides of the circuit board.

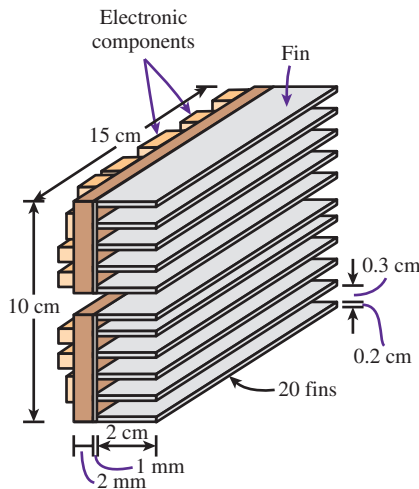


FIGURE P3-202

3-203  In a manufacturing plant, 100-mm by 40-mm thin rectangular electronic devices are assembled in mass quantity. The top surface of the electronic device is made of aluminum and is attached with an HS 5030 aluminum heat sink positioned horizontally. The electronic device dissipates 45 W of heat through the heat sink. Both aluminum surfaces have a roughness of about 10 μ m and an average interface pressure of 1 atm. To prevent the electronic device from overheating,

the top surface temperature should be kept below 85°C in an ambient surrounding of 30°C. Your task as a product engineer is to ensure that the heat sink attached on the device's top surface is able to keep the device from overheating. Determine whether or not the top surface temperature of the electronic device, with an HS 5030 heat sink attached, is higher than 85°C. If so, what action can be taken to reduce the surface temperature to below 85°C? Since your company has a vast quantity of the HS 5030 heat sinks in inventory, replacing the heat sinks with more effective ones is too costly and is not a viable solution.

3-204 A row of 10 parallel pipes that are 5 m long and have an outer diameter of 6 cm are used to transport steam at 145°C through the concrete floor ($k = 0.75$ W/m·K) of a 10-m $\times$ 5-m room that is maintained at 24°C. The combined convection and radiation heat transfer coefficient at the floor is 12 W/m²·K. If the surface temperature of the concrete floor is not to exceed 38°C, determine how deep the steam pipes should be buried below the surface of the concrete floor.

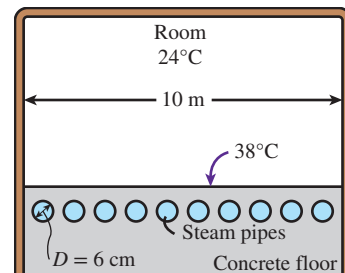


FIGURE P3-204

3-205 A 0.6-m-diameter, 1.9-m-long cylindrical tank containing liquefied natural gas (LNG) at -160°C is placed at the center of a 1.9-m-long 1.4-m $\times$ 1.4-m square solid bar made of an insulating material with $k = 0.0002$ W/m·K. If the outer surface temperature of the bar is 12°C, determine the rate of heat transfer to the tank. Also, determine the LNG temperature after one month. Take the density and the specific heat of LNG to be 425 kg/m³ and 3.475 kJ/kg·°C, respectively.

3-206 In a combined heat and power (CHP) generation process, by-product heat is used for domestic or industrial heating

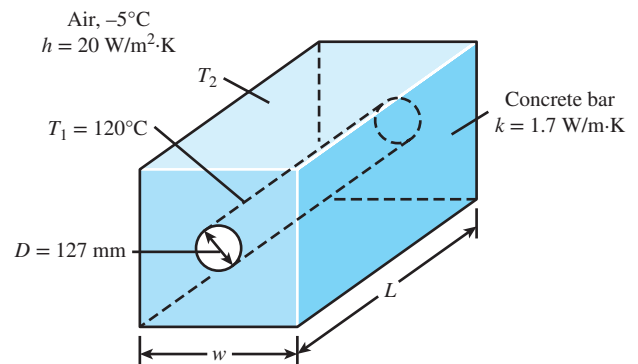


FIGURE P3-206

purposes. Hot steam is carried from a CHP generation plant by a tube with diameter of 127 mm centered at a square cross-section solid bar made of concrete with thermal conductivity of $1.7 \text{ W/m}\cdot\text{K}$. The surface temperature of the tube is constant at 120°C , while the square concrete bar is exposed to air with temperature of -5°C and convection heat transfer coefficient of $20 \text{ W/m}^2\cdot\text{K}$. If the temperature difference between the outer surface of the square concrete bar and the ambient air is to be maintained at 5°C , determine the width of the square concrete bar and the rate of heat loss per meter length. *Answers: 1.32 m, 530 W/m*

3-207 A 2.2-m-diameter spherical steel tank filled with iced water at 0°C is buried underground at a location where the thermal conductivity of the soil is $k = 0.55 \text{ W/m}\cdot\text{K}$. The distance between the tank center and the ground surface is 2.4 m. For a ground surface temperature of 18°C , determine the rate of heat transfer to the iced water in the tank. What would your answer be if the soil temperature were 18°C and the ground surface were insulated?

3-208 A thin-walled spherical tank is buried in the ground at a depth of 3 m. The tank has a diameter of 1.5 m, and it contains chemicals undergoing exothermic reaction that provides a uniform heat flux of 1 kW/m^2 to the tank's inner surface. From soil analysis, the ground has a thermal conductivity of $1.3 \text{ W/m}\cdot\text{K}$ and a temperature of 10°C . Determine the surface temperature of the tank. Discuss the effect of the ground depth on the surface temperature of the tank.

Fundamentals of Engineering (FE) Exam Problems

3-209 Heat is lost at a rate of 275 W per m^2 area of a 15-cm-thick wall with a thermal conductivity of $k = 1.1 \text{ W/m}\cdot\text{K}$. The temperature drop across the wall is

- (a) 37.5°C (b) 27.5°C (c) 16.0°C
(d) 8.0°C (e) 4.0°C

3-210 Consider a wall that consists of two layers, A and B, with the following values: $k_A = 1.2 \text{ W/m}\cdot\text{K}$, $L_A = 8 \text{ cm}$, $k_B = 0.2 \text{ W/m}\cdot\text{K}$, $L_B = 5 \text{ cm}$. If the temperature drop across the wall is 18°C , the rate of heat transfer through the wall per unit area of the wall is

- (a) 56.8 W/m^2 (b) 72.1 W/m^2 (c) 114 W/m^2
(d) 201 W/m^2 (e) 270 W/m^2

3-211 Heat is generated steadily in a 3-cm-diameter spherical ball. The ball is exposed to ambient air at 26°C with a heat transfer coefficient of $7.5 \text{ W/m}^2\cdot\text{K}$. The ball is to be covered with a material of thermal conductivity $0.15 \text{ W/m}\cdot\text{K}$. The thickness of the covering material that will maximize heat generation within the ball while keeping ball surface temperature constant is

- (a) 0.5 cm (b) 1.0 cm (c) 1.5 cm (d) 2.0 cm (e) 2.5 cm

3-212 Consider a 1.5-m-high and 2-m-wide triple pane window. The thickness of each glass layer ($k = 0.80 \text{ W/m}\cdot\text{K}$) is 0.5 cm, and the thickness of each airspace ($k = 0.025 \text{ W/m}\cdot\text{K}$) is 1.2 cm. If the inner and outer surface temperatures of the

window are 10°C and 0°C , respectively, the rate of heat loss through the window is

- (a) 3.4 W (b) 10.2 W (c) 30.7 W (d) 61.7 W (e) 86.8 W

3-213 Consider two metal plates pressed against each other. Other things being equal, which of the measures below will cause the thermal contact resistance to increase?

- (a) Cleaning the surfaces to make them shinier.
(b) Pressing the plates against each other with a greater force.
(c) Filling the gap with a conducting fluid.
(d) Using softer metals.
(e) Coating the contact surfaces with a thin layer of soft metal such as tin.

3-214 A 10-m-long, 8-cm-outer-radius cylindrical steam pipe is covered with 3-cm-thick cylindrical insulation with a thermal conductivity of $0.05 \text{ W/m}\cdot\text{K}$. If the rate of heat loss from the pipe is 1000 W , the temperature drop across the insulation is

- (a) 58°C (b) 101°C (c) 143°C (d) 282°C (e) 600°C

3-215 A 5-m-diameter spherical tank is filled with liquid oxygen ($\rho = 1141 \text{ kg/m}^3$, $c_p = 1.71 \text{ kJ/kg}\cdot^\circ\text{C}$) at -184°C . It is observed that the temperature of oxygen increases to -183°C in a 144-hour period. The average rate of heat transfer to the tank is

- (a) 124 W (b) 185 W (c) 246 W (d) 348 W (e) 421 W

3-216 A 2.5-m-high, 4-m-wide, and 20-cm-thick wall of a house has a thermal resistance of 0.025°C/W . The thermal conductivity of the wall is

- (a) $0.8 \text{ W/m}\cdot\text{K}$ (b) $1.2 \text{ W/m}\cdot\text{K}$ (c) $3.4 \text{ W/m}\cdot\text{K}$
(d) $5.2 \text{ W/m}\cdot\text{K}$ (e) $8.0 \text{ W/m}\cdot\text{K}$

3-217 Consider two walls, A and B, with the same surface areas and the same temperature drops across their thicknesses. The ratio of thermal conductivities is $k_A/k_B = 4$ and the ratio of the wall thicknesses is $L_A/L_B = 2$. The ratio of heat transfer rates through the walls $\dot{Q}_A/\dot{Q}_B$ is

- (a) 0.5 (b) 1 (c) 2 (d) 4 (e) 8

3-218 A hot plane surface at 100°C is exposed to air at 25°C with a combined heat transfer coefficient of $20 \text{ W/m}^2\cdot\text{K}$. The heat loss from the surface is to be reduced by half by covering it with sufficient insulation with a thermal conductivity of $0.10 \text{ W/m}\cdot\text{K}$. Assuming the heat transfer coefficient to remain constant, the required thickness of insulation is

- (a) 0.1 cm (b) 0.5 cm (c) 1.0 cm (d) 2.0 cm (e) 5 cm

3-219 A room at 20°C air temperature is losing heat to the outdoor air at 0°C at a rate of 1000 W through a 2.5-m-high and 4-m-long wall. Now the wall is insulated with 2-cm-thick insulation with a conductivity of $0.02 \text{ W/m}\cdot\text{K}$. Determine the rate of heat loss from the room through this wall after insulation. Assume the heat transfer coefficients on the inner and outer surfaces of the wall, the room air temperature, and the outdoor air temperature remain unchanged. Also, disregard radiation.

- (a) 20 W (b) 561 W (c) 388 W (d) 167 W (e) 200 W

3-220 A 1-cm-diameter, 30-cm-long fin made of aluminum ($k = 237 \text{ W/m}\cdot\text{K}$) is attached to a surface at 80°C . The surface is exposed to ambient air at 22°C with a heat transfer coefficient of $18 \text{ W/m}^2\cdot\text{K}$. If the fin can be assumed to be very long, the rate of heat transfer from the fin is

- (a) 2.0 W (b) 3.2 W (c) 4.4 W (d) 5.5 W (e) 6.0 W

3-221 A 1-cm-diameter, 30-cm-long fin made of aluminum ($k = 237 \text{ W/m}\cdot\text{K}$) is attached to a surface at 80°C . The surface is exposed to ambient air at 22°C with a heat transfer coefficient of $11 \text{ W/m}^2\cdot\text{K}$. If the fin can be assumed to be very long, its efficiency is

- (a) 0.60 (b) 0.67 (c) 0.72 (d) 0.77 (e) 0.88

3-222 A hot surface at 80°C in air at 20°C is to be cooled by attaching 10-cm-long and 1-cm-diameter cylindrical fins. The combined heat transfer coefficient is $30 \text{ W/m}^2\cdot\text{K}$, and heat transfer from the fin tip is negligible. If the fin efficiency is 0.75, the rate of heat loss from 100 fins is

- (a) 325 W (b) 707 W (c) 566 W (d) 424 W (e) 754 W

3-223 A cylindrical pin fin of diameter 0.6 cm and length of 3 cm with negligible heat loss from the tip has an efficiency of 0.7. The effectiveness of this fin is

- (a) 0.3 (b) 0.7 (c) 2 (d) 8 (e) 14

3-224 A 3-cm-long, 2-mm $\times$ 2-mm rectangular cross-section aluminum fin ($k = 237 \text{ W/m}\cdot\text{K}$) is attached to a surface. If the fin efficiency is 65 percent, the effectiveness of this single fin is

- (a) 39 (b) 30 (c) 24 (d) 18 (e) 7

3-225 Two finned surfaces with long fins are identical, except that the convection heat transfer coefficient for the first finned surface is twice that of the second one. Which of the following conditions is accurate for the efficiency and effectiveness of the first finned surface relative to the second one?

- (a) Higher efficiency and higher effectiveness
(b) Higher efficiency but lower effectiveness
(c) Lower efficiency but higher effectiveness
(d) Lower efficiency and lower effectiveness
(e) Equal efficiency and equal effectiveness

3-226 A 20-cm-diameter hot sphere at 120°C is buried in the ground with a thermal conductivity of $1.2 \text{ W/m}\cdot\text{K}$. The distance between the center of the sphere and the ground surface is 0.8 m, and the ground surface temperature is 15°C . The rate of heat loss from the sphere is

- (a) 169 W (b) 20 W (c) 217 W (d) 312 W (e) 1.8 W

3-227 A 25-cm-diameter, 2.4-m-long vertical cylinder containing ice at 0°C is buried right under the ground. The cylinder is thin-shelled and is made of a high-thermal-conductivity material. The surface temperature and the thermal conductivity of the ground are 18°C and $0.85 \text{ W/m}\cdot\text{K}$, respectively. The rate of heat transfer to the cylinder is

- (a) 37.2 W (b) 63.2 W (c) 158 W
(d) 480 W (e) 1210 W

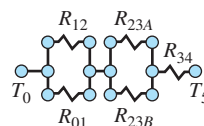
3-228 Hot water ($c_p = 4.179 \text{ kJ/kg}\cdot\text{K}$) flows through an 80-m-long PVC ($k = 0.092 \text{ W/m}\cdot\text{K}$) pipe whose inner diameter is 2 cm and outer diameter is 2.5 cm at a rate of 1 kg/s, entering at 40°C . If the entire interior surface of this pipe is maintained at 35°C and the entire exterior surface at 20°C , the outlet temperature of water is

- (a) 35°C (b) 36°C (c) 37°C (d) 38°C (e) 39°C

3-229 The walls of a food storage facility are made of a 2-cm-thick layer of wood ($k = 0.1 \text{ W/m}\cdot\text{K}$) in contact with a 5-cm-thick layer of polyurethane foam ($k = 0.03 \text{ W/m}\cdot\text{K}$). If the temperature of the surface of the wood is -10°C and the temperature of the surface of the polyurethane foam is 20°C , the temperature of the surface where the two layers are in contact is

- (a) -7°C (b) -2°C (c) 3°C (d) 8°C (e) 11°C

3-230 The equivalent thermal resistance for the thermal circuit shown here is



- (a) $R_{12}R_{01} + R_{23A} + R_{23B} + R_{34}$

- (b) $R_{12}R_{01} + \left(\frac{R_{23A}R_{23B}}{R_{23A} + R_{23B}} \right) + R_{34}$

- (c) $\left(\frac{R_{12}R_{01}}{R_{12} + R_{01}} \right) + \left(\frac{R_{23A}R_{23B}}{R_{23A} + R_{23B}} \right) + \frac{1}{R_{34}}$

- (d) $\left(\frac{R_{12}R_{01}}{R_{12} + R_{01}} \right) + \left(\frac{R_{23A}R_{23B}}{R_{23A} + R_{23B}} \right) + R_{34}$

- (e) None of them

3-231 The 700 m² ceiling of a building has a thermal resistance of $0.52 \text{ m}^2\cdot\text{K/W}$. The rate at which heat is lost through this ceiling on a cold winter day when the ambient temperature is -10°C and the interior is at 20°C is

- (a) 23.1 kW (b) 40.4 kW (c) 55.6 kW
(d) 68.1 kW (e) 88.6 kW

3-232 A 1-m-inner-diameter liquid-oxygen storage tank at a hospital keeps the liquid oxygen at 90 K. The tank consists of a 0.5-cm-thick aluminum ($k = 170 \text{ W/m}\cdot\text{K}$) shell whose exterior is covered with a 10-cm-thick layer of insulation ($k = 0.02 \text{ W/m}\cdot\text{K}$). The insulation is exposed to the ambient air at 20°C , and the heat transfer coefficient on the exterior side of the insulation is $5 \text{ W/m}^2\cdot\text{K}$. The rate at which the liquid oxygen gains heat is

- (a) 141 W (b) 176 W (c) 181 W (d) 201 W (e) 221 W

3-233 A 1-m-inner-diameter liquid-oxygen storage tank at a hospital keeps the liquid oxygen at 90 K. The tank consists of a 0.5-cm-thick aluminum ($k = 170 \text{ W/m}\cdot\text{K}$) shell whose exterior

is covered with a 10-cm-thick layer of insulation ($k = 0.02 \text{ W/m}\cdot\text{K}$). The insulation is exposed to the ambient air at 20°C , and the heat transfer coefficient on the exterior side of the insulation is $5 \text{ W/m}^2\cdot\text{K}$. The temperature of the exterior surface of the insulation is

- (a) 13°C (b) 9°C (c) 2°C (d) -3°C (e) -12°C

3-234 The fin efficiency is defined as the ratio of the actual heat transfer from the fin to

- (a) The heat transfer from the same fin with an adiabatic tip
 (b) The heat transfer from an equivalent fin which is infinitely long
 (c) The heat transfer from the same fin if the temperature along the entire length of the fin is the same as the base temperature
 (d) The heat transfer through the base area of the same fin
 (e) None of the above

3-235 Computer memory chips are mounted on a finned metallic mount to protect them from overheating. A 152-MB memory chip dissipates 5 W of heat to air at 25°C . If the temperature of this chip is not to exceed 60°C , the overall heat transfer coefficient–area product of the finned metal mount must be at least

- (a) $0.14 \text{ W}/^\circ\text{C}$ (b) $0.20 \text{ W}/^\circ\text{C}$ (c) $0.32 \text{ W}/^\circ\text{C}$
 (d) $0.48 \text{ W}/^\circ\text{C}$ (e) $0.76 \text{ W}/^\circ\text{C}$

3-236 In the United States, building insulation is specified by the R -value (thermal resistance in $\text{h}\cdot\text{ft}^2\cdot^\circ\text{F}/\text{Btu}$ units). A homeowner decides to save on the cost of heating the home by adding additional insulation in the attic. If the total R -value is increased from 15 to 25, the homeowner can expect the heat loss through the ceiling to be reduced by

- (a) 25% (b) 40% (c) 50% (d) 60% (e) 75%

3-237 A triangular-shaped fin on a motorcycle engine is 0.5 cm thick at its base and 3 cm long (normal distance between the base and the tip of the triangle), and is made of aluminum ($k = 150 \text{ W/m}\cdot\text{K}$). This fin is exposed to air with a convective heat transfer coefficient of $30 \text{ W/m}^2\cdot\text{K}$ acting on its surfaces. The efficiency of the fin is 75 percent. If the fin base temperature is 130°C and the air temperature is 25°C , the heat transfer from this fin per unit width is

- (a) 32 W/m (b) 57 W/m (c) 102 W/m
 (d) 124 W/m (e) 142 W/m

3-238 A plane brick wall ($k = 0.7 \text{ W/m}\cdot\text{K}$) is 10 cm thick. The thermal resistance of this wall per unit of wall area is

- (a) $0.143 \text{ m}^2\cdot\text{K}/\text{W}$ (b) $0.250 \text{ m}^2\cdot\text{K}/\text{W}$
 (c) $0.327 \text{ m}^2\cdot\text{K}/\text{W}$ (d) $0.448 \text{ m}^2\cdot\text{K}/\text{W}$
 (e) $0.524 \text{ m}^2\cdot\text{K}/\text{W}$

Design and Essay Problems

3-239 The temperature in deep space is close to absolute zero, which presents thermal challenges for the astronauts who do space walks. Propose a design for the clothing of the astronauts that will be most suitable for the thermal environment in space. Defend the selections in your design.

3-240 In the design of electronic components, it is desirable to attach the electronic circuitry to a substrate material that is a very good thermal conductor but also a very effective electrical insulator. If high cost is not a major concern, what material would you propose for the substrate?

3-241 Using cylindrical samples of the same material, devise an experiment to determine the thermal contact resistance. Cylindrical samples are available at any length, and the thermal conductivity of the material is known.

3-242 Find out about the wall construction of the cabins of large commercial airplanes, the range of ambient conditions under which they operate, typical heat transfer coefficients on the inner and outer surfaces of the wall, and the heat generation rates inside. Determine the size of the heating and air-conditioning system that will be able to maintain the cabin at 20°C at all times for an airplane capable of carrying 400 people.

3-243 Repeat Prob. 3-242 for a submarine with a crew of 60 people.

3-244 A house with 200-m² floor space is to be heated with geothermal water flowing through pipes laid in the ground under the floor. The walls of the house are 4 m high, and there are 10 single-paned windows in the house that are 1.2 m wide and 1.8 m high. The house has R -19 (in $\text{h}\cdot\text{ft}^2\cdot^\circ\text{F}/\text{Btu}$) insulation in the walls and R -30 in the ceiling. The floor temperature is not to exceed 40°C . Hot geothermal water is available at 90°C , and the inner and outer diameters of the pipes to be used are 2.4 cm and 3.0 cm. Design such a heating system for this house in your area.

3-245 Using a timer (or watch) and a thermometer, conduct this experiment to determine the rate of heat gain of your refrigerator. First, make sure that the door of the refrigerator is not opened for at least a few hours to make sure that steady operating conditions are established. Start the timer when the refrigerator stops running, and measure the time Δt_1 it stays off before it kicks in. Then measure the time Δt_2 it stays on. Noting that the heat removed during Δt_2 is equal to the heat gain of the refrigerator during $\Delta t_1 + \Delta t_2$ and using the power consumed by the refrigerator when it is running, determine the average rate of heat gain for your refrigerator, in watts. Take the COP (coefficient of performance) of your refrigerator to be 1.3 if it is not available.

Now, clean the condenser coils of the refrigerator and remove any obstacles in the way of airflow through the coils. Then determine the improvement in the COP of the refrigerator.